Infections & Inflammation Rapid Revision

Campbell-Walsh-Wein · Chapters 26–31

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 6h 06min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 26* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
26	Infections of the Urinary Tract	106 min
27	Prostatitis, Orchitis & Pain Conditions	52 min
28	Interstitial Cystitis / Bladder Pain Syndrome	56 min
29	Sexually Transmitted Diseases	37 min
30	Cutaneous Diseases of the External Genitalia	48 min
31	Tuberculosis & Parasitic Infections	61 min
★	Putting it together: infection decisions	6 min
	Total (25776 words, at 70 words/min)	**6h 06min**

26 Infections of the Urinary Tract

Reading time ≈ 106 min (slow pace)

1 · DEFINITIONS AND CLASSIFICATION

- **Urinary tract infection (UTI)**: inflammatory response of the urothelium to bacterial invasion, usually with bacteriuria and pyuria. Urine is **not sterile**, and bacteriuria may even protect against UTI.
- **Bacteriuria** = bacteria in urine (colonisation, infection or contamination). **Pyuria** = white blood cells (WBCs) in urine (infection or inflammation from stone, foreign body, etc.). Neither is synonymous with UTI.
- Bacteriuria without pyuria → colonisation. **Sterile pyuria** → needs evaluation (tuberculosis [TB] → *Ch 31*, stone, urothelial injury, interstitial nephritis).
- **Cystitis**: dysuria, frequency, urgency ± suprapubic pain. Mimics: urethritis, vaginitis, interstitial cystitis/painful bladder syndrome (IC/PBS) → *Ch 28*, bladder carcinoma, calculi.
- **Acute pyelonephritis**: chills, fever, flank pain + bacteriuria and pyuria. Diagnose clinically, not on radiological perinephric stranding alone. Flank symptoms may be absent in spinal cord injury (SCI).
- **Chronic pyelonephritis**: atrophic, fibrotic kidney. Infective scar = focal coarse cortical scar over a **distorted calyx**, radiologically identical to vesicoureteral reflux (VUR) or calyceal-stone scarring. Uniform global cortical thinning is uncommon and suggests obstruction with infection.

	Uncomplicated UTI	**Complicated UTI**
Host	Healthy, non-pregnant woman	Compromised host and/or virulent or resistant organism
Tract	Structurally and functionally normal	Structural or functional abnormality, catheter
Organism	Eradicated by short oral course	Frequently multidrug-resistant (MDR) → culture mandatory

- **Factors making a UTI complicated**: anatomical or functional abnormality (benign prostatic hyperplasia [BPH], diverticula, ureteric obstruction, neurogenic bladder [NGB], bladder outlet obstruction [BOO], concentrating defects), **male sex**, pregnancy, elderly, diabetes mellitus (DM), immunosuppression, childhood UTI, recent antibiotics, catheter, instrumentation, hospital-acquired, symptoms >7 days. Each item either impairs host defence or selects a resistant organism. MCQ Male sex alone makes UTI complicated.
- US Food and Drug Administration (FDA) 2018 complicated UTI: pyuria + pathogen in urine/blood + local or systemic features (fever >38 °C, flank pain, costovertebral angle [CVA] tenderness) with abnormality or catheter.

Term	**Definition**	**Discriminator**
Isolated	First, or one remote UTI	—
Unresolved	Therapy failed; same organism	Never cleared
Reinfection	New event; bacteria reintroduced (~**70%** of recurrences, often different organism)	Cleared, then new inoculation
Bacterial persistence (relapse)	Same organism re-emerging from a focus (infected stone, prostate)	Cleared from urine, focus remains; recurs quickly

- **Antimicrobial prophylaxis (AP)** prevents reinfection and presumes a sterilised tract. **Suppression** prevents growth from a focus that cannot be eradicated.
- **Catheter-associated UTI (CAUTI)**: commonest cause of secondary bacteraemia; ~**80%** of nosocomial UTIs.
- Recurrence risk rises with number of previous UTIs and falls with time since the last. Uncomplicated recurrent UTI does not cause scarring, hypertension or renal failure.

2 · PATHOGENESIS

Infection = virulence × inoculum versus host defence. A strong host is infected only by highly virulent bacteria; a compromised host by organisms of minimal virulence.

Routes

Route	Frequency	Features
Ascending	Commonest	Bowel/skin flora → introitus → urethra → bladder → ureter → papillary collecting ducts. Aided by faecal soiling, spermicide, catheters, and anything antiperistaltic (Gram-negative endotoxin, pregnancy, obstruction, high bladder pressure). Reflux not required: cystitis oedema can distort the vesicoureteric junction.
Haematogenous	Uncommon	*Staphylococcus aureus*, *Candida*; obstruction enhances it
Lymphatic	Rare	Severe bowel infection, retroperitoneal abscess

Organisms

- ***Escherichia coli***: ~**85%** community, ~50% hospital UTIs; share falls from ~three-quarters in uncomplicated cystitis to under half in complicated UTI, pyelonephritis and urosepsis.
- ***Staphylococcus saprophyticus***: ~10% of cystitis in young sexually active women; appears only in uncomplicated cystitis.
- Nosocomial/complicated: *Klebsiella*, *Enterobacter*, *Serratia*, *Pseudomonas aeruginosa*, *Enterococcus faecalis*.
- Culture-negative chronic symptoms: *Ureaplasma urealyticum*, *Mycoplasma hominis* (targeted treatment helps).
- Most uropathogenic *E. coli* (UPEC) are phylogroup B2. **ST131 (O25b:H4)**: emerging MDR clone; first noted for extended-spectrum β-lactamases (ESBLs); hallmark fluoroquinolone resistance.
- MCQ *Enterococcus* is a Gram-positive coccus (streptococcal group), not an Enterobacteriaceae.
- **Anaerobes**: rare in cystitis (common contaminant), but present in most scrotal, prostatic and perinephric abscesses in men and in enteric fistulae. Suspect when microscopy shows organisms but aerobic culture is sterile.
- Urinary TB: haematuria and/or sterile pyuria; **polymerase chain reaction (PCR)** has replaced acid-fast staining.

Bacterial virulence — fimbriae

- UPEC virulence genes lie on pathogenicity-associated islands (PAIs): adhesins, toxins (haemolysin HlyA, secreted autotransporter toxin Sat), capsule, flagella, iron acquisition.
- Adherence is the essential first step. A piliated cell carries 100–400 pili; pili are typed by haemagglutination.

	Type 1 pili (mannose-sensitive)	P pili (mannose-resistant)
Gene / adhesin	*fim* operon; FimA rod, **FimH** tip adhesin	*pap* operon on a PAI
Haemagglutination	Guinea-pig red cells; inhibited by mannose	Not inhibited by mannose
Receptor	Mannosylated **uroplakins Ia/Ib** on umbrella cells	Glycolipid (P blood-group antigen) receptors in kidney
Tropism	Bladder → cystitis	Kidney → **pyelonephritis** (~**90%** of pyelonephritis strains)
Distribution	Pathogenic and commensal *E. coli*	Most upper-tract UPEC

- P-pilus frequency falls stepwise: pyelonephritis > cystitis > asymptomatic bacteriuria (ASB) > faecal isolates.
- **Phase variation**: rapid switching between piliated and non-piliated states. Type 1 pili help bladder adhesion but become a liability in suspension (they promote phagocytosis), so P pili take over in the

kidney.

Intracellular cycle (why cystitis recurs)

ALGORITHM
FimH binds uroplakin → invasion of umbrella cell (15 min–1 h) → intracellular replication → **intracellular bacterial community (IBC)**, biofilm-like (4–16 h) → fluxing and filamentation, escape and reinvasion (16–24 h) → exfoliation exposes deeper cells → **quiescent intracellular reservoirs (QIRs)** (from ~2 weeks)

- QIRs are metabolically inactive and cause no inflammation, so they are invisible between episodes. Urothelial disruption reactivates them → recurrence with the "same" strain without a urological focus.
- Biofilm increases resistance: poor antibiotic penetration, slow growth, inhibitory factors and exclusion of neutrophils.

Host defences and susceptibility

- **Introital receptivity**: *E. coli* adheres more to vaginal (and buccal) cells of women with recurrent cystitis → probably genetic; certain Lewis blood-group phenotypes predispose. This was the first biological difference shown in UTI-prone women.
- Normal periurethral flora (lactobacilli) is the barrier. Antimicrobials and spermicides disturb it acutely; oestrogen loss raises pH.
- Urine: osmolality, urea, organic acids, pH. **Uromodulin (Tamm–Horsfall protein)**, the most abundant urinary protein, competes with uroplakins for type 1 pili.
- Bladder emptying is the key defence. **Obstruction** converts cystitis or pyelonephritis into life-threatening infection.
- **VUR**: in children, high-grade VUR + UTI → scarring, proteinuria, renal failure. In adults, VUR alone does not reduce function without high pressure/stasis + UTI.
- **DM**: glycosuria, neutrophil dysfunction, increased adherence, neuropathic emptying → more pyelonephritis, renal and perinephric abscess, emphysematous infection, fungi, papillary necrosis.
- **Human immunodeficiency virus (HIV)**: UTI ~5× commoner. With CD4 <200, expect atypical (fungal, mycobacterial, viral) infection. Trimethoprim–sulfamethoxazole (TMP-SMX) given as Pneumocystis prophylaxis also reduces UTI.
- **Renal transplant**: more than a third get UTI; graft pyelonephritis independently predicts functional decline.
- **SCI**: no bacteriuric group matches it for morbidity (high pressure → VUR, hydronephrosis).

Renal papillary necrosis

- Causes (impaired medullary blood flow or oxygenation of the papilla): classic group **DM, obstruction/pyelonephritis, analgesic abuse, sickle cell disease**; also transplant rejection, cirrhosis, dehydration, renal vein thrombosis, candidiasis.
- Sloughed papilla → acute ureteric obstruction; retained papillae calcify → infective nidus. Treat with antibiotics + early relief of obstruction.

3 · EVALUATION AND DIAGNOSIS

Syndrome	Features
Cystitis	Dysuria, frequency, urgency, suprapubic pain, haematuria; afebrile (mucosal infection)
Pyelonephritis	Fever, chills, flank pain, CVA tenderness ± nausea, vomiting; lower urinary tract symptoms (LUTS) may be absent
Fistula	Pneumaturia, faecaluria
Renal/perirenal abscess	Indolent fever, flank mass

Syndrome	Features
Catheter	Usually ASB; fever with bacteraemia can develop fast

	Cystitis	Vaginitis	Urethritis
Onset	Acute	Subacute	Subacute
Frequency/urgency	Prominent	Absent; no haematuria or suprapubic pain	Less marked; no fever
Clues	—	Discharge, odour, new partner	Urethral discharge; initial-stream pyuria in men
Causes	*E. coli*	Herpes simplex virus (HSV), gonorrhoea, *Chlamydia*, *Trichomonas*, yeast, bacterial vaginosis	*N. gonorrhoeae*, *Chlamydia*, HSV, *Trichomonas* → *Ch 29*

- Dysuria + frequency without vaginal discharge → ~**90%** probability of UTI. Haematuria + positive nitrite → ≥90%. Dysuria is the most reliable single symptom; vaginal discharge lowers probability.
- **Rule**: painless gross haematuria, or microhaematuria with negative culture or persisting after infection clears → full upper and lower tract evaluation for malignancy.
- Examination: suprapubic distension (post-void residual [PVR]); CVA tenderness; in women vaginal atrophy, pelvic organ prolapse (POP), urethral diverticulum or Skene's gland; in men digital rectal examination (DRE); perineal sensation.

Urine collection

Method	Accuracy	Use
Midstream clean catch	Contamination common in women (squamous cells, lactobacilli)	Default
Catheterised (in-and-out)	More accurate	Contaminated voided sample, pessary, obesity, poor dexterity, cannot void; roughly halves unnecessary prescribing
Suprapubic aspirate	Most accurate	Rarely; cannot void on command

- Indwelling catheter: replace the catheter and sample from the new one. Stoma: catheterise the stoma, never sample the bag. Ignore bedpan or non-sterile cup samples.

Urinalysis

- **Leucocyte esterase** = pyuria (not bacteria). **Nitrite**: Enterobacteriaceae reduce dietary nitrate. Non-producers: all Gram-positives (enterococci, staphylococci), *Pseudomonas*, *Acinetobacter*. Low-nitrate diet or short bladder dwell → false negative.
- Nitrite = high specificity, low sensitivity; it is the single best rule-in test with symptoms. Best combination: nitrite positive, or leucocyte esterase + blood. Dipsticks are better at **ruling out**.
- Microscopy: centrifuge 6–8 mL; pyuria = ≥**5–10 WBC/high-power field (HPF)**; squamous cells mark contamination. No pyuria → question the diagnosis. Culture-positive without pyuria → repeat with catheterised sample.
- A negative automated urinalysis reliably excludes UTI. Positives overdiagnose because healthy women have abnormal parameters.

Culture

Threshold	Use
≥10^5 colony-forming units (CFU)/mL	Kass (originally for pyelonephritis); ASB definition
≥**10^2 CFU/mL** of a known pathogen	Symptomatic women with cystitis; catheterised specimens

- Why counts are low in cystitis: bacterial doubling time 30–45 min, but irritated bladder voids every 15–30 min, so bacteria are washed out before reaching 10^5.

- Gram-negative rods in a symptomatic patient → treat. Enterococci and group B streptococci in a midstream sample from premenopausal women predict nothing; catheter samples usually grow *E. coli*.
- Susceptibility: S / I / R + minimum inhibitory concentration (MIC). "I" may still work in urine because urinary levels are often several hundred times serum levels. All common oral agents achieve inhibitory urine levels except macrolides. Choose by site and route, not lowest MIC.
- **Microbiome**: sequencing and expanded quantitative urine culture (EQUC) show a commensal urinary microbiome; "no growth" ≠ sterile and organism ≠ infection. Commercial sequencing gives no susceptibilities. Decisions rest on symptoms.

Localisation

- **Ureteric catheterisation (Stamey)** localises persistence to one kidney, ectopic ureter or ureteric stump (also native kidneys in transplant recipients).
- **Stone culture** predicts post-procedure sepsis and systemic inflammatory response syndrome (SIRS) better than bladder urine. Culture the stone separately; results may differ from urine.

Imaging — indications

Imaging answers two questions only: is there obstruction or a collection needing drainage, and is there a focus of persistence? Most UTIs need none.

- **Indications in acute pyelonephritis**: possible obstruction (stone, especially struvite; stricture; tumour), possible papillary necrosis (sickle cell, DM, analgesics), prior surgery (reimplant, diversion), **no response after 72 h** (box says 5–6 days), DM, polycystic kidneys on dialysis, NGB, unusual organisms (TB, fungi, urea-splitters).
- Also: most men, febrile infections, compromised hosts, rapid recurrence suggesting persistence.

Question	Best test
Stone	Non-contrast computed tomography (CT)
Abscess, acute focal bacterial nephritis (AFBN), drainage planning	Contrast CT
Bedside hydronephrosis, pyonephrosis, PVR	Ultrasound (US)
Extrarenal extension	Magnetic resonance imaging (MRI)
Urethral diverticulum	Pelvic MRI with contrast (double-balloon voiding study is historical)
VUR; non-refluxing stump	Voiding cystourethrogram (VCUG): febrile UTI in women, childhood VUR, recurrent adult pyelonephritis
Split function, scarring	Dimercaptosuccinic acid (DMSA) scan

Correctable causes of bacterial persistence (Box 26.6)
Infection stones; chronic bacterial prostatitis; unilateral infected atrophic kidney; duplication/ectopic ureter; foreign body; urethral diverticulum and periurethral glands; unilateral medullary sponge kidney; non-refluxing infected ureteric stump; infected urachal cyst; infected communicating calyceal cyst; papillary necrosis; perivesical abscess with fistula. **Each is a nidus antibiotics cannot sterilise → cure only by removal or drainage.**

4 · ANTIMICROBIAL PRINCIPLES

- Uncomplicated cystitis: **urinary** concentration matters. Febrile parenchymal infection (kidney, prostate) and bacteraemia: **serum and tissue** levels matter.
- Renal insufficiency and obstruction lower urinary drug levels → eradication harder; adjust doses of renally cleared drugs.
- Uncomplicated cystitis rarely progresses; the goal is symptom relief. Non-steroidal anti-inflammatory drugs (NSAIDs) alone → longer symptoms and more pyelonephritis than antibiotics.

- **Collateral damage**: ecological effects (resistance, MDR colonisation, *Clostridioides difficile*, candidiasis), mainly from broad cephalosporins and quinolones. It matters most in cystitis, a low-risk infection in healthy people.

Resistance

Type	Mechanism	Implication
Intrinsic	Species lacks target	Predictable
Acquired chromosomal	Resistant mutants selected when urine level ≈ or < MIC	Full dose, compliance, drug with urine level ≫ MIC
Plasmid (extrachromosomal)	Horizontal transfer in **bowel** flora; far commoner	One drug selects multi-resistance (R factors), so post-antibiotic UTIs are often MDR

Enzyme	Encoded	Inactivates	Notes
ESBL	Plasmid	Most β-lactams incl. 3rd-generation cephalosporins	Co-resistance to quinolones and aminoglycosides; carbapenem, fosfomycin, nitrofurantoin remain options
AmpC	Chromosomal; now plasmid	Broad-spectrum cephalosporins	—
Klebsiella pneumoniae carbapenemase (KPC)	—	Carbapenems	Commonest US carbapenemase; spreading to other Gram-negatives

- Resistance scales with use; reducing use can restore susceptibility. Prior quinolone use is the strongest determinant of quinolone resistance, which heralds MDR.
- **Stewardship**: narrow spectrum, shortest effective course, culture before treating recurrent or complicated UTI, use local antibiograms. Stewardship programmes improve guideline compliance and shorten courses.

Agents

Class	Action	Resistance
β-lactams	Cell wall	β-lactamase, altered penicillin-binding protein, porins
Fosfomycin	Cell wall (different step) → little cross-resistance	Transporter loss
Vancomycin	Cell wall (another step)	Altered peptidoglycan target
Aminoglycosides	Ribosome, bactericidal	Modifying enzymes, reduced uptake
Quinolones	DNA gyrase	Gyrase mutation, efflux, porins
TMP-SMX	Folate antagonism	Enterococci scavenge environmental folate (ineffective even if "sensitive")
Nitrofurantoin	Multiple enzyme systems	Develops slowly; very low

- Bactericidal: aminoglycosides, quinolones, β-lactams, vancomycin. Bacteriostatic: TMP, sulfonamides, tetracycline, macrolides, clindamycin, chloramphenicol. Nitrofurantoin is static, but cidal at high dose.
- **Anti-pseudomonal**: antipseudomonal penicillins, **ceftazidime** (not ceftriaxone), aztreonam, aminoglycosides, fluoroquinolones. **Enterococcal** cover: aminopenicillins, piperacillin, nitrofurantoin, fosfomycin; **never cephalosporins**. Aztreonam and pivmecillinam: no Gram-positive cover. Vancomycin: no Gram-negative cover. Ertapenem misses *Pseudomonas* and *Enterococcus*; carbapenems miss *E. faecium* and methicillin-resistant *S. aureus* (MRSA).

Agent	Key points
Nitrofurantoin	Urine-only levels → useless for pyelonephritis, prostate, bacteraemia. Spares flora, minimal resistance. Toxicity: pulmonary reactions/fibrosis (long-term), neuropathy, haemolysis in glucose-6-phosphate dehydrogenase (G6PD) deficiency. Renal cut-off: Beers 2015 creatinine clearance (CrCl) **<30 mL/min** (older 50–60). Beers advises against long-term suppression in the elderly.
TMP-SMX	Good tissue levels; misses *Enterococcus*, *Pseudomonas*. Commonest antibiotic cause of Stevens–Johnson syndrome/toxic epidermal necrolysis; potentiates warfarin.
Fosfomycin	Single 3 g dose; active against ESBL and vancomycin-resistant *Enterococcus* (VRE) → oral option for MDR cystitis.
Fluoroquinolones	Broad incl. *Pseudomonas*; spare anaerobes. Adjust at CrCl <30. FDA warnings: tendon rupture (risk ↑ age >60, steroids, renal failure), neuropathy, disability, dysglycaemia, psychiatric effects, aortic dissection; **2016: reserve for no alternative in uncomplicated UTI**. Avoid in children and pregnancy. Interactions: antacids/iron (separate ≥2 h), theophylline, warfarin.
Cephalosporins	Good on Enterobacteriaceae, poor on enterococci. 3rd generation most reliable β-lactam for Gram-negatives. For cystitis only when first-line agents cannot be used.
Carbapenems	Stable to ESBL and AmpC; penicillin cross-sensitivity <1%.
Aminoglycosides	Once-daily 5–7 mg/kg: concentration-dependent killing, post-antibiotic effect, less nephrotoxicity. Interval from a nomogram level taken 6–14 h after the dose (redose 24/36/48-hourly). Oto- and nephrotoxic; monitor.
Aztreonam	Aminoglycoside-like spectrum without nephrotoxicity; for penicillin allergy and renal impairment.
Pivmecillinam	~2% *E. coli* resistance; Nordic first-line; not in USA.

> **KEY POINT** Narrow, urine-confined, flora-sparing agents (nitrofurantoin, fosfomycin, pivmecillinam) lead in cystitis. Broad, tissue-penetrating agents (quinolones, cephalosporins, carbapenems) are kept for invasive infection. Resistance is made mainly in the bowel, so an antibiotic's breadth, duration and repetition matter more than its MIC.

***C. difficile* infection (CDI)**

- Culprits: cephalosporins, clindamycin, carbapenems, quinolones, TMP-SMX. Diagnosis: ≥3 unformed stools/24 h + toxigenic *C. difficile*/toxin.

Episode (IDSA/SHEA 2021)	Treatment
Initial	**Fidaxomicin** 200 mg twice daily × 10 d; alternative oral vancomycin 125 mg four times daily × 10 d; metronidazole only if non-severe and others unavailable
First recurrence	Fidaxomicin or tapered vancomycin ± **bezlotoxumab** (anti-toxin B)
≥2 recurrences	As above, or **faecal microbiota transplantation**
Fulminant	Oral/rectal vancomycin 500 mg four times daily + IV metronidazole

IDSA = Infectious Diseases Society of America; SHEA = Society for Healthcare Epidemiology of America.

5 · ASYMPTOMATIC BACTERIURIA

Group (IDSA)	Specimen	Threshold
Women	**Two** consecutive voided, same strain	$\geq 10^5$ CFU/mL
Men	One voided	$\geq 10^5$ CFU/mL
Either sex	One catheterised	$\geq \mathbf{10^2}$ CFU/mL

- Prevalence rises with age, DM, institutionalisation and catheters: premenopausal women 1–5%, long-term care 25–50%, long-term indwelling catheter **100%**.
- Untreated ASB does not cause hypertension, renal failure, cancer or recurrent UTI. Treating it gives more symptomatic UTI, MDR and adverse effects. ASB *E. coli* are low-virulence and may be protective (bacterial interference).

Do NOT screen or treat (IDSA 2019)	Screen and treat
Premenopausal non-pregnant women; DM; community and institutionalised elderly; SCI; indwelling catheters; ASB with pyuria; children; renal transplant >1 month; other solid-organ transplant; **non-urological surgery (incl. arthroplasty, cardiac)**	**Pregnancy**; **urological procedures with anticipated mucosal bleeding** (transurethral resection, ureteroscopy, PCNL). No consensus: neutropenia.

- Arthroplasty: ASB marks risk but is not the source (different organisms); treating it does not reduce joint infection.

> **KEY POINT** ASB is treated only where bacteria are about to gain a route into blood or tissue: pregnancy (hydroureter → pyelonephritis) and mucosa-breaching urological surgery. Elsewhere the resident strain is harmless or protective.

6 · CYSTITIS

Acute uncomplicated cystitis in women

- Risk factors by defence defeated. **Washout**: obstruction, NGB, dehydration. **Flora/binding**: intercourse, spermicide, oestrogen loss, antibiotics. **Ascent**: catheter, incontinence, residual urine.
- Diagnosis: acute dysuria + change from baseline voiding.
- Exclude early pyelonephritis (fever, flank pain); if present, or if oral therapy is not possible, treat as pyelonephritis.

Tier (IDSA/ESCMID)	Regimen	Caveat
First-line	**Nitrofurantoin** 100 mg twice daily × **5 d**	Not if pyelonephritis suspected
First-line	**TMP-SMX** 160/800 mg twice daily × **3 d**	Avoid if local resistance >**20%** or used for UTI in past 3 months
First-line	**Fosfomycin** 3 g single dose	Slightly lower efficacy
First-line	**Pivmecillinam** 400 mg twice daily × 3–7 d	Not in North America
Second-line	Fluoroquinolone × 3 d	Collateral damage and toxicity; last resort (FDA 2016)
Second-line	β-lactam (amoxicillin–clavulanate, cefpodoxime, cefadroxil) × 3–7 d	Lower efficacy; never amoxicillin alone

ESCMID = European Society for Microbiology and Infectious Diseases. Quinolones are demoted for ecology and toxicity, not inefficacy. **3 days** is optimal: better than a single dose and equal to 7 days, with fewer adverse effects. Use 7 days if complicating factors. No post-treatment culture if asymptomatic.

Cystitis in men

- Considered complicated; about half have an abnormality (commonly incomplete emptying) → evaluate. Consider prostatitis → *Ch 27*.
- Afebrile men: **7 days = 14 days** (Drekonja trial). European Association of Urology (EAU): TMP-SMX 160/800 twice daily × 7 d; quinolone per local susceptibility.

Complicated UTI

Category (CUA)	Examples
Anatomical	Cystocele, fistula, diverticulum
Iatrogenic	Nosocomial, surgery, catheter
Obstruction	BOO, pelviureteric junction obstruction, ureteric stricture
Voiding dysfunction	Neurological disease, VUR, incontinence, high PVR
Other	Pregnancy, stones, DM, immunosuppression

CUA = Canadian Urological Association. Management: culture always; correct the abnormality (otherwise therapy fails); give **10–14 days** culture-directed treatment; switch IV → oral once afebrile; repeat culture if no response.

Emphysematous cystitis

- Gas in the bladder wall; elderly woman with poorly controlled DM ± stasis. *E. coli* ~60%, *Klebsiella*. Abdominal pain (~80%), gross haematuria; fever in only 30–50%.
- Plain film: curvilinear lucency outlining the bladder. **CT confirms** and excludes fistula or adjacent gangrene.
- ~**90%** medical: broad Gram-negative cover (add ampicillin if Gram-positive cocci), catheter drainage, glucose control. Debridement or partial/total cystectomy is rarely needed (necrosis, non-response).

Unresolved bacteriuria

Cause (descending frequency)	On-treatment culture	Action
1. Resistant from outset (recent antibiotics)	Resistant	Change drug; no further work-up
2. Emergent resistance (underdosing)	Same species, now resistant	
3. Unsuspected second pathogen	New species present from start	
4. Rapid reinfection with new resistant species	New species → suspect enterovesical fistula	
5–7. Azotaemia; papillary necrosis; giant staghorn ("critical mass")	Same species, still **sensitive**	Renal function + imaging
8. Non-compliance, self-inflicted (Munchausen)	—	—

7 · RECURRENT UTI

- Definition: **2 in 6 months or ≥3 in 12 months**, culture-documented. About a quarter of young women recur within 6 months.
- Persistence = same strain, recurs quickly, from a focus (uncommon). Reinfection = different organism or same organism >2 weeks after treatment; commoner (faecal flora or QIRs). In men, reinfection also raises suspicion of an anatomical abnormality.
- Risk factors: intercourse, new partner, spermicide, first-degree female relative with UTI, UTI before menopause, menopause, recent antibiotics; postmenopausal incontinence, cystocele, raised PVR.
- Evaluation: history (fistula clues: pneumaturia, faecaluria, diverticulitis, pelvic radiotherapy), pelvic examination, **culture each episode**, PVR ± flow rate. **Imaging and cystoscopy are not routine** in uncomplicated recurrent UTI (AUA, EAU, CUA).

Indications for imaging ± cystoscopy in recurrent UTI (CUA)
Prior urinary tract surgery/trauma; stones; gross haematuria or microhaematuria persisting after infection clears; obstructive symptoms, low flow, high PVR; urea-splitters; prior abdominopelvic malignancy; persistence despite sensitivity-based therapy; DM/immunocompromise; pneumaturia, faecaluria, anaerobes, diverticulitis; repeated pyelonephritis. MCQ Frequency of recurrence alone is **not** an indication.

- **Struvite stones**: urease (*Proteus mirabilis*) alkalinises urine → magnesium ammonium phosphate. Bacteria live within the stone, so urine may be sterile after treatment. AUA: remove infection stones completely in all fit patients. If unfit: acetohydroxamic acid + suppressive antibiotics.

Prevention algorithm (women, AUA 2019 / authors)

ALGORITHM
Confirm true recurrent UTI (symptoms + positive culture) → same organism at short intervals? → yes: evaluate for persistence (upper tract imaging + cystoscopy) and remove nidus → reinfection: behavioural measures (≥1.5 L water, frequent and postcoital voiding, stop spermicide) → cranberry ≥36 mg proanthocyanidin (PAC) → intercourse-related: **postcoital antibiotic**; postmenopausal: **vaginal oestrogen**; neither: **methenamine hippurate** (if renal function normal) → continuous antibiotic prophylaxis as last resort

Measure	Mechanism	Evidence / use
Water ≥1.5 L/day	Washout	randomised controlled trial (RCT) (premenopausal): episodes roughly halved (1.7 vs 3.2/yr)
Vaginal oestrogen (cream, tablet, ring)	Restores lactobacilli, lowers vaginal pH	Strongest non-antibiotic evidence; ~90% reduction (Raz–Stamm); benefit after ≥12 weeks. **Oral oestrogen ineffective.** Minimal systemic absorption; no excess breast cancer or cardiovascular risk; usable in breast cancer survivors with oncologist agreement.
Cranberry ≥36 mg PAC (twice daily)	Type A PAC blocks **P-fimbrial** adhesion	Cochrane 2023 favourable (women with recurrent UTI, children, post-intervention); safe with warfarin
Methenamine hippurate 1 g twice daily	Converts to **formaldehyde in acid urine (pH <6)**	RCT non-inferior to daily antibiotic prophylaxis. Ineffective with urease producers or NGB/anatomical abnormality. Contraindicated CrCl <50, liver disease; antacids, carbonic anhydrase inhibitors and alkalinising diets (citrus) reduce effect.
D-mannose	Competitive inhibitor of **FimH** (type 1 pili)	Promising; Cochrane 2022 inconclusive
Immunostimulants: OM-89 (oral, 18 *E. coli* strains), Urovac (vaginal), Uromune MV140 (sublingual)	Bacterial lysate vaccines	Some benefit; small, biased trials
Probiotics, oral oestrogen, vitamin C	—	Not effective

Antibiotic strategy	Evidence	Position
Continuous low dose 6–12 months (nitrofurantoin, TMP, TMP-SMX, cephalexin)	Effective only while taken; old trials; recurrence returns to baseline on stopping; MDR	AUA "may prescribe"; EAU/authors: last resort
Postcoital single dose	Effective; preferable to daily for coital UTI	Acceptable
Self-start	Old data; perpetuates overuse	Prefer analgesic + urine sample + observe, then culture-directed agent

KEY POINT Recurrent UTI in women is reinfection in a receptive host, not a hidden focus. Investigation is reserved for features of persistence or complication. Prevention targets flora, pH, adhesion and washout rather than repeatedly sterilising urine.

8 · KIDNEY INFECTIONS

Acute pyelonephritis

- Clinical diagnosis: abrupt chills, fever ≥38 °C, flank/CVA pain; LUTS often absent; may mimic gastrointestinal disease. Symptoms correlate poorly with the true site of infection.
- Labs: leucocytosis, raised C-reactive protein (CRP); **WBC casts** suggest renal origin (bacteria-laden casts specific). Blood cultures if toxic, admitted or pregnant.
- Usually P-fimbriated *E. coli*. Suspect resistant organisms after recurrent UTI, hospitalisation, catheter or instrumentation.

Setting (IDSA)	Regimen	Days
Outpatient, no vomiting	Ciprofloxacin 500 mg twice daily if community quinolone resistance **≤10%**; levofloxacin 750 mg daily; TMP-SMX DS twice daily only if known sensitive. ± one initial IV dose of ceftriaxone 1 g or aminoglycoside.	Cipro 7; levo 5; TMP-SMX 14
Inpatient, severe/possible sepsis	Ampicillin + gentamicin; levofloxacin; ceftriaxone 1 g daily; carbapenem (ESBL risk). Gram-positive cocci → ampicillin–sulbactam ± aminoglycoside.	10–14 total
Pregnancy	Ampicillin + gentamicin, aztreonam, ceftriaxone → oral cephalexin. **No quinolone.**	10–14

- IV indications: high fever, leucocytosis, vomiting, dehydration, sepsis, complicated infection, outpatient failure.
- Positive blood cultures → IV until stable; negative → 2–3 days IV, then oral to complete 10–14 days.
- **Obstruction** → urgent decompression by the simplest, safest route: **ureteric stent or percutaneous nephrostomy**. A solitary obstructed kidney causes acute renal failure → dose by glomerular filtration rate (GFR).

ALGORITHM
Pyelonephritis → no sepsis/vomiting: urine culture ± US → oral outpatient 7–10 d | sepsis: blood + urine cultures, imaging → IV inpatient → improved by **72 h**: oral step-down | not improved: repeat cultures + **CT** → drain obstruction or abscess

KEY POINT Urine sterilises within hours but fever may lag for days. Fever beyond 72 h means obstruction or an undrained collection until CT proves otherwise. Antibiotics treat the infection; drainage treats the complication.

Acute focal (lobar) bacterial nephritis (AFBN)

- Dense leucocyte infiltrate in one lobe (focal) or several (multifocal); midway between pyelonephritis and abscess and probably an early abscess phase. About half are diabetic and half bacteraemic.
- Treat with IV antibiotics ≥7 days then 7 days oral. Non-response → exclude abscess, obstruction, renal cell carcinoma (RCC), renal vein thrombosis.

	AFBN	Renal abscess
CT shape	Wedge-shaped, poorly marginated	Round, well defined
Content	No wall, no liquefaction	Liquid centre; chronic → enhancing **ring sign**
Visibility	Only after contrast	Before and after contrast
Treatment	Antibiotics	Size-dependent

Emphysematous pyelonephritis (EPN)

- Urological emergency: necrotising parenchymal and perirenal infection with gas. Mostly **diabetic women**; many have obstruction (stone, papillary necrosis). Non-diabetics usually have obstruction and less extensive disease.
- Mechanism proposed: tissue glucose fermented to CO_2 in ischaemic tissue. Organisms: *E. coli* > *Klebsiella*, *Proteus*; urine culture almost always positive.

- Triad: fever, vomiting, flank pain. Plain film: mottled gas over kidney or upper-pole gas crescent. **CT is the imaging of choice.**

CT pattern	Prognosis
Streaky/mottled gas, **no fluid**	Rapid destruction; 50–60% mortality
Renal/perirenal **fluid** with bubbly/loculated gas, or gas in collecting system	<20% mortality

Huang–Tseng class †	Extent of gas	Management †
1	Collecting system only (emphysematous pyelitis)	Antibiotics + percutaneous catheter drainage (PCD)/stent
2	Renal parenchyma, no extrarenal extension	Antibiotics + PCD
3A	Extension to perinephric space	PCD if <2 risk factors; nephrectomy if ≥2 risk factors or PCD fails
3B	Extension to pararenal space	
4	Bilateral EPN or solitary kidney	PCD (nephron-sparing)

† Huang–Tseng risk factors: thrombocytopenia, acute kidney injury, altered consciousness, shock. PCD = percutaneous catheter drainage.

- Management: resuscitation, glucose/electrolyte control, broad-spectrum antibiotics, relief of obstruction. Practice has moved from emergency nephrectomy (40–50% mortality) to **PCD as definitive treatment**, with nephrectomy for extensive diffuse gas with destruction. Assess contralateral function with a nuclear scan.
- **Emphysematous pyelitis** (gas confined to the collecting system) is often non-diabetic and less serious, and usually responds to antibiotics + drainage.

> **KEY POINT** Fluid is a good sign and streaky gas a bad one. A liquefied, contained collection can be drained. Diffuse streaky gas without fluid means parenchyma is necrosing faster than the host can wall it off.

Renal abscess

- Pre-antibiotic era: haematogenous staphylococcal abscess in young men with normal kidneys. Now: mostly **ascending Gram-negative** infection in kidneys with prior disease, stones or obstruction. Normal kidneys resist blood-borne Gram-negatives. Risk: DM, stones, UTI, IV drug use, HIV, steroids, pregnancy.
- Vague symptoms; look for a skin source 1–8 weeks earlier. Urine may be sterile (non-communicating or haematogenous abscess).
- US first (hypo-/anechoic lesion with through-transmission). **CT is diagnostic**. Aspirate if abscess vs cystic tumour is uncertain.

Size	Management
≤**3 cm**	IV antibiotics alone, even with abnormal tract if stable
3–5 cm	Variable: antibiotics or drainage
>**5 cm**	Percutaneous drainage first line; may need several drains, open drainage or nephrectomy
Any size, immunocompromised or not responding	Drainage

Empirical therapy: haematogenous source → penicillinase-resistant penicillin (vancomycin if allergic); ascending → 3rd-generation cephalosporin, antipseudomonal penicillin or aminoglycoside. Follow with serial US or CT until resolution.

Infected hydronephrosis and pyonephrosis

- **Pyonephrosis** = infected hydronephrosis with suppurative parenchymal destruction and (near-)total loss of function. Patients are usually very ill. **Bacteriuria may be absent** if the ureter is completely obstructed.
- US: dependent echoes and fluid–debris level in a dilated system.
- Management: antibiotics + **immediate drainage** (stent or nephrostomy) → treat the cause once stable; nephrectomy if non-functioning.

Perinephric abscess

- Beyond the renal capsule but within Gerota's fascia. Causes: rupture of a cortical abscess, extravasation from an obstructed infected kidney (pyonephrosis + calculus), haematogenous (~a third, skin). About a third are diabetic. Rupture through Gerota's → paranephric/psoas abscess (limp, flexed externally rotated hip). Organisms: *E. coli*, *Proteus*, *S. aureus*.
- Insidious; flank mass in half; a third afebrile; cultures unreliable. Only a third are diagnosed at presentation, and delay accounts for nearly all deaths.

	Acute pyelonephritis	**Perinephric abscess**
Symptoms before admission	≤5 days	**>5 days**
Fever on appropriate antibiotics	Settles within 4 days	**≥5 days** (median 7)

- CT defines extent, route and obstruction. Antibiotics: aminoglycoside + antistaphylococcal penicillin (vancomycin or cephalosporin if allergic).
- <3 cm in immunocompetent → antibiotics may suffice. >3 cm or non-responding → **percutaneous drainage** (lower threshold than renal abscess). Surgical drainage or nephrectomy for non-functioning kidney, simultaneously or staged. Treat the cause (enteric fistula, cortical abscess).

Xanthogranulomatous pyelonephritis (XGP)

- Rare chronic destructive infection: **triad of stone (~80%, half staghorn), obstruction and infection**; inadequate acute response in an obstructed, ischaemic kidney. Lipid-laden foamy macrophages (xanthoma cells) with giant cells and plasma cells.
- Middle-aged women, DM; unilateral; diffuse (>80%) or segmental (focal/tumefactive). Flank pain, fever, flank mass (~60%), anaemia, hepatic dysfunction (up to half). Renal failure is uncommon because disease is unilateral.
- Organisms: ***Proteus*** (commonest) > *E. coli*. Urine sterile in about a third; tissue culture positive.
- **CT** (most useful): enlarged non-functioning kidney + central pelvic calculus with a **contracted (non-dilated) pelvis**; parenchyma replaced by low-density cavities whose walls enhance ("**bear-paw**" †), but whose contents do not. Extrarenal extension to psoas, bowel and skin.
- Mimics RCC on imaging and clear cell carcinoma on **frozen section**; it may coexist with RCC, urothelial or squamous carcinoma.
- Differentials: pyonephrosis, malakoplakia (no stones), lymphoma (bilateral, no stones).
- Management: antibiotics to stabilise, then **nephrectomy with removal of perinephric fat and the entire inflammatory mass**; partial nephrectomy for focal disease. Drainage alone → protracted illness and **nephrocutaneous fistula**. Laparoscopy is feasible but difficult, with high conversion rates. Expect adhesion to diaphragm, great vessels and bowel.

Malakoplakia

- "Soft plaque": macrophages ingest *E. coli* but cannot digest it (defective phagolysosomal function, low cyclic guanosine monophosphate [cGMP]).
- Histology: von Hansemann histiocytes + **Michaelis–Gutmann bodies** (laminated calcium-phosphate inclusions; pathognomonic but may be absent early). α1-antitrypsin immunostaining helps early diagnosis.
- Female:male 4:1, >50 years, immunosuppressed (renal transplant). Bladder commonest; also kidney (masses, may be bilateral), testis, prostate.

- Treatment: antibiotics active inside macrophages (**fluoroquinolones**, TMP-SMX, rifampicin, doxycycline) + bethanechol and ascorbic acid (raise cGMP); reduce immunosuppression. Nephrectomy for symptomatic unilateral renal disease. Bilateral or transplant-kidney disease is usually fatal within 6 months.

	XGP	Malakoplakia
Cell	Foamy lipid-laden macrophages	Von Hansemann cells, Michaelis–Gutmann bodies
Stones/obstruction	Usually	Absent
Organism	*Proteus*, *E. coli*	*E. coli*
Laterality	Unilateral	Uni- or multifocal, may be bilateral
Treatment	Nephrectomy	Intracellular antibiotics ± nephrectomy

KEY POINT Across renal infections the axis is diffuse inflammation versus contained collection. Pyelonephritis and AFBN need antibiotics. Pyonephrosis and large or obstructed abscesses need drainage. When the kidney itself is the infected tissue (XGP, diffuse EPN), the answer is nephrectomy.

9 · FOURNIER GANGRENE AND PERIURETHRAL ABSCESS

- **Fournier gangrene**: necrotising fasciitis of genitalia, perineum and/or perianal region. Fournier's original idiopathic disease of young men has given way to contemporary disease: a source is found in ~**95%**, age 50–79, male:female ~10:1.
- Sources: **colorectal** (perianal abscess, fissure), **urogenital** (stricture with extravasation, instrumentation, catheter), **dermal** (hidradenitis, balanitis, scrotal infection).
- Risk factors: **DM**, obesity, alcoholism, immunosuppression, local trauma, paraphimosis, periurethral extravasation, surgery (circumcision, herniorrhaphy).
- Polymicrobial **aerobic–anaerobic synergy** (*E. coli*, *Klebsiella*, enterococci + *Bacteroides*, *Clostridium*, microaerophilic streptococci) → obliterative endarteritis, thrombosis → necrosis.
- Fascial spread: Buck's → dartos → Colles' → Scarpa's fascia (abdominal wall).
- Course: cellulitis → oedema and pain out of proportion → crepitus → dusky purple skin → gangrene. **Skin underestimates fascial extent.** Systemic toxicity out of proportion to local signs is the warning sign.
- Labs: anaemia, raised creatinine, hyponatraemia, **hypocalcaemia** (bacterial lipases → fatty acids chelate calcium).
- CT has the highest specificity for fascial gas and extent. Imaging must not delay surgery when disease is clinically obvious.

Score	Components	Use
Laboratory Risk Indicator for Necrotising Fasciitis (LRINEC)	CRP, WBC, haemoglobin, sodium, creatinine, glucose	≥**6** suspicious; ≥**8** strongly predictive (diagnostic adjunct)
Fournier's Gangrene Severity Index (FGSI) †	Temperature, heart rate, respiratory rate, sodium, potassium, creatinine, haematocrit, WBC, bicarbonate (each 0–4)	>9 predicts mortality

ALGORITHM
Suspicion → resuscitation + electrolytes + broad IV antibiotics (piperacillin–tazobactam; or carbapenem/vancomycin + clindamycin or metronidazole) → **immediate radical debridement** to bleeding normal fascia, leave open → **re-look at 24–48 h** → suprapubic diversion (urethral source) / diverting colostomy (rectal source) → hyperbaric oxygen (adjunct) → reconstruction (grafts, flaps)

- **Orchidectomy is almost never needed**: testicular blood supply (testicular artery) is independent of the scrotal skin circulation.

- Mortality ~20–40%; worse with liver failure, alcoholism, DM, age, cardiac failure, coagulopathy, renal failure.
- **Periurethral abscess**: complicates urethral stricture or catheterisation. Treatment: surgical debridement/drainage + suprapubic diversion + antibiotics.

10 · UROSEPSIS †

***†** This section is not in the source notes; it is added for completeness.*

Term (Sepsis-3)	Definition
Sepsis	Life-threatening organ dysfunction from dysregulated host response to infection: acute rise in Sequential Organ Failure Assessment (SOFA) score ≥**2**
Quick SOFA (qSOFA)	Bedside screen, ≥2 of: respiratory rate ≥22/min, altered mentation, systolic blood pressure (BP) ≤100 mmHg
Septic shock	Vasopressors needed to keep mean arterial pressure ≥65 mmHg **and** lactate >2 mmol/L despite adequate fluids

ALGORITHM

Hour-1 bundle: lactate → blood + urine cultures → broad-spectrum IV antibiotic within 1 h → 30 mL/kg crystalloid if hypotension or lactate ≥4 → vasopressor (noradrenaline) for MAP ≥65 → **source control**: decompress obstruction (stent or nephrostomy; drain only, no definitive stone surgery) → de-escalate by culture

- Manipulating infected stones releases endotoxin → sepsis; so drain first, operate on stones later.

11 · SPECIAL POPULATIONS

Pregnancy

- ASB prevalence is **not** raised by pregnancy (2–10%), but it persists and progresses: untreated bacteriuria → pyelonephritis historically in 15–45%, mostly in the 3rd trimester. Treatment reduces pyelonephritis, preterm birth and low birth weight (Cochrane).
- Changes: kidney +1 cm; ureteric dilatation (right > left) from progesterone and uterine compression, resolving by 6 weeks postpartum; GFR +30–50% → creatinine falls (0.7 → **0.5 mg/dL**), so a "normal" 0.8–0.9 may mean renal impairment. Proteinuria is abnormal >300 mg/24 h.
- **Screen by culture** at the first visit or **12–16 weeks** (dipstick misses cases). Treat with a standard-length course (4–7 days beats single dose).
- After one pyelonephritis or recurrent cystitis → daily nitrofurantoin or cephalexin suppression to delivery (stop nitrofurantoin near term), or close surveillance.
- Pyelonephritis: admit, IV β-lactam (± gentamicin), carbapenem if severe/MDR; no improvement in 72 h → renal US.

Safe	Avoid / restrict
Penicillins, cephalosporins (throughout)	Fluoroquinolones (cartilage)
Nitrofurantoin: avoid at term (neonatal haemolysis, G6PD); not for pyelonephritis	TMP/TMP-SMX: 1st trimester (folate, neural tube); late 3rd trimester (kernicterus)
Fosfomycin (lower UTI)	Tetracyclines (teeth/bone, maternal liver)
Aztreonam; gentamicin (category C, used for pyelonephritis)	Chloramphenicol ("grey baby")

American College of Obstetricians and Gynecologists (ACOG): nitrofurantoin and sulfonamides first-line in 2nd/3rd trimesters; in the 1st only if no alternative.

Elderly

- Bacteriuria is the background, not the signal. **Do not screen or treat ASB** in community or long-term care residents.
- Bacteriuria + confusion ≠ UTI. Without urinary symptoms → hydrate and reassess. Do not culture for non-specific or chronic symptoms, and never to document clearance.
- Pyuria is common and does not predict bacteriuria, but its **absence excludes** it.
- **Revised McGeer criteria** (surveillance): non-catheterised residents need **localising** signs + $\geq 10^5$ CFU/mL of ≤2 species; catheterised residents may qualify on **systemic** signs (fever, rigors, hypotension, delirium + leucocytosis) + $\geq 10^5$ CFU/mL.
- Treat 3–6 days for uncomplicated, 10–14 if febrile. Watch toxicity (renal dosing, quinolone tendon and central nervous system effects).

Catheter-associated UTI

- Definition: catheter in place >2 days + symptom (fever, suprapubic/CVA pain) + $\geq 10^5$ CFU/mL; in catheterised patients $\geq 10^2$ is significant. Bacteriuria is inevitable: **5–10% per catheter-day** (intermittent catheterisation 1–3% per catheterisation); 100% long term.
- Routes: **extraluminal** (periurethral; predominant, especially women) vs **intraluminal** (bag, clustered patients). **Biofilm** forms within minutes, shelters bacteria and encrusts (*Proteus*, *Pseudomonas*). Catheters >30 days carry polymicrobial flora (4–5 species).
- Prevention: avoid or limit catheter (daily review), aseptic insertion, closed dependent drainage, meatal cleaning with water. "6-C" bundle: Consider, Connect, Clean, Closed, Call (irrigate only if needed), Culture only if indicated.
- Silicone = latex. Silver alloy reduces bacteriuria short-term (Cochrane). Systemic prophylaxis delays bacteriuria only 3–4 days and selects resistance → not recommended.
- Treat **only symptomatic** CAUTI: culture first, **change the catheter** (biofilm), stop antibiotics within 48 h of resolution.

Spinal cord injury

- All UTIs are complicated. UTI is the commonest cause of fever. Risk: overdistension, high pressure, VUR, stones, instrumentation.
- **Clean intermittent catheterisation (CIC, Lapides 1972)** has the lowest long-term complication rate; indwelling urethral catheter the highest. A suprapubic catheter delays but does not reduce bacteriuria.
- Symptoms are atypical: fever, increased spasticity, leakage between catheterisations, autonomic dysreflexia †, malaise, cloudy malodorous urine, back/abdominal discomfort.

Drainage (NIDRR)	Significant bacteriuria
Indwelling urethral or suprapubic catheter	Any detectable count
Intermittent catheterisation	$\geq 10^2$ CFU/mL
Catheter-free men (clean voided)	$\geq 10^4$ CFU/mL

NIDRR = National Institute on Disability and Rehabilitation Research consensus.

- Do not screen or treat ASB. Culture before therapy. Afebrile: 5–7 days (recurrent 7–10); febrile: admit, parenteral therapy, **14 days**. Change the indwelling catheter. No response in 24–48 h → image for obstruction, stone or abscess.
- No routine post-treatment culture except for urea-splitters (must document eradication).
- Prophylaxis and methenamine are ineffective in NGB (prophylaxis doubles resistance). For recurrent UTI: imaging, urodynamic studies (UDS), lower storage pressure, review CIC technique.
- Catheter >**10 years** → bladder cancer risk (often squamous †) → surveillance.

Diabetes, HIV, renal transplant

- DM: do not treat ASB. Low threshold for imaging in pyelonephritis (EPN, papillary necrosis, abscess).

- Transplant: do not screen or treat ASB >1–2 months after transplant. Early UTI: uncomplicated 5–10 days (not 3); pyelonephritis 14–21 days. Stents raise UTI by about half → remove within **4 weeks**. Recurrent UTI → PVR, imaging → VCUG (graft VUR) → UDS.

12 · FUNGAL UTI

- *Candida* UTI is linked to catheters, DM, broad-spectrum antibiotics, immunosuppression; also haematogenous seeding of the kidney (candidaemia). AIDS: *Candida*, *Aspergillus*, *Cryptococcus*.
- **Asymptomatic candiduria**: do not treat; remove or change the catheter †. Exceptions: neutropenia, very-low-birth-weight infants †, and before urological procedures.
- Symptomatic *Candida* cystitis/pyelonephritis: oral fluconazole 200–400 mg/day for 2 weeks †. Fluconazole-resistant *C. glabrata*/*C. krusei* → amphotericin B †.
- **Fungal balls (mycetoma)** obstruct → drainage (nephrostomy) ± antifungal irrigation and endoscopic or percutaneous removal †, plus systemic therapy. AUA: preoperative fungal culture + 5–7 days periprocedural antifungal.

Funguria before procedures	Position
IDSA 2016	Treat all candiduria before any urological procedure: fluconazole or amphotericin B for several days before and after
AUA Best Practice Statement (BPS) 2019	Antifungal prophylaxis only for high-pressure endoscopy (TURBT, TURP, therapeutic ureteroscopy) and open entry into the tract; full treatment for symptomatic infection at tube changes; longer course with neutropenia + obstruction

13 · PERIOPERATIVE ANTIMICROBIAL PROPHYLAXIS

Principle (AUA BPS 2019)	Rule
Indication	Consider for every procedure breaching a tissue barrier, weighing host defence and consequences of infection
Timing	Within **60 min** of incision; vancomycin and quinolones within 120 min
Duration	**Single dose** default; ≤**24 h** only for PCNL, prostheses, special cases
Agent	Cover likely local flora per antibiogram
Known infection	Treat to sterile urine before elective surgery (that is treatment, not prophylaxis); operating on infected urine → bacteraemia in about half

- Host risk factors: age, anomaly, malnutrition, smoking, steroids, immunodeficiency, indwelling hardware, infected stones, distant infection, prolonged stay.

Wound class	Examples	AP
I clean	Scrotal/inguinal, RPLND, nephrectomy without entering tract	Not needed (except groin/perineal incision, prosthesis)
II clean-contaminated	Any endoscopy; opening the urinary tract	Single dose
III contaminated	Infected stone PCNL, **transrectal biopsy**, bowel segments	Strongly recommended; add anaerobic cover
IV dirty	Abscess, open trauma	Treatment, then tailor to cultures

RPLND = retroperitoneal lymph node dissection.

Procedure	AP?	Agent	Duration
Diagnostic cystoscopy / UDS (no risk factors)	No (EAU strongly against)	—	—

Procedure	AP?	Agent	Duration
Cystoscopy + biopsy/fulguration	Consider host factors	TMP-SMX, amoxicillin–clavulanate, cephalosporin	Single
Transurethral resection of prostate (TURP) / bladder tumour (TURBT), laser prostate	All (TURBT: EAU only if high risk)	Cefazolin, TMP-SMX	Single (TURP short course with catheter acceptable)
Transrectal prostate biopsy	All	Fluoroquinolone or cephalosporin ± aminoglycoside; targeted by rectal swab	Single (≥1 day)
Ureteroscopy	All (diagnostic: uncertain)	TMP-SMX, 1st/2nd-gen cephalosporin	Single
Percutaneous nephrolithotomy (PCNL)	All	1st/2nd-gen cephalosporin; aminoglycoside + metronidazole	≤24 h
Shock wave lithotripsy (SWL)	Only with risk factors (EAU: no)	TMP-SMX, cephalosporin	Single
Open/robotic entering tract (prostatectomy, pyeloplasty)	All	Cefazolin	Single
Colon segment	All	Cefazolin + metronidazole / cefoxitin / ertapenem + oral neomycin with erythromycin or metronidazole	Single
Artificial urinary sphincter, penile prosthesis, sacral neuromodulation	All	Aminoglycoside + cephalosporin or vancomycin	≤24 h
Vaginal surgery incl. slings	All	Cefazolin/cefoxitin	Single
Penile (circumcision), clean scrotal	Likely none (scrotal: high-risk only)	Cefazolin	Single

- Aztreonam replaces aminoglycosides in renal impairment. Routine vancomycin is not recommended (inferior to β-lactams for methicillin-sensitive *S. aureus*).
- **Transrectal ultrasound (TRUS) biopsy**: sepsis 0.3–3.1%, mainly quinolone-resistant *E. coli*; strongest risk factor is antibiotics within 6 months. Options: augmented (quinolone + aminoglycoside), **targeted** by rectal swab culture, fosfomycin. AUA: use alternatives if local resistance >20%. EAU: add povidone-iodine rectal preparation.
- **UDS (SUFU 2017)**: AP if neurogenic LUT dysfunction, raised PVR, ASB, immunosuppression, age >70, catheter use, recent joint implant. DM, prior UTI, valve disease and obesity alone do not need it. SUFU = Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction.
- **Catheter or stent removal**: consider a single dose with risk factors. No AP for chronic catheter changes or long-term CIC. Stent colonisation far exceeds urine colonisation → culture the stent in high-risk patients.
- **Endocarditis** (American Heart Association): no AP solely for prevention; but high-risk valve + colonised urine before manipulation → sterilise urine (amoxicillin for enterococcus).
- **Orthopaedic hardware**: pins, plates and screws never need AP. Joint <2 years or high-risk host + high-bacteraemia procedure (stone manipulation, upper tract endoscopy, bowel, transrectal biopsy) → AP (quinolone, or ampicillin + gentamicin). Do not treat ASB before arthroplasty.

> **KEY POINT** Prophylaxis is justified by wound contamination × host vulnerability. Where neither is raised (diagnostic cystoscopy, sterile-urine SWL, index UDS), ecological cost outweighs benefit. Otherwise give one well-timed, flora-matched dose; anything longer is treatment.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Raz & Stamm 1993	RCT, intravaginal oestriol vs placebo, postmenopausal recurrent UTI	0.5 vs 5.9 UTIs/patient-year; lactobacilli returned, pH fell	Vaginal oestrogen is first-line prevention after menopause
Hooton 2018	RCT, +1.5 L water/day vs none, premenopausal women	Cystitis 1.7 vs 3.2/year	Hydration halves recurrences
Cai (ASB in recurrent UTI)	RCT, treat vs observe ASB in young women	Treatment ↑ symptomatic recurrence and resistant *E. coli*	Do not treat ASB; *E. faecalis* may protect
ALTAR (methenamine)	Methenamine hippurate vs daily antibiotic prophylaxis, 12 months	Non-inferior	Non-antibiotic alternative to prophylaxis
Drekonja 2021	RCT, afebrile male UTI, 7 vs 14 days ciprofloxacin/TMP-SMX	7 days non-inferior	7 days for afebrile men

15 · MCQ ONE-LINERS

1. *S. saprophyticus*: ~10% of cystitis in young sexually active women.
2. P pili bind glycolipid receptors in the kidney and are found in ~90% of pyelonephritis strains.
3. Type 1 pili adhesin FimH binds uroplakin Ia; mannose inhibits it.
4. Most abundant urinary protein blocking type 1 pili: uromodulin (Tamm–Horsfall).
5. Quiescent intracellular reservoirs explain same-strain recurrence without a urological focus.
6. Nitrite-negative uropathogens: enterococci, staphylococci, *Pseudomonas*.
7. Cystitis culture threshold in symptomatic women: 10^2 CFU/mL.
8. ASB in women needs two consecutive voided $\geq 10^5$ specimens; catheter specimen $\geq 10^2$.
9. Only screen-and-treat ASB groups: pregnancy and mucosa-breaching urological procedures.
0. TMP-SMX empirical for cystitis only if local resistance ≤20%.
1. Quinolone for outpatient pyelonephritis only if community resistance ≤10%.
2. Cranberry acts via type A proanthocyanidin blocking P-fimbrial adhesion; dose ≥36 mg.
3. Methenamine requires urine pH <6; ineffective against urease producers.
14. Oral oestrogen does not prevent recurrent UTI; vaginal oestrogen does.
15. Enterococci are never covered by cephalosporins.
16. Ceftazidime covers *Pseudomonas*; ceftriaxone does not.
17. Perinephric abscess: symptoms >5 days or fever ≥5 days on antibiotics.
18. Renal abscess ≤3 cm: antibiotics alone; >5 cm: percutaneous drainage.
19. XGP commonest organism: *Proteus*; pelvis contracted around a staghorn.
20. Michaelis–Gutmann bodies: pathognomonic of malakoplakia.
21. EPN without fluid, streaky gas: worst prognosis.
22. Fournier hypocalcaemia: lipase-released fatty acids chelate calcium.
23. LRINEC ≥6 suggests necrotising fasciitis.
24. Testes are spared in Fournier gangrene because of independent blood supply.
25. CAUTI: bacteriuria 5–10% per catheter-day; change catheter before treating.
26. SCI bladder management with fewest long-term complications: clean intermittent catheterisation.
27. Surgical prophylaxis: within 60 min; single dose; ≤24 h for PCNL and prostheses.
28. Pins, plates and screws never require urological antibiotic prophylaxis.

27 Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis, Orchitis, Epididymitis and Scrotal Pain

Reading time ≈ 52 min (slow pace)

1 · PROSTATITIS: DEFINITIONS AND CLASSIFICATION

- **Drach (1978)** gave the first systematic classification, built on the segmented cultures of **Meares and Stamey (1968)**: acute bacterial, chronic bacterial, nonbacterial (inflammation, negative culture), prostatodynia (neither). MCQ
- **National Institutes of Health (NIH) classification (1999)** is the current standard.

NIH category	Entity	Defining feature
I	Acute bacterial prostatitis (ABP)	Acute urinary tract infection (UTI) of prostate, often febrile
II	Chronic bacterial prostatitis (CBP)	Recurrent UTI with the **same organism**; well between episodes
IIIA	Chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS), inflammatory	Leucocytes in expressed prostatic secretions (EPS), post-massage urine (VB3) or semen
IIIB	CP/CPPS, non-inflammatory	No leucocytes
IV	Asymptomatic inflammatory prostatitis	No pain; incidental on biopsy (for cancer) or semen analysis (infertility)

- Categories I + II = **5–10%**; category III = **90–95%**. Old nonbacterial prostatitis + prostatodynia were merged into III.
- **IIIA vs IIIB has no clinical or therapeutic significance**; leucocyte microscopy of VB3/EPS is no longer recommended.
- "Prostatitis" is a misnomer in III: the prostate may not be the pain source, hence "chronic pelvic pain syndrome".
- **Category III definition (NIH Chronic Prostatitis Collaborative Research Network, CPCRN)**: pelvic pain/discomfort for **≥3 of the previous 6 months**. Exclusions: uropathogens on standard culture, urogenital cancer, prior radiation/chemotherapy, urethral stricture, neurological bladder disease.
- Multidisciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) research network term: urologic chronic pelvic pain syndrome.

Histopathology

- Histological inflammation is near-universal (autopsy: chronic inflammation in **>70%**) and incidental in benign prostatic hyperplasia (BPH) chips and biopsies, so on its own it cannot define a clinical disease.
- **Nickel histological grading** (location / extent / grade):
  - Location: glandular; periglandular (within ~50 µm of a gland); stromal (>50 µm).
  - Extent: focal <10%, multifocal 10–50%, diffuse >50%.
  - Grade: 1 = separated cells; 2 = confluent sheets, no destruction; 3 = confluent with tissue destruction/nodules.
- **Prostatic calculi** obstruct and act as a bacterial nidus.
- **Granulomatous prostatitis** — tissue diagnosis (epithelioid granulomas). Epstein–Hutchins types: specific, non-specific, post-transurethral resection of the prostate (TURP), allergic. Classically after **intravesical bacillus Calmette–Guérin (BCG)**: causes a **transient, benign rise in prostate-specific antigen (PSA)** — do not read as cancer. Tuberculosis (TB) also causes it.

2 · CATEGORY I: ACUTE BACTERIAL PROSTATITIS

Routes and risk factors

- Routes: **ascending urethral** (main); intraprostatic reflux of infected urine into ducts; direct seeding at biopsy; lymphatic (rectum, rare); haematogenous (sepsis, TB).
- Risks: insertive anal intercourse; phimosis; condom or indwelling catheter; clean intermittent catheterisation (up to one-third); instrumentation/biopsy; obstruction (BPH, stricture, dysfunctional voiding); preceding cystitis or epididymo-orchitis.

Microbiology

- ***Escherichia coli* 65–80%**; then *Pseudomonas*, *Proteus*, *Klebsiella*, *Serratia*; *Enterococcus* up to 10%; consider *Neisseria gonorrhoeae* in young sexually active men.
- **Post-instrumentation**: *Pseudomonas* ~20%, more abscess. **Post-transrectal biopsy**: fluoroquinolone (FQ)-resistant, extended-spectrum β-lactamase (ESBL)-producing, positive blood cultures more likely.
- Human immunodeficiency virus (HIV): also *Staphylococcus aureus*, *Salmonella*, mycobacteria.

> **KEY POINT** **Why young healthy men get ABP:** the organism, not the host, explains it — virulent uropathogenic *E. coli* that forms **biofilm** (bacteria in a polysaccharide matrix) more often than cystitis/pyelonephritis strains; biofilm blocks antibiotic penetration, which is why prostatic infection needs longer, well-penetrating therapy.

Clinical features

- Frequency, dysuria, perineal/suprapubic pain; fever, rigors, myalgia; retention (oedema, pain, or bladder-neck spasm); severe cases — sepsis.
- Spontaneous ABP: younger, lower sepsis risk. Post-instrumentation/biopsy: older, more septicaemia (worse with ESBL).
- In febrile male UTI, **>90%** show rise in PSA and/or prostate volume even if the picture suggests pyelonephritis — prostatic involvement is the rule.

Evaluation

- First question: **spontaneous or after manipulation/transrectal biopsy?** — it decides empirical antibiotics. Seek diabetes, HIV, neurological disease, recent antibiotics.
- Digital rectal examination (DRE): tender, swollen, "boggy" prostate (60–90%). **Gentle only; prostatic massage is contraindicated** (bacteraemia/sepsis).
- Full blood count, urinalysis, midstream urine culture (positive 60–85%); **blood cultures** if febrile (positive usually >38.4 °C; same organism); renal function.
- Nucleic acid amplification test (NAAT) for gonorrhoea/chlamydia only if urethral discharge (sexually transmitted infection, STI, ~2%).
- Post-void residual (PVR) by **ultrasound**, not catheter.
- **PSA not recommended** acutely; if raised, follow to normal (up to **6 months**) before attributing to cancer.
- Imaging only if abscess suspected (transrectal ultrasonography, TRUS, or computed tomography, CT). **No biopsy**.

Treatment

- Treat fully: ~10% progress to category II and ~10% to III (about one-quarter after post-biopsy ESBL infection).
- **Outpatient only if all three**: no systemic illness, tolerating orals, no retention. Otherwise admit for parenteral therapy and fluids.
- **European Association of Urology (EAU)**, systemic illness: amoxicillin + aminoglycoside; 2nd-generation cephalosporin + aminoglycoside; or 3rd-generation cephalosporin. Ciprofloxacin only if local resistance **<10%**, fully oral course, not hospitalised, or β-lactam anaphylaxis.

- **Post-transrectal-biopsy febrile prostatitis** (American Urological Association, AUA, White Paper): **no FQ or trimethoprim–sulfamethoxazole (TMP-SMX)**; resuscitate and give **carbapenem**, amikacin, or 2nd/3rd-generation cephalosporin after cultures.
- Step down to oral FQ if susceptible (ciprofloxacin 500 mg bd or levofloxacin 500 mg od). Reculture at 1 week.
- **Duration**: traditionally 4 weeks; **2 weeks** ciprofloxacin gives equivalent cure. Relapse lower with FQ/parenteral β-lactams than co-trimoxazole/oral β-lactams. TB prostatitis: ≥**6 months** anti-TB therapy.
- Adjuncts: non-steroidal anti-inflammatory drugs (NSAIDs); α-blockers for lower urinary tract symptoms (LUTS).
- **Retention**: suprapubic catheter traditionally (avoids prostatic irritation and bacteraemia); brief urethral/intermittent catheterisation acceptable short-term to some; **suprapubic for long-term drainage** (settled).

Prostatic abscess

- Suspect if high fever, immunosuppression, or **no response after 48 h**. Risks: catheters, instrumentation/biopsy, outlet obstruction, CBP, diabetes, cirrhosis, HIV, dialysis.
- TRUS: hypoechoic area. **CT**: location, extraprostatic spread, planning drainage.

ALGORITHM
Abscess <1–2 cm → antibiotics (drainage may shorten course) → progression mandates intervention → peripheral/localised: US-guided percutaneous drainage (recurrence 15–33%) → large or failed: **transurethral unroofing** (intra-op US to locate; massage after unroofing) → extraprostatic extension: open drainage (rare)

3 · CATEGORY II: CHRONIC BACTERIAL PROSTATITIS

- Recurrent UTI with the **same organism** = organism harboured in prostate, not reacquired. Patients are **relatively asymptomatic between episodes** (CP/CPPS is continuous).
- Organisms: *E. coli*, *Pseudomonas*, *Proteus*, *Klebsiella*, *Enterobacter*; *Enterococcus faecalis* (dominant Gram-positive). Debated: staphylococci, streptococci, *Corynebacterium*, *Ureaplasma*, *Mycoplasma*. Need special culture: *M. tuberculosis*, *Candida*. *Trichomonas vaginalis* possible. Biofilm explains poor response.
- **Chlamydia**: role debated because urethral contamination of EPS/semen/tissue cannot be excluded (perineal lateral-lobe biopsies found none). Support: when VB3 positive but VB1 negative, levofloxacin + azithromycin for 4 weeks lowered PSA and symptoms. Verdict: some evidence for a causal role.

Diagnosis: localisation cultures

Test	Specimens	Interpretation
Meares–Stamey 4-glass (previous standard)	VB1 first 10 mL (urethra); VB2 midstream (bladder); EPS after massage; VB3 first 10 mL after massage (prostate)	Prostatic source if EPS/VB3 counts exceed VB1/VB2
2-glass pre/post-massage (current standard)	Midstream pre-massage + first 10 mL post-massage	Predicts 4-glass result in **>95%**; simpler

- No agreed bacterial cut-off; some use a **10-fold rise** in post-massage specimen. **Semen culture not recommended** (poor yield, skin contamination). HIV: culture also for anaerobes, fungi, TB.
- Examine abdomen, scrotum (epididymo-orchitis), DRE (nodule = think cancer). Recheck haematuria 4–6 weeks after infection clears.
- Exclude outlet obstruction/retention: PVR **>180 mL** raises infection risk.

- Men **<45** with first UTI: imaging normal — but assess for **urethral stricture** (up to 41%). Image if diabetes, stones, voiding or neurological problems, poor response, persistent haematuria.

Treatment

> **KEY POINT** **Why fluoroquinolones?** No active transport into the prostate; entry is by passive diffusion, governed by lipid solubility, ionisation, protein binding and size. FQs are lipid-soluble **zwitterions** and concentrate in the alkaline CBP fluid; they also cover Gram-negatives (incl. *Pseudomonas*), Gram-positives and atypicals. Tetracyclines, macrolides and trimethoprim also penetrate well.

Situation (EAU)	Regimen
First line	FQ (ciprofloxacin = levofloxacin) **4–6 weeks**
FQ-resistant, TMP-SMX-sensitive	TMP-SMX **3 months** (cure 30–50% vs 60–80% for FQ)
Chlamydia trachomatis	Azithromycin 500 mg daily 3 weeks, or doxycycline 100 mg bd 10 days; azithromycin > ciprofloxacin (Cochrane)
T. vaginalis	Metronidazole 500 mg tds 14 days

- Adding a macrolide helps — **macrolides penetrate biofilm**.
- HIV/persistent immunosuppression: low-dose suppression; lifelong suppression if immunosuppressed despite antiretrovirals (prevents abscess).
- Refractory cases: daily suppressive antibiotics or TURP.

4 · CATEGORY III: CP/CPPS — AETIOLOGY

Unifying hypothesis: an insult (infection, stress, trauma) in a genetically susceptible man initiates **neurogenic inflammation**, which is then maintained by other factors — the maintaining mechanism need not be the initiating one.

Domain	Key evidence
Infection	Same bacterial counts (~70% have bacteria) in patients and controls; polymerase chain reaction (PCR) finds no causative organism. Mouse **CP1 strain** caused pain persisting after clearance — bacteria may initiate pain that becomes independent of them.
Inflammation/ autoimmunity	Only ~1/3 have biopsy inflammation; degree does not correlate with symptoms. Symptoms correlate with **chemokines/cytokines** (macrophage inflammatory protein-1α, interleukin-17), not histology. T-cell responses to PSA, prostatic acid phosphatase.
Neurological	**Central sensitisation** (activity without/with minimal stimulus); normal sensory thresholds (no peripheral neuropathy); brain (insula) changes.
Pelvic floor	Muscle tenderness and **failure to relax** (not weakness); electromyography: high resting tone, poor endurance. Tender in **81%** on standardised examination.
Psychosocial	Stress, **catastrophising and helplessness predict** pain, urinary symptoms and depression; abuse history raises risk.
Endocrine	Low baseline, blunted-stress adrenocorticotropic hormone (ACTH).

5 · CP/CPPS — CLINICAL FEATURES AND PHENOTYPING

- **Pain is the defining symptom** (distinguishes from BPH). Commonest site **perineum** (63%), then testes, suprapubic, penis; ejaculatory pain ~half. More pain sites = more severe disease.
- **NIH Chronic Prostatitis Symptom Index (NIH-CPSI)**: pain, urinary, quality of life (QOL) domains; pain and QOL subscales are responsive (trial endpoints). †Total 0–43 (pain 0–21, urinary 0–10, QOL 0–12). A fall of **≥6 points** = clinically meaningful response. Limitation: no bladder-pain items → **Genitourinary Pain Index (GUPI)** adds pain on bladder filling/emptying.

- MAPP: pain and urinary symptoms are **separate factors** (depression tracks pain, not urinary symptoms) — assess separately.
- **Bladder pain in men**: ~3/4 have painful urgency or pain on filling; ~42% meet interstitial cystitis/bladder pain syndrome (IC/BPS) criteria → *Ch 28*. Bladder phenotype is more severe. **Hunner lesions** (~12%, ↑ over 50) respond to fulguration.
- Pain beyond the pelvis or overlapping pain conditions (irritable bowel syndrome, IBS; fibromyalgia) = more severe disease.
- Course largely stable; ~70% flare yearly (predicted only by recent sex and UTI-like symptoms).
- Erectile dysfunction (ED) 15–40%; anxiety/depression ~2×.

UPOINT phenotyping

Monotherapy trials failed because patients differ in active domains ("snowflake hypothesis"); UPOINT maps each patient's domains and directs simultaneous multimodal therapy. Domain count correlates with severity. Recommended by UK and International Consultation on Urologic Disease (ICUD) guidance.

Domain	Identify by	Therapy
Urinary	LUTS, PVR	α-blocker; †antimuscarinic
Psychosocial	Depression, catastrophising	Psychology, cognitive therapy, stress reduction
Organ-specific	Prostate tenderness, leucocytes; bladder pain	†Quercetin/pollen extract; bladder-directed therapy; 5α-reductase inhibitor in BPH
Infection	Positive VB3 with negative VB2	Antibiotics
Neurologic/systemic	Pain beyond pelvis, IBS, fibromyalgia	Neuropathic agents (tricyclics, pregabalin), acupuncture
Tenderness	Pelvic floor trigger points	Pelvic floor physiotherapy, myofascial release

6 · CP/CPPS — EVALUATION

A **diagnosis of exclusion**. Onset history points to domain: stress → pelvic floor; physical trauma → neurological/musculoskeletal; after UTI/STI → seek persistent infection.

- **Questionnaires**: NIH-CPSI or GUPI; AUA Symptom Index for voiding; Sexual Health Inventory for Men (SHIM) for ED.
- **Examination**: neurological screen, abdomen, hernia/hydrocele/testis. DRE: prostate tender in <half (not needed for diagnosis); severe tenderness suggests ABP; **nodule = cancer work-up**. Palpate muscles lateral to prostate to coccyx (leg abduction helps); extended exam at perineal body, levator, obturator, urogenital diaphragm — tenderness is **muscular, not prostatic** (47% vs 6%).
- **Labs**: urinalysis (confirm haematuria ≥3 red cells/high-power field); midstream culture (positive VB2 = category II, not III); ICUD recommends 2-glass test (positive VB3 alone → UPOINT "I"). NAAT only if discharge; no semen culture. **Cytology** if irritative symptoms (carcinoma in situ can mimic). PVR. **PSA not indicated**; if raised, never ascribe to CP/CPPS.
- **Imaging (optional)**: CT if abdominal pain; scrotal US for testicular pain; lumbosacral magnetic resonance imaging (MRI) if radiculopathy. **No routine pelvic MRI** (prostate normal; non-specific periprostatic venous dilatation). TRUS of limited use.
- **Functional**: uroflow (young man, poor stream → stricture). Videourodynamics if medical failure with voiding symptoms/poor flow/high PVR: shows bladder-neck obstruction or **pseudodyssynergia** (external sphincter activity during voiding *without* abdominal straining) → physiotherapy, not surgery.
- **Cystoscopy**: low flow/high PVR (stricture) or pain on filling relieved by voiding (Hunner lesion). **No prostate biopsy**.

7 · CP/CPPS — TREATMENT

> **KEY POINT** **Governing fact:** no monotherapy has a positive definitive trial — every failure tested one agent in an unphenotyped population. Start conservative, then give **simultaneous, phenotype-directed multimodal therapy** with a multidisciplinary team (urology, physiotherapy, psychology, pain, gastroenterology).

Modality	Evidence / role
Antibiotics	Placebo-controlled ciprofloxacin and levofloxacin trials negative. EAU: quinolone or tetracycline ≥**6 weeks** only if **treatment-naive and <1 year**. Benefit may be anti-inflammatory (↓ interleukin-6, tumour necrosis factor-α) or due to VB3-only bacteria. **Repeated courses without positive culture not accepted**.
α-blockers	Rationale: dysfunctional voiding + central α-receptors in pain. NIH tamsulosin and alfuzosin trials negative. Use in newly diagnosed, α-blocker-naive men with voiding symptoms (ICUD) / <1 year (EAU). Best meta-analytic combination: **antibiotic + α-blocker**.
Anti-inflammatories	Celecoxib helped pain, not voiding; steroids negative. Combination use only.
5α-reductase inhibitors	Dutasteride in REDUCE improved NIH-CPSI (≥6-point fall 49% vs 37%). Use in older men with BPH; counsel ejaculatory/sexual side effects.
Neuropathic agents	Pregabalin missed primary endpoint (subset benefit); amitriptyline — escalate to ≥50 mg (underdosing causes apparent failure).
Conservative	Individualised diet; **aerobic exercise**; stress reduction, cognitive therapy.
Acupuncture	Strongest conservative evidence: large sham-controlled trial responders 61.5% vs 38.3% at 32 weeks. Offer after first-line failure or if drugs declined.
Pelvic floor physiotherapy	Myofascial trigger point release + paradoxical relaxation (Stanford protocol); myofascial therapy > global massage (57% vs 21%). Refer if pelvic tenderness; works in pseudodyssynergia. Urodynamics do not predict response. Use experienced therapists.
Neuromodulation	Percutaneous tibial nerve stimulation (PTNS) for CPPS; sacral nerve stimulation (SNS) when pain coexists with frequency/urgency (off-label for pain).
Surgery	TURP/radical prostatectomy **not** recommended (pain centralised). **Bladder neck incision** only for proven bladder-neck obstruction after excluding pseudodyssynergia. Hunner lesion fulguration.

8 · ACUTE ORCHITIS

- **Bacterial** (usually >35 y): Gram-negatives (*E. coli*, *Pseudomonas*); also syphilis, TB, brucellosis, filariasis, candiduria.
- **Mumps** = commonest viral cause; outbreaks occur in vaccinated people. Prodrome → parotitis 2–3 weeks after exposure → orchitis with fever, settles in ~1 week. **Atrophy in ~half**, possible oligospermia, rarely complete infertility. Interferon may be used but does not always prevent atrophy.
- Other viruses: HIV, Epstein–Barr, SARS-CoV-2 (usually benign).
- **Autoimmune**: primary (anti-sperm antibodies, asymptomatic, infertility work-up); secondary (vasculitis — Behçet, polyarteritis nodosa, Henoch–Schönlein) → steroids, azathioprine/cyclophosphamide, intravenous immunoglobulin.
- Evaluation: **pain duration** (torsion is the key differential); ask about parotitis, brucellosis symptoms, vasculitis. Urine culture (positive ~40%); STI testing if sexually active; **scrotal US** for torsion, tumour, abscess.
- Treat Gram-negatives (adjust to culture) or STI; HIV — maintenance suppression (esp. *Salmonella*). **Follow-up mandatory** to catch missed torsion or tumour; counsel pain settles in 1–3 days, swelling in 2–4 weeks.

9 · ACUTE EPIDIDYMITIS

- Pain, swelling, inflammation of the epididymis for **<6 weeks**. Route: **ascending from urethra** (~80% urine–epididymis concordance).
- Classic: **<35 years sexually active → *C. trachomatis, N. gonorrhoeae***; **>35 and children → enteric (*E. coli*)**. But >40% under 35 grow enteric organisms, so **screen all sexually active men for STI**; 13% no organism; viruses/mumps not significant.
- Pain starts in the **tail (cauda)** and ascends; unilateral 96%; fever ~25%; orchitis in half; abscess <10%.
- Tests: full blood count, urine culture, first-void urine leucocyte esterase/microscopy, STI NAAT; PVR in older men (treat the cause). **Doppler US**: hyperaemia, enlargement; excludes torsion.

Likely pathogen	Regimen
Chlamydia/gonorrhoea	Ceftriaxone **500 mg IM** single dose (1 g if ≥150 kg) + doxycycline 100 mg bd **10 days**
STI + enteric (insertive anal intercourse)	Ceftriaxone 500 mg IM + levofloxacin 500 mg od 10 days
Enteric only	Levofloxacin 500 mg od 10 days

- Supportive: rest, scrotal elevation, cooling, NSAIDs; admit if fever with leucocytosis. Confirmed STI → test for HIV/other STIs and treat partners.
- **No improvement in 72 h** → re-evaluate and repeat US for abscess or testicular infarction (→ surgery).

10 · CHRONIC ORCHITIS AND EPIDIDYMITIS

- Testicular pain >3 months without known infection is better termed **chronic orchialgia/chronic scrotal content pain**.
- **Granulomatous orchitis**: TB, brucellosis, syphilis, leprosy, sarcoidosis. **Idiopathic** (non-caseating, giant cells) usually diagnosed after orchidectomy for suspected tumour (identical on US); IgG4-related disease may contribute. **Malakoplakia**: histiocytes with **Michaelis–Gutmann bodies** (pathognomonic).
- **Chronic epididymitis**: epididymal pain ≥**3 months**. **Nickel classification** (maps to examination):
  - Inflammatory (swollen/indurated): infective (chlamydia), post-infective, granulomatous (TB), drug-induced (**amiodarone**), idiopathic.
  - Obstructive (may be indurated): congenital, post-infective, post-vasectomy.
  - Epididymalgia: epididymis palpably normal.
- Work-up: urine culture, scrotal US (exclude tumour); then reassurance, scrotal support; drugs lack evidence.

Tuberculous epididymitis

- Epididymis is the commonest genital site (sole site in 45%); testis by contiguous spread. Subacute, often painless swelling, scrotal thickening, **sinus/fistula**; AIDS predisposes.
- Four-drug anti-TB therapy ≥**6 months**. Surgery (drain abscess, epididymectomy/orchidectomy) **only while on anti-TB drugs**; histology decides further duration.
- **Mass not shrinking or enlarging after 3 weeks of therapy → inguinal exploration** to exclude malignancy.

11 · CHRONIC SCROTAL CONTENT PAIN (CSCP)

Definition: uni- or bilateral pain in intrascrotal structures for ≥**3 months** interfering with daily life; includes orchialgia and epididymalgia. No evidence-based guidelines.

Causes and pathophysiology

- Peak 20–30 years. Post-procedural: **inguinal hernia repair** (commonest; ↑ with mesh/recurrent repair), vasectomy (long-term pain ~**1 in 1000** — consent figure), scrotal surgery.
- Others: varicocele, hydrocele, spermatocele, intermittent torsion, referred pain (mid/distal ureteric stone, disc disease, aneurysm, hernia), trauma. Idiopathic 18–25%.
- **Pelvic floor dysfunction**: 93% have a symptom, 88% raised resting tone — same mechanism as CP/CPPS.
- **Neuropathic**: central sensitisation + **Wallerian degeneration** (84% vs 20% controls) in the **"trifecta nerve complex"** — cremasteric muscle, perivasal sheath, posterior periarterial/lipomatous tissue; ↑ calcitonin gene-related peptide in cord.

Evaluation

- **First decision**: isolated scrotal pain, or part of wider pelvic pain (CP/CPPS)? This sets the whole pathway.
- Ask about torsion, infection, surgery, back pain, IBS. Check **testosterone and vitamin B12** (majority deficient). Chronic Orchialgia Symptom Index (COSI, 12 items).
- Examine testis, epididymis, cord, hernia, prostate and pelvic floor. Urinalysis; **scrotal duplex Doppler US**; CT if stone suspected.
- **Spermatic cord block (SCB)**: **20 mL 0.25% bupivacaine** into the cord at the **pubic tubercle**. **>50% temporary relief** = cord nerves involved and independently predicts success of microsurgical denervation of the spermatic cord (MDSC).

Non-surgical treatment

- Antibiotics rarely indicated (doxycycline or quinolone, up to 4 weeks, if used). NSAIDs for analgesia. **Neuropathic agents**: amitriptyline, nortriptyline, gabapentin, pregabalin; avoid chronic opioids.
- **Therapeutic SCB**: serial bupivacaine + triamcinolone blocks for those unfit/unwilling for surgery (~70% sustained relief).
- **Pelvic floor physiotherapy** if abnormal tone on DRE (DRE is the selection test).
- OnabotulinumtoxinA cord injection: open-label benefit vanished in a double-blind randomised controlled trial (RCT) — placebo effect. Pulsed radiofrequency (42 °C) in block responders: promising, no long-term data.

Surgical treatment

ALGORITHM
Identifiable lesion → directed surgery (hydrocelectomy, spermatocelectomy, orchidopexy for intermittent torsion, **microsurgical varicocelectomy**) → post-vasectomy pain → **vasectomy reversal** (improvement 82–93%; relief may take ≥3 months) → no lesion + positive SCB → **MDSC** → localised epididymal pathology → epididymectomy → last resort: **inguinal orchidectomy**

- **Varicocelectomy** best results microsurgical; predictors: >7 veins ligated, pain score >6, duration >9 months. Embolisation recurs more.
- **MDSC** (Devine and Schellhammer, 1978): †subinguinal, under the microscope: divide cord nerves; **preserve arterial supply and lymphatics** (prevents hydrocele); ligate internal spermatic veins; preserve cremaster and fascia. **Targeted "trifecta" MDSC** treats only cremasteric fibres, vasal sheath and posterior perivascular fat — equal efficacy, ~21 vs 53 min.
- MDSC: cure 50–70%, improving with time; better if several structures painful; prior failed surgery does not reduce success. Complications few — **testicular atrophy** must be consented.
- **Epididymectomy**: 50–75% complete relief; best with focal tenderness, US lesion or cyst; for post-vasectomy pain include the distal vasectomy site (removes sperm granuloma/fibrosis).
- **Orchidectomy**: least reliable (0–75%); **inguinal > scrotal** (75% vs 55%); better for nociceptive (ischaemic) than neuropathic pain; "phantom" pain may persist. Persistent pain afterwards → resect

genital branch of genitofemoral nerve at external ring and bury proximal end in pelvis.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
NIH CPCRN ciprofloxacin/tamsulosin	RCT, 6 weeks, NIH-CPSI ≥15	No benefit vs placebo	Antibiotic/α-blocker monotherapy fails in unselected CP/CPPS
REDUCE (CP/CPPS analysis)	Dutasteride vs placebo, 4 years	≥6-point NIH-CPSI fall 49% vs 37%	5α-reductase inhibitor for older men with BPH
MAPP myofascial physiotherapy	Randomised feasibility vs global massage	57% vs 21% responders	Pelvic floor physiotherapy for tender pelvic floor
Acupuncture vs sham	Large RCT	Responders 61.5% vs 38.3% at 32 weeks	Acupuncture is evidence-based option
Botulinum toxin cord injection	Double-blind RCT	No better than local anaesthetic	Open-label scrotal pain series overestimate benefit

13 · MCQ ONE-LINERS

1. NIH category IV prostatitis: no pain; found on biopsy or semen analysis.
2. IIIA vs IIIB: leucocytes in EPS/VB3/semen; no therapeutic relevance.
3. CP/CPPS definition: pelvic pain ≥3 of previous 6 months.
4. Commonest organism in ABP: *E. coli* (65–80%).
5. Febrile prostatitis after transrectal biopsy: carbapenem; avoid FQ and TMP-SMX.
6. Retention in ABP needing long-term drainage: suprapubic catheter.
7. No response to ABP therapy at 48 h: image for prostatic abscess.
8. CBP hallmark: recurrent UTI, same organism, asymptomatic between episodes.
9. Current localisation standard: 2-glass pre/post-massage test (predicts 4-glass >95%).
0. FQs penetrate prostate because they are lipid-soluble zwitterions.
1. CBP first line: FQ 4–6 weeks.
2. Macrolides help in CBP because they penetrate biofilm.
3. BCG-induced granulomatous prostatitis: transient benign PSA rise.
4. PSA in CP/CPPS: not indicated; raised value must not be ascribed to it.
15. Pseudodyssynergia: sphincter activity during voiding without straining → physiotherapy.
16. Clinically meaningful NIH-CPSI response: ≥6-point fall.
17. UPOINT: Urinary, Psychosocial, Organ, Infection, Neurologic/systemic, Tenderness.
18. Best combination in CP/CPPS meta-analyses: antibiotic plus α-blocker.
19. Commonest viral orchitis: mumps; atrophy in about half.
20. Epididymitis <35 y: chlamydia/gonorrhoea; >35 y: *E. coli*.
21. STI epididymitis: ceftriaxone 500 mg IM + doxycycline 10 days.
22. Drug causing chronic epididymitis: amiodarone.
23. Michaelis–Gutmann bodies: malakoplakia.
24. TB epididymal mass enlarging after 3 weeks of therapy: inguinal exploration.
25. Spermatic cord block: 20 mL 0.25% bupivacaine at pubic tubercle.
26. >50% relief after cord block predicts MDSC success.
27. Trifecta nerve complex: cremaster, perivasal sheath, posterior lipomatous tissue.
28. Orchidectomy for scrotal pain: inguinal approach better than scrotal.

28 Interstitial Cystitis/Bladder Pain Syndrome and Related Disorders

Reading time ≈ 56 min (slow pace)

1 · DEFINITION AND NOMENCLATURE

- **Interstitial cystitis/bladder pain syndrome (IC/BPS)** — umbrella term for a heterogeneous chronic pelvic pain syndrome, ranging from gross inflammatory bladder disease with **Hunner lesions (HLs)** to bladder pain with many non-urological complaints and a normal-looking bladder.
- **American Urological Association (AUA) definition** (clinical): "an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the urinary bladder, associated with lower urinary tract symptoms (LUTS) of **>6 weeks**' duration, in the absence of infection or other identifiable causes." Pressure and discomfort are included deliberately because patients describe them so.
- † **European Society for the Study of Interstitial Cystitis (ESSIC) definition (2008)**: chronic (**>6 months**) pelvic pain, pressure or discomfort perceived to be related to the bladder, with ≥1 other urinary symptom (persistent urge or frequency), confusable diseases excluded. † ESSIC types by cystoscopy with hydrodistension (1 normal, 2 glomerulations, 3 HL; X not done) and biopsy (A normal, B inconclusive, C inflammatory infiltrate/detrusor mastocytosis/granulation tissue/fibrosis) — e.g. **type 3C** = classic HL disease.
- **National Institutes of Health/National Institute of Diabetes and Digestive and Kidney Diseases (NIH/NIDDK) criteria (1987)**: a *research* definition for homogeneous trial populations; applied clinically it missed **~60%** of patients → definitions broadened.
- **International Classification of Diseases, 11th Revision (ICD-11)** "chronic primary bladder pain syndrome": chronic primary pain in the bladder region + ≥1 symptom (pain worse on filling, day/night frequency), not explained by infection or secondary visceral pain — classes the pain itself as the disease.
- **Urological chronic pelvic pain syndrome (UCPPS)** — used by the Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) Research Network to include IC/BPS (both sexes) and chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS, males) → *Ch 27*.
- **Current trend**: "**IC**" = HL disease (end-organ inflammation); "**BPS**" = non-HL symptom complex. BPS has never been shown to progress to HL disease → probably distinct disorders.

2 · EPIDEMIOLOGY (HEADLINE)

- RAND Interstitial Cystitis Epidemiology (RICE) study: **3.3–7.9 million US women** and 2.1–4.6 million men have IC/BPS-type symptoms. Mean onset 30–40 years.
- Female:male **5:1** in clinic series but **2–3:1** in population studies → men are under-diagnosed. Many men labelled CP/CPPS report painful bladder filling/painful urgency → **ask men about bladder pain** and reconsider IC/BPS when CP/CPPS treatment fails.
- **HL prevalence**: working figure **~20%** (range 3.5–56%, method- and centre-dependent). Rises with age: uncommon <30, significantly higher ≥50, >half of patients over 60–70. Same rate in men.
- **Natural history**: chronic but **not progressive** — stable course to 9 years with flares (commonest trigger: sexual activity; also diet, abdominal exercise). About one-third have a remission ≥3 months.
- Childhood abuse/adversity is a risk factor; catastrophising and depression predict suicidal ideation → cognitive behavioural therapy (CBT).

> **KEY POINT** Why uncontrolled series "work" but randomised controlled trials (RCTs) fail: regression to the mean, placebo effect and more attention in early follow-up. Expect improvement in the first 3–4 weeks of any treatment.

3 · AETIOLOGY AND PATHOPHYSIOLOGY

Unknown and probably multifactorial; several mechanisms may combine into a self-perpetuating cycle of chronic bladder pain and voiding dysfunction.

Theory	Mechanism / evidence	Treatment link
Glycosaminoglycan (GAG) layer / "leaky" urothelium	Defective GAG barrier lets urinary solutes (e.g. potassium, "toxic cations") reach suburothelial nerves and muscle → depolarisation, pain, inflammation. Luminal chondroitin deficit shown in IC.	Pentosan polysulfate (PPS), heparin, hyaluronic acid, chondroitin; basis of potassium test
Mast cells	Activation/degranulation releases histamine and mediators → pain, hyperaemia, fibrosis. Marked in HL disease (detrusor + an epithelial/lamina propria population absent in controls); sparse in BPS. Mast-cell counts have **no diagnostic role**.	Hydroxyzine, cimetidine, amitriptyline (H1 block)
Neurogenic	Neurogenic inflammation, altered neuroplasticity, pelvic organ cross-sensitisation (convergent afferents in dorsal root ganglion, cord, brain), central sensitisation and altered brain connectivity; hypothalamic–pituitary–adrenal axis/stress dysregulation. Small-fibre polyneuropathy found in some (causal role unknown).	Tricyclics, gabapentinoids, neuromodulation, CBT
Autoimmune	Not a classic primary autoimmune bladder disease, but large relative excess of systemic lupus erythematosus (SLE), Sjögren syndrome (SS), rheumatoid arthritis (RA), inflammatory bowel disease (IBD). Lymphoplasmacytic infiltrate in HL disease.	Ciclosporin A (CyA) for HL disease
Infection	No consistent organism (cf. *Helicobacter pylori*) even with culture-independent/microbiome techniques. 18-week sequential antibiotic RCT: no significant benefit, more adverse events.	No antibiotics without culture-proven infection

> **KEY POINT** Central sensitisation explains why many BPS patients have no physical findings and do not respond to local bladder therapy — the mechanism is systemic, not vesical.

4 · PATHOLOGY: HUNNER LESION VS NON-HUNNER PHENOTYPE

- **No pathognomonic histology**; cold-cup biopsy is normal in about half.
- **Role of biopsy**: only in HL/suspicious lesions — to exclude carcinoma in situ (CIS), bladder cancer, eosinophilic, tuberculous and ketamine cystitis.

Feature	HL disease ("IC")	Non-HL ("BPS")
Cystoscopy	Stellate erythematous lesion with radiating vessels, central scar/coagulum; bleeds on filling ("waterfall"); rare at trigone	Normal, or glomerulations after distension
Histology	Ulceration, granulation tissue, dense **lymphoplasmacytic** infiltrate, ↑mast cells, perineural infiltrate; epithelial denudation beyond lesion (field change)	Near-normal mucosa, sparse inflammation; small ruptures/haemorrhages (possibly iatrogenic)
Pain source	Bladder — true end-organ disease	Often regional/widespread pain
Specific treatment	Fulguration / triamcinolone injection; CyA; DMSO responds better; cystectomy succeeds	Phenotype-directed; surgery often fails

5 · ASSOCIATED CONDITIONS

- **75–80%** have pain beyond the pelvis; ~38% widespread; only 20–25% pelvis-only. True somatisation is rare.
- **Chronic overlapping pain conditions (COPCs)**: **irritable bowel syndrome (IBS)** — commonest (>30%), shares visceral hypersensitivity; **fibromyalgia syndrome (FMS)** — prototype of centralised (nociplastic) pain; chronic fatigue syndrome (CFS); migraine; temporomandibular joint disorder (TMJD); low back pain.
- **Regional**: **high-tone pelvic floor dysfunction (HTPFD)** in **~80%**; vulvodynia (vulvar pain ≥3 months); endometriosis.
- **Autoimmune** — rare in absolute terms but largest relative excess: IBD/Crohn (~100-fold, single study), SLE (tens-fold), SS and RA (10–15-fold); allergy and asthma modestly more common.
- **Trap**: in SLE consider **lupus cystitis** — fibrotic bladder with hydroureteronephrosis.

> **KEY POINT** Commonest ≠ most over-represented: IBS is commonest; autoimmune disease shows the greatest relative excess. Associated disorders are phenotype markers — widespread pain predicts failure of bladder-directed therapy.

6 · CLINICAL FEATURES

- Suprapubic pain **worse with filling, temporarily relieved by voiding** → frequent day and night voiding to avoid pain. Dyspareunia common; isolated dysuria/urethral pain uncommon.
- Urgency is **painful urgency** — unlike overactive bladder (OAB):

Feature	OAB	IC/BPS	HTPFD
Urge	Sudden; fear of leakage	Persistent, mounting; driven by pain	Persistent; pelvic pressure
Pain	Absent	Present	Present
Incontinence	Urgency urinary incontinence (UUI)	Not a feature	—
Voiding	—	Frequency, low volumes	Obstructive voiding

Pain and UUI discriminate IC/BPS from OAB.

7 · DIAGNOSIS — EXCLUSION PLUS LOCALISATION

IC/BPS is a **clinical diagnosis of exclusion**. Failure to identify other pain generators is among the commonest causes of treatment failure.

Basic evaluation (AUA 2022) — enough in almost all uncomplicated cases

- **History**: pain–filling relationship, location, triggers (diet), sexual pain, bowel function, prior pelvic surgery/irradiation, **cyclophosphamide, ketamine**, associated syndromes, response to previous therapy.
- **Examination**: abdomen (hernia, tender muscles); women — urethral diverticulum, atrophy, lichen sclerosus, mesh complications, **cotton-swab (Q-tip) test** for vulvodynia, single-digit vaginal palpation of bladder/urethra, bimanual; men — genital examination and digital rectal examination (DRE) with pain mapping; **pelvic-floor palpation**; focused neurological examination.
- **Tests**: post-void residual (PVR), urinalysis, urine culture; cytology if malignancy risk (smoking, haematuria, age >40); microhaematuria per AUA guideline.
- **Frequency–volume chart** (24 h suffices): frequent low-volume voids. Large voids → seek a non-bladder source.
- **Body map**: widespread pain = ≥3 of 12 regions besides the pelvis.

Symptom scores

- **O'Leary–Sant Interstitial Cystitis Symptom Index and Problem Index (ICSI/ICPI)** — most used; composite score; trial endpoint (≥30% fall in ICSI).
- Genitourinary Pain Index (GUPI); Bladder Pain/IC Symptom Score (BPIC-SS); pain visual analogue scale (VAS) 0–10; Global Response Assessment (GRA).
- **† Pelvic Pain and Urgency/Frequency (PUF) symptom scale** — patient questionnaire of pain, urgency, frequency and bother, devised to screen for IC/BPS; not diagnostic.
- Score pain and urinary symptoms **separately**: individual pain and urinary trajectories diverge over time and after treatment.

Confusable disorders (ESSIC-modified)

Group	Disorder	Excluded by
Malignancy	Bladder cancer/CIS; prostate, gynaecological, colon cancer	Cystoscopy ± biopsy; prostate-specific antigen (PSA); pelvic examination
Infection	Bacterial, atypical organisms, **tuberculosis (TB)**, herpes; prostatitis	Culture; **sterile pyuria → culture for TB**
Iatrogenic/drug	Radiation, cyclophosphamide, tiaprofenic acid, **ketamine**	History; ketamine — upper-tract change
Outflow/emptying	Bladder neck obstruction, benign prostatic obstruction, retention	PVR, uroflow, (video)urodynamics
Structural/other	Stones, urethral diverticulum; endometriosis, vulvodynia, atrophy; IBS, IBD	Imaging, cystoscopy, examination, laparoscopy, colonoscopy
Neuromuscular	Pudendal nerve entrapment; pelvic-floor myalgia	Nerve block (diagnostic); pelvic-floor examination

Ketamine cystitis

- Recreational N-methyl-D-aspartate (NMDA) antagonist; mimics IC/BPS; continued use → widespread HL-like inflammation → irreversible fibrosis, **hydronephrosis (~30%)**, renal insufficiency (~10%).
- Suspect in the **young (<30) patient with visible bladder inflammation**; use is rarely volunteered. Treatment: **cessation** (with rehabilitation) improves LUTS.

High-tone pelvic floor dysfunction

- Pelvic pressure, persistent urge, obstructive voiding, constipation, next-day dyspareunia — often since childhood.
- Signs: hypertonic, tender muscle, banding, **myofascial trigger points** reproducing pain. Standardised six-site examination: right/left anterior levator (pubococcygeus), posterior levator (iliococcygeus), obturator internus — vaginally in women, transrectally in men.
- Missed HTPFD → poor response because treatment differs → refer to pelvic-floor physiotherapist.

Urodynamics (UDS)

- Not part of the standard work-up (AUA). Use for complicated cases: failed therapy, significant incontinence, high PVR, suspected obstruction.
- Findings: reduced capacity, early strong desire (hypersensitivity), **normal compliance** (pain prevents filling to non-compliance), pain on filling reproducing symptoms. End-stage fibrotic bladder → low capacity and compliance. Detrusor overactivity (DO) in 15–19% — **does not exclude IC/BPS**.

Office cystoscopy

- **AUA 2022**: cystoscopy when **HLs are suspected — especially age >50**. HLs cannot be predicted from symptoms or urine; recognising them opens phenotype-specific therapy. Also for haematuria, failed therapy, suspected stone/cancer/obstruction.

- Office flexible cystoscopy shows most HLs **without hydrodistension**; overdistension bleeding obscures them. Biopsy CIS-like lesions and **any trigonal lesion** (HLs rarely occur there). Narrow-band imaging may reveal subtle lesions.

Local anaesthetic challenge vs potassium sensitivity test

- **Anaesthetic challenge**: e.g. 15 mL 0.5% bupivacaine + 15 mL 2% lidocaine intravesically. Relief → bladder-centric pain and selects intravesical therapy; no relief does not exclude IC/BPS.
- **Potassium sensitivity test (PST)**: intravesical **0.4 M KCl** vs sodium (water); provocation of pain/urgency = positive, thought to reflect epithelial permeability (GAG defect) or hypersensitivity. **Not recommended**: painful, and neither sensitive nor specific — negative in up to ¼ of NIDDK patients; positive in ~⅓ of normal men, ¼ of OAB, most CP/CPPS, virtually all radiation/acute bacterial cystitis. Does not predict PPS response.

Cystoscopy with hydrodistension (CHD)

- Under anaesthesia: fill to **80 cm H_2O for 1–3 min** → drain (terminal effluent blood-tinged) → refill → **glomerulations** (pinpoint submucosal petechiae); mucosal tears, often near bladder neck.
- **Glomerulations are non-specific**: seen after radiotherapy, with carcinoma, chemotherapy, in dialysis/diverted (long-unfilled) bladders, prostate pain syndromes, and normal bladders after overdistension; absent in some IC/BPS; number does not correlate with severity. Hence CHD has limited diagnostic value.
- Biopsy (cold-cup) only **after** distension (perforation risk) and only for suspicious lesions — chiefly HL vs CIS. CIS has been misdiagnosed as IC/BPS.

8 · PHENOTYPE-DIRECTED MANAGEMENT

Phenotype	Identified by	Pain centre	Treatment
Hunner lesion	Cystoscopy (>50)	Bladder — inflammation	Fulguration ± triamcinolone; refractory → CyA
Anaesthetic-responsive bladder pain	Lidocaine challenge	Bladder	Intravesical therapy
Pelvic-floor tenderness/HTPFD	Six-site examination	Pelvic floor	Myofascial physical therapy
Widespread pain	Body map, COPCs	Central	Tricyclics, gabapentinoids, CBT, multidisciplinary care

> **KEY POINT** MAPP reanalysis (2023) of two negative NIH RCTs: hydroxyzine/PPS helped **localised pelvic pain** only; amitriptyline helped **widespread pain** only. A trial negative in all-comers can hide a positive result in the right phenotype.

9 · TREATMENT — AUA 2022 PRINCIPLES

- **No "home run" therapy**; goal is acceptable quality of life (QoL) and coping, not cure. Set expectations, shared decision-making, agree a flare plan (e.g. anaesthetic instillation).
- **No more first- to sixth-line tiers**: options are grouped (behavioural, oral, intravesical, procedures, major surgery). Patients may move to any category **except major surgery**, or to multimodal therapy, without first failing "lower" options. Severe symptoms may justify initial multimodal therapy.
- Stop ineffective treatment; not improving → re-evaluate for other pain generators.
- **Guideline Standard**: manual (myofascial) physical therapy for pelvic-floor tenderness. **Recommendation**: fulguration and/or triamcinolone for HLs. Most drugs are **Options**.
- Multidisciplinary: CBT; pudendal block; trigger-point injections (anaesthetic or onabotulinumtoxinA [BTX-A]). Chronic opioids not routinely recommended.

ALGORITHM

Symptoms ≥6 weeks → history, examination (pelvic floor, Q-tip, DRE), PVR, urinalysis, culture ± cytology → exclude confusable disease → cystoscopy if HL suspected (>50) → HL: fulguration/triamcinolone → refractory: CyA

No HL → match to phenotype: pelvic floor → physiotherapy; bladder-centric (lidocaine responder) → intravesical; widespread → systemic/CBT → behavioural + oral + intravesical as needed → BTX-A or SNM → (rare, bladder-centric, all else failed) diversion ± cystectomy

10 · TREATMENT OPTIONS (AUA 2022)

Category	Option	Key point
Behavioural / non-pharmacological	Education, fluid control, elimination diet, bladder training/urge suppression, stress management, heat/cold, CBT	All patients; no risk
	Myofascial physical therapy	Standard for HTPFD; **avoid Kegel** strengthening
Oral	Amitriptyline	Widespread pain; titrate to ≥50 mg
	PPS	Only US Food and Drug Administration (FDA)-approved oral drug; maculopathy
	Hydroxyzine	Allergy subset; sleep
	Cimetidine	One positive placebo RCT
	Ciclosporin A	Refractory HL
Intravesical	DMSO 50%	FDA-approved; better in HL
	Heparin	GAG replacement; slow
	Lidocaine (alkalinised)	Immediate — flares
Procedures	HL fulguration/triamcinolone	Recommendation; 75–90% respond
	Short, low-pressure hydrodistension	Brief benefit
	Intradetrusor BTX-A	Must accept clean intermittent catheterisation (CIC)
	Sacral neuromodulation (SNM)	HTPFD/obstructive voiding
Major surgery	Urinary diversion ± cystectomy; augmentation	Last resort, bladder-centric only

10.1 Behavioural and conservative

- Stress reduction (mindfulness, yoga); avoid individual triggers.
- **Diet**: triggers in most patients — **caffeine, alcohol, carbonated drinks, artificial sweeteners, spicy food, citrus/acidic juices including cranberry, tomato**. Safe: water, milk, bland non-acidic foods. **Elimination diet**: diary → bland diet → reintroduce one item at a time → avoid only confirmed triggers (prevents needless restriction).
- **Education and behavioural modification programme (EBMP)** (control arm of NIH amitriptyline RCT): 45% moderately/markedly improved alone.

10.2 Pelvic-floor physical therapy

- Relaxes trigger points and lengthens contractures in pelvic floor, abdomen and hip girdle.
- Landmark RCT: 10 sessions myofascial therapy vs global massage → **59% vs 26%** improved.
- Adjuncts: reverse Kegels, heat, muscle relaxants, constipation control, trigger-point injections when physiotherapy plateaus.

10.3 Oral therapy

- **Amitriptyline** (tricyclic antidepressant, TCA): anticholinergic, serotonin/noradrenaline reuptake block, potent H1 antagonism, sedation, ?β-adrenergic detrusor relaxation. Dose: **10 mg at night, +10 mg weekly to ~50 mg** (25–75 mg). NIH RCT: 55% vs 45% (EBMP alone), not significant overall, but **77% vs 53%** in those tolerating ≥50 mg. Efficacy independent of HL status. Side-effects: sedation, anticholinergic effects, weight gain; caution with incomplete emptying. **Contraindications**: long QT, conduction block, recent myocardial infarction, heart failure, ventricular arrhythmia.
- **Hydroxyzine** (H1, blocks neuronal mast-cell activation): 25 mg nocte → 50 mg nocte + 25 mg mane; NIH RCT negative; useful for allergy and sleep, often with amitriptyline.
- **Cimetidine** (H2): 400 mg twice daily; positive double-blind RCT; mechanism unexplained (histology unchanged).
- **PPS**: synthetic sulfated polysaccharide resembling GAGs; **100 mg three times daily**; only **3–6%** excreted in urine; effect takes **3–6 months**. Only FDA-approved oral drug for IC. Trials conflicting; phase IV RCT negative (≈40% response in both arms); probably less effective in HL disease. AUA Option. Adverse effects (<4%): reversible alopecia, diarrhoea, nausea, rash, rare bleeding; caution in high breast-cancer risk.
- **Ciclosporin A**: for HL disease refractory to fulguration/triamcinolone; far superior to PPS in an open-label RCT; works in most HL but few non-HL patients. Dose **1–3 mg/kg/day in two doses** (below transplant dose); judge at 3–4 months; relapse on stopping responds to restarting. Toxicity in 30–55%: **nephrotoxicity, hypertension**, infection, malignancy, gingival hyperplasia. Monitor blood pressure, renal/hepatic function, drug level (1 month, then 3-monthly).
- **Others**: gabapentinoids (no IC/BPS RCTs; centralised pain); sildenafil, quercetin (small studies).
- **Antibiotics**: AUA — **long-term antibiotics should not be offered**. One empirical course is reasonable if never treated; a flare thought infective needs urinalysis and culture. Recurrent proven cystitis triggering flares → consider prophylaxis.

10.4 PPS pigmentary maculopathy

- Unique retinal pigmentary maculopathy; risk rises with **cumulative dose**; may be irreversible and progress after stopping. Symptoms: difficulty reading, slow dark adaptation, blurred vision, rarely blindness.
- **FDA label (2020)**: ophthalmic history in all before starting → comprehensive baseline retinal examination if pre-existing eye disease → retinal examination for all **within 6 months** of starting, then periodically → if changes develop, re-weigh benefit and risk. Counsel long-term users.

10.5 Intravesical therapy

Treats the end-organ without systemic effects; suits **bladder-centric** pain (lidocaine responders). A poor lidocaine response predicts failure of every instillation. AUA Option, alone or combined.

Agent	Regimen	Key points
DMSO 50% (dimethyl sulfoxide)	50 mL, **15–20 min dwell**, every 1–2 weeks × 4–6	FDA-approved as stand-alone (often used in cocktails). Anti-inflammatory, analgesic, muscle relaxant, collagen-dissolving, enhances absorption. Positive crossover RCTs; better in HL disease; DO patients do not respond. Garlic-like breath; initial flare of urgency/frequency; longer dwell worsens pain. Enhances absorption of co-instilled lidocaine → toxicity risk.
Heparin	10,000–40,000 U, held 30–60 min, self-instilled	Mimics GAG layer, anti-inflammatory; negligible systemic absorption; benefit may take months; better with alkalinised lidocaine.
Alkalinised lidocaine	200 mg + 8.4% bicarbonate daily × 5	Alkalinisation allows urothelial penetration. RCT: 30% vs ~10% at day 3. Immediate effect → flares, maintenance. Watch tachyphylaxis.

Agent	Regimen	Key points
Hyaluronic acid, chondroitin, intravesical PPS	Weekly then monthly	GAG replenishment; weak/negative trials

- Cocktail (e.g. DMSO + triamcinolone + bicarbonate + heparin ± lidocaine) weekly × 6, monthly maintenance in responders; no combination proved superior to DMSO alone.
- Observe early instillations for lidocaine toxicity (perioral numbness) → drain and wash out.

10.6 Hunner lesion treatment

- **AUA Recommendation**: fulguration and/or submucosal triamcinolone; biopsy if never biopsied (exclude CIS). **75–90%** improve; re-treatment needed at mean 7–12 months (triamcinolone) or 11–29 months (fulguration) — counsel about periodic repeat.
- Resection risks perforation (anterior lesions); laser risks bowel injury by forward scatter. Treating glomerulations does not help.

ALGORITHM
General anaesthesia → on entry, map HL site/number/size before distension bleeding → mild distension (lesion centres bleed) → biopsy suspicious and any trigonal lesion → fulgurate at **low volume** (Bugbee) from the **outer margin inwards** → triamcinolone 40 mg/mL (or diluted 1:1 with bupivacaine/lidocaine) in 0.5-mL aliquots, 2–3 mm submucosally → fulgurate bleeding sites → empty, instil lidocaine–bupivacaine

> **KEY POINT** Why fulgurate from the margin inward? Cautery-induced erythema and oedema keep spreading peripherally, so the true lesion edge is lost if you start centrally.

10.7 Hydrodistension

- AUA Option: **short-duration (1–3 min), low-pressure (60–80 cm H_2O)** under anaesthesia, inspecting throughout; compress urethra against scope if leaking; instil anaesthetic at end.
- Benefit is brief: ~½ at 1 month, ¼ at 3 months, almost none at 6 months. Capacity <600 mL responds better. Counsel: post-procedure flare is common.
- Anaesthetic capacity **<400 mL** marks a bladder-centric phenotype (worse pain, more inflammation, often HL).

10.8 Intradetrusor onabotulinumtoxinA

- AUA Option after failed conservative, oral and intravesical therapy; not FDA-approved for IC/BPS. Acts on sensory pathways (↓ substance P, glutamate release) — not just muscle.
- **Dose 100 U** in 5 mL saline (20 U/mL); 200 U caused unacceptable dysuria/retention. Injected into the **trigone** (densest sensory innervation; spares the detrusor body → less retention). Trigone-including vs sparing: no difference in efficacy or complications. Response lasts ~5–9 months; repeat injections remain effective.
- **Prerequisite**: willing and able to do CIC — hard in IC/BPS (urethral pain, low capacity). Caution with incomplete emptying. Toxin cannot cross urothelium → must be injected.

10.9 Neuromodulation

- Transcutaneous electrical nerve stimulation (TENS): open-label, adjunct only. Posterior tibial nerve stimulation (PTNS): sham-controlled negative. **Sacral neuromodulation (SNM, S3)**: uncontrolled long-term pain reduction with and without HL — AUA Option.
- SNM best when pain coexists with **HTPFD/levator tenderness**; poor without pelvic-floor dysfunction.
- Counsel: IC/BPS has the highest SNM adverse-event and explantation rates.

	BTX-A	SNM
Favoured when	Bladder-centric pain; can do CIC	Obstructive voiding, incomplete emptying, HTPFD pain, CIC impossible

	BTX-A	SNM
Main risk	Retention → CIC	Adverse events, explantation

10.10 Major surgery — last resort

- For **rare (<10%), carefully selected, bladder-centric** patients who have failed all other therapy; experienced surgeons only. Pain may persist even after cystectomy.
- **Pre-operative sequence**: history → body map → pelvic-floor examination → intravesical anaesthetic trial (profound relief localises to bladder) → treat other pain generators. **Differential spinal block**: pain persisting after spinal to T10 is generated higher in the cord — surgery will fail.

Predicts success	Predicts failure
HLs (especially refractory)	No HLs
Small fibrotic bladder; low anaesthetic capacity (<200–400 mL)	High anaesthetic capacity
Profound relief with intravesical anaesthetic	Equivocal/no anaesthetic response
Painful filling, pain reproduced by scope touch	Urethral or neuropathic pain; HTPFD; widespread pain

- **Options**: conduit with bladder in situ (~⅕ later need cystectomy); cystectomy + conduit (~¾ pain-free); continent pouch (pouch pain 40–50%); supratrigonal augmentation (pain/HL recurrence at trigonal cuff, CIC).
- **Diversion ± cystourethrectomy** removes urine–mucosa contact and distension; avoids CIC and has lower metabolic risk than augmentation → prefer the **simplest diversion**. Retained bladder risks pain, pyocystis, haemorrhage, spasm.

11 · WHAT NOT TO DO (AUA)

Do not offer	Why
Long-term oral antibiotics	No organism; RCT negative with frequent adverse events (80% vs 40%)
Intravesical bacille Calmette–Guérin (BCG)	Ineffective; DMSO superior in crossover trial; † risk of BCG sepsis
High-pressure, long-duration hydrodistension †	No added benefit; risk of bladder rupture and sepsis †
Long-term systemic (oral) glucocorticoids	No benefit; serious systemic toxicity
† Intravesical resiniferatoxin	Ineffective in RCT †
Kegel strengthening exercises	Exacerbate pain in HTPFD
Major surgery before all other therapy / in non-bladder-centric pain	Irreversible; pain often persists

> **KEY POINT** Nothing is worse than irreversible surgery followed by continued pain and a surgeon with nothing more to offer — every predictor of surgical success marks bladder-generated pain.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
Myofascial physical therapy RCT (FitzGerald 2012)	10 sessions myofascial therapy vs global massage	59% vs 26% improved	Guideline Standard for pelvic-floor tenderness
NIH amitriptyline RCT (Foster 2010)	EBMP ± amitriptyline	55% vs 45% (not significant); 77% vs 53% if ≥50 mg	Titrate to ≥50 mg; behavioural therapy itself effective

Trial	Design	Result	Take-home
PPS phase IV RCT	PPS vs placebo	No difference (≈40% each)	PPS efficacy doubtful; remains Option
MAPP phenotype reanalysis (2023)	Two negative NIH RCTs stratified by pain distribution	Hydroxyzine/PPS help localised; amitriptyline widespread	Treat by phenotype

13 · MCQ ONE-LINERS

1. AUA IC/BPS definition requires symptoms for >6 weeks; ESSIC requires >6 months.
2. NIDDK criteria were research criteria and missed ~60% of clinically diagnosed patients.
3. ESSIC type 3C = Hunner lesion with inflammatory infiltrate on biopsy.
4. About 20% of IC/BPS patients have HLs; prevalence rises with age (>50).
5. Commonest associated condition: IBS; largest relative excess: autoimmune disease (IBD, SLE).
6. Pain and urgency incontinence distinguish IC/BPS from OAB.
7. Detrusor overactivity on UDS (15–19%) does not exclude IC/BPS.
8. AUA 2022: office cystoscopy when HL suspected, especially age >50.
9. Glomerulations are non-specific and do not correlate with severity.
0. Potassium sensitivity test uses 0.4 M KCl; no longer recommended.
1. Local anaesthetic challenge localises pain to the bladder and selects intravesical therapy.
2. Sterile pyuria in suspected IC/BPS: culture for tuberculosis.
3. Young patient, inflamed bladder, hydronephrosis: think ketamine cystitis.
14. Myofascial PT beat global massage 59% vs 26% — Guideline Standard.
15. Amitriptyline: start 10 mg nocte, increase 10 mg weekly to ~50 mg.
16. PPS: only FDA-approved oral IC drug; 3–6% reaches urine; 3–6 months to act.
17. PPS causes cumulative-dose pigmentary maculopathy; retinal examination within 6 months of starting.
18. Ciclosporin A 1–3 mg/kg/day is for refractory HL disease.
19. DMSO 50%: FDA-approved intravesical agent; dwell 15–20 min; garlic breath.
20. HL treatment: fulguration/triamcinolone — 75–90% respond; fulgurate from margin inward.
21. Therapeutic hydrodistension: 60–80 cm H_2O for 1–3 min; benefit gone by 6 months.
22. Intradetrusor BTX-A 100 U into trigone; patient must accept CIC.
23. IC/BPS has the highest SNM explantation rate among indications.
24. AUA: no long-term antibiotics, BCG, long high-pressure hydrodistension, or chronic oral steroids.
25. Pain persisting after spinal block to T10 predicts failure of bladder surgery.

29 Sexually Transmitted Diseases

Reading time ≈ 37 min (slow pace)

1 · SCREENING, PREVENTION AND PARTNER NOTIFICATION

- **Sexually transmitted diseases (STDs)**: chlamydia is the commonest reportable STD; syphilis is the fastest rising.
- **Nucleic acid amplification test (NAAT)** is the standard screen. Sites follow anatomy and exposure: men who have sex with men (MSM) need urethral, rectal and pharyngeal tests. MCQ Test the pharynx for gonorrhoea, not chlamydia.
- **Condoms** cut HIV (human immunodeficiency virus) acquisition by about 80%. They protect only the skin they cover. Lambskin condoms let viruses through.
- **Partner notification**: partners from the previous **60 days** for urethritis, chlamydia and *Mycoplasma genitalium*. All partners for trichomoniasis; the previous month for scabies and lice. Abstain for 7 days and until partners are treated.
- Test everyone with, or at risk of, an STD for HIV and syphilis.

2 · URETHRITIS

- Causes discharge and dysuria, or no symptoms. Complications: epididymitis, orchitis, prostatitis.
- **Gonococcal** (*Neisseria gonorrhoeae*): 5–20% of cases.
- **Non-gonococcal urethritis (NGU)**: *Chlamydia trachomatis* > *M. genitalium*. *M. genitalium* is the leading cause of **persistent or recurrent** urethritis. Less common: *Trichomonas vaginalis*, herpes simplex virus (HSV), adenovirus. No pathogen is found in up to half.
- **Diagnosis** needs any one of the following:
  - discharge;
  - Gram stain with **≥2 white blood cells (WBC) per oil-immersion field**;
  - positive leucocyte esterase on first-void urine;
  - ≥10 WBC per high-power field in first-void urine.

  The old threshold of ≥5 WBC missed many infections.
- Intracellular gram-negative diplococci in a symptomatic man confirm gonorrhoea. A negative Gram stain does not exclude it.
- **First-void urine NAAT** is preferred for gonorrhoea and chlamydia because it is more sensitive. Use **culture** for treatment failure, resistance surveillance and children. Take the culture before other swabs.

Pathogen	First line	Alternatives / notes	Follow-up
Gonorrhoea (Centers for Disease Control and Prevention [CDC] 2021)	**Ceftriaxone 500 mg intramuscular (IM) once** (1 g if ≥150 kg)	Gentamicin 240 mg IM + azithromycin 2 g (β-lactam allergy); cefixime 800 mg. Europe: ceftriaxone 1 g + azithromycin 2 g. Add chlamydia cover unless excluded	Test of cure only for **pharyngeal** infection; retest at 3 months
Chlamydia / NGU	**Doxycycline 100 mg twice daily (bd) 7 days**	Azithromycin 1 g once	No test of cure; retest at 3 months
M. genitalium	Doxycycline 7 days, then **azithromycin** 1 g + 500 mg daily × 3 if macrolide-sensitive, or **moxifloxacin** 400 mg × 7 days if resistant or untested	Single-dose azithromycin is inadequate	Abstain until cured
T. vaginalis (male)	Metronidazole 2 g once	Tinidazole 2 g once	Treat all partners

- **Why ceftriaxone monotherapy**: resistance is rising, so CDC uses a higher ceftriaxone dose and has dropped azithromycin.
- Third-generation cephalosporins are safe unless there was an immunoglobulin E (IgE)-mediated penicillin reaction in the last 10 years (cross-reactivity <1%).
- **Gonorrhoea treatment failure**: symptoms beyond 3–5 days, or a positive test of cure, without re-exposure. Send culture + susceptibility testing.
- ***M. genitalium*** has no cell wall, so it is invisible on Gram stain, and culture takes months. NAAT + macrolide-resistance assay is the only useful test. Doxycycline first lowers the organism load; the resistance result then picks the second drug.
- **Trichomonas** is not in the first NGU work-up (low prevalence): test partners of infected women and men in high-prevalence areas or after empiric failure.
- **Aftercare**: test for chlamydia/gonorrhoea, syphilis and HIV; no chlamydia test of cure because residual nucleic acid persists for about 3 weeks.

ALGORITHM
Persistent NGU → check adherence and untreated partner → NAAT for *M. genitalium* ± *T. vaginalis* → negative → think chronic pelvic pain syndrome

KEY POINT Men with gonorrhoea usually have symptoms; women are usually silent and present with pelvic inflammatory disease (PID) or infertility. Treating the man protects his partner. Chlamydia can also cause reactive arthritis.

3 · EPIDIDYMITIS BY AGE

Acute epididymitis lasts <6 weeks. Chronic disease (≥6 weeks) is most often due to tuberculosis (TB) → *Ch 27*. Always exclude **torsion**. Ultrasound (US) is only for suspected torsion. Investigate with urethritis criteria + NAAT, and urine culture if enteric infection is likely.

Group	Organisms	Empiric regimen (10 days)
Young, sexually active	*C. trachomatis*, *N. gonorrhoeae*	Ceftriaxone 500 mg IM + doxycycline 100 mg bd
Insertive anal sex	Above + enterics	Ceftriaxone + levofloxacin 500 mg daily
>35 years, outlet obstruction or instrumentation	*Escherichia coli*	Levofloxacin alone

The course is 10 days, not the 7 used for urethritis. Rule of thumb: doxycycline for chlamydia, ceftriaxone for gonorrhoea, levofloxacin for enterics. Abstain until the patient and partners are treated and symptom-free; test for other STDs and HIV.

4 · GENITAL ULCER DISEASE

Herpes is the commonest cause, then syphilis. Appearance misleads, so **every ulcer gets the same tests**:

- syphilis serology ± darkfield or polymerase chain reaction (PCR);
- HSV NAAT + type-specific serology;
- a *Haemophilus ducreyi* test if endemic;
- HIV test.

Biopsy atypical or non-healing ulcers, because carcinoma can mimic them.

Disease (organism)	Lesion	Nodes	Test	Treatment
Primary syphilis (*Treponema pallidum*)	**Painless**, indurated, clean, single	Painless, rubbery, bilateral	Serology; darkfield	Benzathine penicillin G 2.4 million units (MU) IM once

Disease (organism)	Lesion	Nodes	Test	Treatment
Herpes (HSV-2 > HSV-1)	**Painful** multiple vesicles → shallow ulcers	Tender, bilateral	NAAT of the unroofed base	Aciclovir and related drugs
Chancroid (*H. ducreyi*)	**Painful, ragged, undermined, purulent**	Tender **suppurative buboes**	Special culture; diagnosis by exclusion	Azithromycin 1 g once, or ceftriaxone 250 mg IM once, or ciprofloxacin 500 mg bd 3 days, or erythromycin 7 days
Lymphogranuloma venereum (LGV) (*C. trachomatis* L1–L3)	Small, painless, transient	Painful, matted, with fistulae; †groove sign	Chlamydia NAAT	Doxycycline bd **21 days**
Donovanosis (*Klebsiella granulomatis*)	**Painless, beefy red**, bleeds easily	**Absent** (†pseudobuboes)	**Donovan bodies** on crush preparation	Azithromycin 1 g weekly or 500 mg daily, ≥3 weeks **and until healed**

Genital herpes

- HSV is latent in the **sacral ganglia**. It reactivates with trauma (including surgery), ultraviolet light, immunosuppression or fever.
- Asymptomatic shedding occurs on about 10% of days, so most transmission comes from people unaware of infection.
- HSV-2 recurs more often than HSV-1, so type it for counselling. A negative NAAT does not exclude herpes. The Tzanck smear is insensitive.
- Complications: aseptic meningitis and **urinary retention** (autonomic involvement). HSV-2 raises HIV risk 2–3-fold.

Drug	First episode (7–10 days)	Suppressive	Episodic
Aciclovir	400 mg three times daily (tds)	400 mg bd	800 mg bd 5 days
Valaciclovir	1 g bd	500 mg–1 g daily	500 mg bd 3 days
Famciclovir	250 mg tds	250 mg bd	1 g bd 1 day

- Treat every first episode.
- **Suppressive** therapy also reduces transmission.
- **Episodic** therapy must start within 1 day of lesions, or in the prodrome.

Chancroid, donovanosis, LGV

- **Chancroid** needs all four:
  - a painful ulcer;
  - a typical clinical picture;
  - syphilis excluded;
  - HSV excluded.

  Aspirate or drain fluctuant buboes. Men with HIV or a foreskin respond less well.
- **Donovanosis**: if antibiotics fail, biopsy, because **penile squamous cell carcinoma mimics it**.
- **LGV**: proctocolitis in MSM is now the commonest presentation. Untreated LGV leads to genital elephantiasis and rectal stricture. Groin nodes are commoner in men because the female genital tract drains to retroperitoneal nodes.

5 · SYPHILIS STAGING AND SEROLOGY

Stage	Features	Treatment (penicillin-allergic)
Primary (10–90 days)	Chancre + nodes; heals in 3–8 weeks	Benzathine penicillin **2.4 MU IM once** (doxycycline 14 days)
Secondary (3–5 months)	Rash on **palms and soles**, **condylomata lata**, alopecia, hepatitis, nephritis	
Early latent (<1 year)	Positive serology only	
Late latent / unknown	Positive serology only	Benzathine **weekly × 3** (doxycycline 28 days)
Tertiary	Gumma, aortitis, tabes dorsalis	Weekly × 3 if cerebrospinal fluid (CSF) is normal
Neurosyphilis (any stage)	Central nervous system (CNS), eye, ear	**Aqueous penicillin G 18–24 MU/day intravenous (IV) 10–14 days**

- *T. pallidum* cannot be cultured. Diagnosis needs **two serological tests**, because either alone misses early disease or gives false positives:
  - **Nontreponemal test (NTT)**, e.g. venereal disease research laboratory (VDRL) or rapid plasma reagin (RPR). False positives are common (HIV, autoimmune disease, pregnancy). The titre falls with treatment, so the NTT is used **to monitor treatment**.
  - **Treponemal test (TT)**, e.g. fluorescent treponemal antibody absorption (FTA-ABS), *T. pallidum* particle agglutination (TP-PA) or enzyme immunoassay. It usually stays positive for life.
- A significant change is **fourfold (two dilutions)**. Use the same NTT every time.

ALGORITHM
Reverse algorithm: TT positive → NTT → NTT negative → second, different TT → positive without past treatment = treat as late latent

- **Jarisch–Herxheimer reaction**: fever and myalgia within 24 h of treatment. It is caused by treponemal lysis, not allergy, and is treated with non-steroidal anti-inflammatory drugs (NSAIDs).
- Penicillin allergy with doubtful adherence: desensitise and give benzathine. Benzathine is IM only; neurosyphilis needs IV aqueous penicillin.
- Reverse-algorithm TT positive with documented past treatment: no action unless re-exposure (then repeat NTT in 2–4 weeks).
- All primary or secondary syphilis: test for HIV and offer pre-exposure prophylaxis (PrEP) if negative.
- **Follow-up**: serology at 6 and 12 months, plus 24 months for latent disease. A fourfold rise means failure or reinfection: retreat weekly × 3, retest HIV, consider CSF.
- In HIV, serology may be atypical: use biopsy or darkfield if suspicion persists.

6 · HPV AND ANOGENITAL WARTS

- **Human papillomavirus (HPV)** types:
  - **6 and 11**: about 90% of warts;
  - **16 and 18**: >70% of cervical cancer, plus penile and anal cancer.
- **Bowenoid papulosis** (HPV-16) is now called high-grade squamous intraepithelial lesion (HSIL).
- **Giant condyloma** (†Buschke–Löwenstein) is a verrucous carcinoma caused by low-risk HPV-6. It invades deeply despite bland histology.
- Diagnosis is clinical; HPV typing is not needed. Inspect the anal canal and meatus.
- **Biopsy if**:
  - the diagnosis is uncertain;

- there is no response or worsening on treatment;
- the patient is immunocompromised;
- the lesion is pigmented, indurated, fixed or ulcerated.

Treatment	Key points
Imiquimod 5% (patient)	3 nights a week for up to 16 weeks; wash off after 6–10 h. Works by local immune stimulation. Weakens condoms
Podofilox 0.5% (patient)	Antimitotic. Twice daily for 3 days, then 4 days off. **Contraindicated in pregnancy**
Sinecatechins 15% (patient)	Green-tea extract; three times daily; do not wash off. Weakens condoms. Not for use in HIV, herpes or pregnancy
Cryotherapy, trichloroacetic acid 80–90% (provider)	Cryotherapy every 1–2 weeks (cytolysis). Trichloroacetic acid weekly (chemical coagulation): let it dry before the patient stands, and neutralise with bicarbonate if painful
Excision, curettage, CO_2 laser	For large warts; use smoke extraction

Change treatment if there is no response by 3 months. Observation is acceptable. Treatment does not clear HPV.

Vaccine:

- **Nonavalent** covers 6, 11, 16, 18, 31, 33, 45, 52 and 58.
- Give at **11–12 years** (possible from 9), with catch-up to **26 years**, for both sexes.
- Started <15 years: **2 doses**. Started ≥15 years or immunocompromised: **3 doses**.
- Prophylactic only. Not given in pregnancy or yeast allergy.

7 · MOLLUSCUM AND ECTOPARASITES

- **Molluscum contagiosum**: **umbilicated** dome-shaped papules. They regress in 6–12 months, so observation is acceptable. Options: cryotherapy, curettage, imiquimod. With HIV, lesions are extensive (≥100, confluent) and number rises as CD4 falls, so start highly active antiretroviral therapy (HAART) first. Biopsy atypical lesions.
- **Pubic lice** (*Pthirus pubis*): **permethrin 1%** rinse for 10 min. For resistance: malathion 0.5% or ivermectin 250 µg/kg repeated in 7–14 days. Diagnosis needs live lice or viable eggs. Condoms do not protect. Review at 1 week.
- **Scabies**: itchy burrows, worse at night. Symptoms come from delayed hypersensitivity, so they appear 3–6 weeks after the first infestation but 1–3 days after re-infestation.
  - Treat with **permethrin 5%** on all skin, or ivermectin 200 µg/kg repeated at 2 weeks.
  - **Crusted scabies** (immunocompromised): topical agent + repeated ivermectin; never lindane.
  - Lindane 1% is a last resort (age ≥10 years) because of toxicity.
  - Scratching → streptococcal infection → post-streptococcal glomerulonephritis.
- Treat contacts from the past month. Wash bedding at 50 °C.

8 · HIV FOR UROLOGISTS

- **Acquired immunodeficiency syndrome (AIDS)**: CD4 <200/mm³ or an AIDS-defining illness.
- Acute infection is flu-like and highly infectious, and the antibody test may still be negative.
- Screen everyone aged 15–65 once. Diagnose with an antigen–antibody immunoassay, then a differentiation assay. If these disagree, test HIV RNA.

Postexposure prophylaxis (PEP) and PrEP

- **PEP**: start within **72 h** after a substantial-risk exposure, i.e. mucosal, broken-skin or percutaneous contact with blood or genital fluid from an HIV-positive source. Do not give PEP after 72 h or for urine, saliva or tears.

- **PEP regimen**: **28 days** of tenofovir disoproxil fumarate (TDF) + emtricitabine + raltegravir or dolutegravir. Test at 6 weeks, 3 months and 6 months.
- **Needlestick**: risk about **0.3%**. Give three-drug PEP even if the source viral load is low.
- **PrEP**: daily emtricitabine + TDF, or tenofovir alafenamide (TAF) in men only, for people at sexual or injection risk.
- Ulcerative STDs (herpes, syphilis, chancroid) increase both HIV acquisition and transmission. Syphilis raises viral load and lowers CD4.
- **Circumcision** cuts men's heterosexual HIV acquisition by **50–60%**. The thin inner foreskin is rich in Langerhans cells. It does not protect partners.

Problem	Key points
Urinary tract infection (UTI)	Risk rises below CD4 500. Culture first. *Pseudomonas* in up to 30%. If no response, look for fungi, TB and cytomegalovirus (CMV). *Salmonella* UTI needs lifelong prophylaxis
Renal TB	Acid-fast bacilli (AFB) culture of 3–6 early-morning urines. Rifampicin induces cytochrome P450 and lowers protease inhibitor (PI) levels. Annual urine AFB for 10 years
Skin	Cellulitis: cover methicillin-resistant *Staphylococcus aureus* (MRSA). Fasciitis is commoner. Warts in 20%
Prostate	Abscesses and fungi (CD4 <200); relapse in up to 70% despite prolonged therapy. Cryptococcal prostatitis is a reservoir for meningitis
Testis	Recurrent epididymo-orchitis. The testis is a reservoir for opportunistic organisms, so give maintenance suppression (especially *Salmonella*). Atrophy is common
Stones	**Indinavir** stones are **radiolucent** and form in alkaline urine. **Stop the drug and hydrate**, which succeeds in up to 70%. Do a full metabolic work-up
Kidney	HIV-associated nephropathy (HIVAN): nephrotic-range proteinuria, collapsing glomerulosclerosis. Treat with HAART + angiotensin-converting enzyme (ACE) inhibitor. Tenofovir is nephrotoxic
Voiding	**Retention with sacral sensory loss may mark seroconversion**. Bladder and bowel dysfunction with sciatic pain suggests CMV polyradiculopathy: do a lumbar puncture
Erectile dysfunction (ED)	Affects about half of men under 50. PIs inhibit CYP3A and raise sildenafil levels 3–10-fold, so **start at a low dose**
Cancer	On HAART, non-AIDS cancers (prostate, anal, testis) now make up about 70%

After sexual assault (men): ceftriaxone 500 mg once + doxycycline 7 days (add metronidazole for women). Give HBV (hepatitis B virus) vaccine, plus hepatitis B immune globulin if the assailant is hepatitis B surface antigen-positive. Give HPV vaccine if aged 9–26, and nPEP (non-occupational PEP) within 72 h.

9 · MCQ ONE-LINERS

1. Gram-stain urethritis threshold: ≥2 WBC per oil-immersion field.
2. Gonorrhoea: ceftriaxone 500 mg IM once.
3. Gonorrhoea test of cure: pharyngeal infection only.
4. Recurrent NGU: *M. genitalium*.
5. Resistant *M. genitalium*: doxycycline, then moxifloxacin.
6. Chlamydia: doxycycline 7 days.
7. Epididymitis in a man >35: levofloxacin 10 days.
8. Commonest genital ulcer: herpes.
9. Painful undermined ulcer with bubo: chancroid.
0. Donovan bodies: granuloma inguinale.
1. LGV: doxycycline 21 days.
2. HSV latency: sacral ganglia.
13. Palmar rash and condylomata lata: secondary syphilis.
14. Late latent syphilis: benzathine weekly × 3.
15. Neurosyphilis: IV aqueous penicillin.
16. Fever after syphilis treatment: Jarisch–Herxheimer reaction.
17. Significant NTT change: fourfold.
18. HPV-6: giant condyloma.
19. Podofilox: contraindicated in pregnancy.
20. HPV vaccine before 15: two doses.
21. Scabies: permethrin 5%.
22. PEP: within 72 h, for 28 days.
23. Needlestick HIV risk: 0.3%.
24. Indinavir stones: radiolucent.

25. Circumcision: 50–60% less HIV.
26. Male trichomonas: metronidazole 2 g once.
27. Chancroid diagnosis: by exclusion of syphilis and HSV.
28. Commonest LGV presentation now: proctocolitis.
29. Molluscum: umbilicated papules; HAART first in HIV.
30. HIV ED: low-dose sildenafil with protease inhibitors.
31. Salmonella UTI in HIV: lifelong prophylaxis.

30 Cutaneous Diseases of the External Genitalia

Reading time ≈ 48 min (slow pace)

1 · APPROACH TO GENITAL DERMATOSES

- Diagnosis rests on **history + examination**. Skin has a limited morphological repertoire, so mimics abound → **low threshold for biopsy**.
- History: duration, onset, symptoms (pruritus, burning, pain), allergies, occupation, prior treatment, over-the-counter products (lubricants, spermicides, anti-itch creams). A painless lesion is itself informative.
- **Full skin survey**: extragenital lesions often give the diagnosis (e.g. extensor silvery plaques → psoriasis).
- Identify the **primary lesion** (colour + morphology). On mucosal genital skin, papules and macules may present as erosions.

Primary lesion	Definition	Secondary lesion	Definition
Macule	Flat discolouration	Scale / crust	Shed keratin / dried serum
Papule	Solid, raised, **≤0.5 cm**	Erosion	Epidermal loss only → **no scar**
Nodule / plaque	Solid >0.5 cm (plaque = superficial)	Ulcer	Epidermis + dermis → **scars**
Vesicle / bulla	Free fluid ≤ / >0.5 cm	Fissure	Linear dermal loss, vertical walls
Pustule	Pus	Atrophy / scar	Thinning / dermal damage
Wheal	Oedematous dermis (not free fluid); transient	Dermoepidermal junction decides scarring	

Bedside / adjunct test	Use
Potassium hydroxide (KOH) prep; periodic acid–Schiff (PAS)	Dermatophyte (branching hyphae), *Candida* (yeast + pseudohyphae); KOH dissolves keratin
Mineral oil prep	Scabies mites, eggs
Tzanck smear	Herpes simplex virus (HSV), varicella zoster, molluscum
Wood's lamp	Erythrasma (coral-red); vitiligo (accentuated)
Serology / culture	Syphilis / *Pseudomonas*

Biopsy (difficult cases, suspected malignancy): punch 2–3 mm is often diagnostic, closed with 6-0/7-0 nylon or left to granulate. Small shaft/scrotal lesions in office under local anaesthesia; large or meatal lesions in theatre.

2 · PRINCIPLES OF THERAPY

- **Topical first**; escalate to systemic only when needed. Urologists overprescribe systemic antibiotics → resistance, drug interactions, prolonged disruption of bowel/vaginal flora.
- **Systemic antifungals** only for: very extensive dermatophytosis, endemic mycoses, Majocchi granuloma (deep follicular), severe immunocompromise, topical failure. Avoid systemic **ketoconazole** (hepatotoxicity, adrenal insufficiency).
- **Systemic glucocorticosteroid (GCS)**: prednisone is agent of choice; short courses (≤3 weeks) as single morning dose to limit hypothalamic–pituitary–adrenal (HPA) suppression. Long-term: osteoporosis, avascular necrosis of femoral head, cataract, diabetes, hypertension.
- **Topical steroids**: genital skin has a thin dermis → atrophy, striae (inner thigh), telangiectasia, hypopigmentation. The **prepuce acts as an occlusive dressing**, enhancing penetration — atrophy

can follow 8 weeks under the foreskin. Rule: low potency, short courses (exception: lichen sclerosus).
- Emollients: ointments preferred (creams sting); apply to moist skin; ceramides restore barrier. Topical Janus kinase (JAK) inhibitor **ruxolitinib** avoids steroid atrophy.

Physical modality	Indication	Caveat
Narrow-band ultraviolet B (UVB); 308-nm excimer laser	Psoriasis, vitiligo, atopic/seborrhoeic dermatitis	Excimer: localised genital disease, no non-melanoma skin cancer risk
Psoralen + UVA (PUVA)	Psoriasis, vitiligo, lichen planus	**Genital squamous cell carcinoma (SCC)** risk → shield genitals; supplanted
Photodynamic therapy	Large/resistant condyloma, warts in pregnancy	Oxygen radicals; off-label
CO_2 laser; cryosurgery	Condyloma, pearly penile papules; molluscum	Cryo more widely used

3 · ECZEMATOUS DERMATOSES

A pruritic red genital eruption is not automatically allergic: exclude *Candida*, herpes, impetigo and drug reaction first.

Atopic dermatitis (AD, eczema)

- Chronic relapsing, flexural, intensely pruritic; ill-defined border (papulosquamous lesions are sharp). **90%** present before age 5; asthma/rhinitis association.
- Mechanism: **filaggrin defect** → leaky barrier → irritant entry and immune activation; scratching → *Staphylococcus aureus* superinfection whose toxins act as superantigens. Every treatment either repairs the barrier or breaks the itch–scratch cycle.
- No pathognomonic test; spongiotic dermatitis on biopsy; atopic history is the key clue.
- Treatment: soap substitutes, frequent emollients (avoid hot baths), short tapering topical steroid, topical calcineurin inhibitors (tacrolimus, pimecrolimus — steroid-sparing), ruxolitinib, bedtime sedating antihistamine. Oral antistaphylococcal agents give no benefit (randomised trial). Severe: **dupilumab** (T helper 2, first monoclonal antibody for AD), tralokinumab (anti-interleukin-13 [IL-13]); rarely systemic immunosuppressants.

Contact dermatitis

Feature	Irritant (ICD)	Allergic (ACD)
Mechanism	Direct chemical cytotoxicity; anyone; dose-driven	Type IV delayed hypersensitivity in sensitised
Share	~**80%**	Minority; genital rare (2.4%)
Agents	Soaps, solvents, acids/alkalis	**Nickel** (commonest; genital piercing), neomycin, bacitracin, fragrances, preservatives, latex, topical steroids
Acute picture	Burning, sharply limited to exposure zone; minutes–hours	Well-demarcated pruritic, weeping eruption
Confirm	Exposure history	**Patch testing**
Treat	Avoidance, barrier preparations	Allergen removal, antihistamine; severe → **3-week** prednisone taper (short course rebounds)

Seborrhoeic dermatitis (SD)

- Pink-yellow plaques with greasy adherent scale in sebaceous areas (scalp, nasolabial, chest, glans, pubis); commoner in uncircumcised.
- Possible response to *Malassezia furfur*. Up to **83%** of acquired immunodeficiency syndrome (AIDS) patients; also Parkinson disease → **extensive SD should prompt human immunodeficiency virus (HIV) testing**.

- Treatment: topical **antifungal** (ketoconazole) is the backbone; antidandruff shampoos; hydrocortisone 2.5% only briefly, never primary.

Lichen simplex chronicus (itch–scratch lichenification; differential of lichen sclerosus): † break the itch–scratch cycle (potent topical steroid short-term, night sedation).

4 · PAPULOSQUAMOUS DISORDERS

Shared lesion: scaly papules/plaques. Differential includes tinea, erythrasma, secondary syphilis and neoplasia (Bowen, extramammary Paget, mycosis fungoides) — a "psoriasis" that fails treatment needs KOH, syphilis serology and biopsy.

Psoriasis

- ~2% of population; polygenic; one-third familial; triggers: trauma (Koebner), streptococcal infection, stress, drugs. Bimodal onset 20–30 and 50–60 years.
- Sharply demarcated erythematous plaque with silvery scale (rapid epidermal turnover). Look at elbows, knees, nails, scalp, **umbilicus**, natal cleft.
- **Inverse psoriasis**: folds lose scale → exclude tinea/*Candida* by KOH. Uncircumcised: plaques under prepuce.
- Histology: **"sandwich sign"** — alternating neutrophils and parakeratosis in stratum corneum.

Step	Genital psoriasis treatment
First-line	Low-potency steroid (hydrocortisone butyrate 0.1%; tar in 1% hydrocortisone) — **≤2 weeks** continuous; triamcinolone 0.1% for thick plaque
Steroid-sparing	Vitamin D_3 analogues (calcipotriene), topical calcineurin inhibitors, low-potency retinoids (may irritate)
Localised	308-nm excimer laser
Contraindicated	**PUVA** for genital psoriasis (dose-dependent genital SCC)
Extensive	Methotrexate, cyclosporine, retinoids, apremilast (phosphodiesterase-4), tumour necrosis factor-α (TNF-α) inhibitors, ustekinumab (IL-12/23), **ixekizumab (IL-17) — best data for genital psoriasis**

Reactive arthritis (formerly Reiter)

- Urethritis (*Chlamydia*, gonococcus, *Ureaplasma*) or enteric infection (*Yersinia, Salmonella, Shigella, Campylobacter*) in human leukocyte antigen (HLA)-B27 hosts; conjunctivitis, polyarthritis, sacroiliitis, oral ulcers. Only one-third have all features.
- **Balanitis circinata** (penis) and **keratoderma blennorrhagicum** (soles) look and stain like psoriasis — diagnosis is from **context**, not histology.
- Genital lesions self-limit over weeks–months; low-potency steroid. Sole lesions: tazarotene or acitretin.

Lichen planus (LP) and lichen nitidus

- LP: autoimmune attack on basal keratinocytes (lichenoid interface reaction). **Polygonal, violaceous, flat-topped papules**; on glans: papules, white reticular or annular (± ulcer). Koebner phenomenon. Oral LP commoner in women; search for drug/contact triggers.
- Differential: SCC in situ/invasive, Zoon balanitis, psoriasis, secondary syphilis, herpes, extramammary Paget. Biopsy small, multiple ulcerated lesions.
- Papular LP resolves in two-thirds by 1 year; **erosive mucosal LP persists, scars, and has SCC risk** → regular skin checks.
- Treatment (symptomatic only): **ultrapotent** topical steroid (clobetasol 0.05%) ± calcineurin inhibitor; severe → systemic steroid; others lack randomised evidence. Warn that erosive LP outcomes disappoint.
- **Lichen nitidus**: tiny flesh-coloured clustered papules; histologically distinct; resolves within a year — reassure ± mild steroid.

5 · LICHEN SCLEROSUS (LS) / BALANITIS XEROTICA OBLITERANS (BXO)

Pathology and clinical features

- Chronic inflammatory **scarring** disease, cause unknown, familial tendency; female 6–10×; in males usually >60 years. **15–20%** extragenital.
- Pallor, loss of architecture, hyperkeratosis, purpura (may mask whiteness), fissures. Male: glans and prepuce, **perianal spared** (unlike females).
- Male course: preputial scarring → **phimosis** → circumcision (usually curative; may recur in scar). Late stage = **BXO**, extends to meatus and penile urethra → **strictures**. Symptoms: pain on voiding or erection.
- Histology: basal vacuolation, epidermal atrophy, dermal oedema, **collagen homogenisation**, follicular plugging.
- **Premalignant**: associated with **HPV-independent** (human papillomavirus) penile and vulvar SCC. Biopsy to confirm and exclude malignancy; differential includes erosive LP, lichen simplex, morphoea, pemphigoid, child abuse.

Management

ALGORITHM
Biopsy → **potent steroid (clobetasol 0.05%) for 3 months** → taper → twice-weekly maintenance (or mometasone 0.1% / calcineurin inhibitor) → lifelong surveillance of non-healing fissures, ulcers, nodules

- Deliberately inverts the short-course rule because the disease scars and transforms. Evidence strongest in women and children; recurrence **40–50%** → proactive maintenance.
- Calcineurin inhibitors inferior to steroid; refractory → retinoids, photodynamic therapy.
- Surgery: circumcision for phimosis. † Meatal/urethral disease: meatotomy/meatoplasty; BXO strictures are best reconstructed with buccal mucosa graft (genital skin flaps recur) → *urethral reconstruction chapter*.

Disease	Steroid potency	Duration
AD / psoriasis / reactive arthritis	Low	Short (psoriasis ≤2 weeks)
Seborrhoeic dermatitis	Low	Initial severe phase only
Lichen planus	Ultrapotent	Until controlled
Lichen sclerosus	Potent	3 months → maintenance

KEY POINT Steroid strength tracks the scarring and malignant stakes of the disease; LS is the single exception to the short-course rule.

6 · VESICULOBULLOUS AND DRUG ERUPTIONS

Feature	Erythema multiforme (EM) minor	Stevens–Johnson syndrome (SJS, EM major)	Fixed drug eruption (FDE)
Trigger	**HSV** (lesions 10–14 days before)	**Drugs**: sulfonamides (co-trimoxazole), NSAIDs, penicillins, cephalosporins, anticonvulsants	Sulfonamides, NSAIDs, tetracyclines, barbiturates, carbamazepine; rarely finasteride, tadalafil, fluconazole
Lesion	Symmetric **target lesions**; palms, soles, mouth	Prodrome → red macules → blisters → necrosis; genital ~**60%**	Solitary violaceous plaque with halo, often glans; may blister
Course	Self-limited, recurs	4–6 weeks; mortality to ~**30%** (TEN overlap)	**Recurs at the same site within 24 h** of re-exposure

Feature	Erythema multiforme (EM) minor	Stevens–Johnson syndrome (SJS, EM major)	Fixed drug eruption (FDE)
Sequelae	None	Meatal stenosis, labial synechiae, vaginal/anal stricture	Brown pigmentation only
Treatment	Antihistamine; **suppressive acyclovir prevents** (useless once targets appear)	See algorithm	Stop drug

NSAIDs = non-steroidal anti-inflammatory drugs; TEN = toxic epidermal necrolysis.

ALGORITHM
SJS: stop offending drug → burn-type supportive care → intensive care / burn unit (lowers mortality) → multispecialty wound care; options with limited data: cyclosporine 3–5 mg/kg/day, etanercept, plasmapheresis, intravenous immunoglobulin (IVIG); systemic steroids controversial

KEY POINT The same drug class can cause a trivial FDE or a lethal SJS: extent and systemic illness, not the culprit, separate them. Genital FDE mimics herpes — the "same spot each time" history is the clue.

† **Autoimmune blistering** (not detailed in source): pemphigus vulgaris — intraepidermal (anti-desmoglein), flaccid bullae, erosions, Nikolsky positive; bullous/mucous membrane pemphigoid — subepidermal (anti-BP180/230), tense bullae; mucous membrane pemphigoid scars (genital synechiae, meatal stenosis). Diagnose by perilesional biopsy with direct immunofluorescence; treat with steroids/immunosuppressants (dermatology).

7 · NONINFECTIOUS GENITAL ULCERS

A genital ulcer is infectious (syphilis, chancroid, herpes, granuloma inguinale, lymphogranuloma venereum) until proven otherwise → *Ch 29*.

Behçet disease (BD)

- Relapsing autoimmune mucocutaneous vasculitis; **HLA-B51**; neutrophil superoxide excess. Highest in Turkey; equal sex ratio but **males more severe**.
- Triad: oral + genital ulcers + uveitis (blindness). Genital ulcers larger and more painful. Also epididymitis, thrombophlebitis, **pulmonary artery aneurysm**, gastrointestinal, neurological disease.
- Oral ulcer = most sensitive; **genital ulcer = most specific**. No confirmatory test.

International Criteria for BD (2014) — ≥4 points	Points
Ocular lesions; oral aphthosis; genital aphthosis	2 each
Skin lesions; central nervous system; vascular; positive pathergy (optional)	1 each

Treatment (rheumatology): colchicine, dapsone, steroids, cyclosporine, 5-aminosalicylates, **anti-TNF (infliximab, adalimumab)**; early aggressive therapy for organ involvement.

Pyoderma gangrenosum (PG) and trauma

- PG: neutrophilic dermatosis with inflammatory bowel disease, arthritis, myeloproliferative disease, HIV. Painful purulent ulcer with **undermined violaceous border** at trauma site (pathergy). Biopsy 5–10 mm from edge (neutrophils); no pathognomonic feature. Differential includes Fournier gangrene, calciphylaxis, metastatic Crohn. Treat: systemic + local steroids ± cyclosporine, anti-TNF, IVIG; topical calcineurin inhibitors for genital PG. † Avoid debridement (pathergy worsens it).
- **Factitial dermatitis / neurotic excoriation**: oddly angulated erosions; psychiatric association (borderline personality, obsessive-compulsive, depression); consider Munchausen, malingering.

8 · INFECTIONS

Balanitis and balanoposthitis

- Balanitis = glans; balanoposthitis = glans + prepuce (uncircumcised). Children: bacterial. Adults: hygiene, trauma, irritant dermatitis, *Candida*, bacteria, viruses.

ALGORITHM
Remove irritants → hygiene → topical antibiotic/antifungal → short low-potency steroid → **failure: biopsy** (neoplasia, Zoon, psoriasis, HPV, HSV) → recurrent with phimosis: **circumcision**

Type of balanitis	Clue	Management
Candidal	Satellite lesions; diabetes	Topical imidazole; check glucose
Irritant/allergic	Products, soaps	Avoidance
Circinate	Urethritis, HLA-B27	Low-potency steroid
Zoon (plasma cell)	Shiny moist red plaque, uncircumcised elderly	Biopsy → circumcision
LS / BXO	White sclerotic glans, phimosis	Potent steroid; circumcision
PEKMB	Mica-like scaly glans plaque, adult-circumcised	Biopsy; excision

Soft-tissue infection by depth

Entity	Level / organism	Sign	Management
Folliculitis	Follicle; *S. aureus*; hot tub → *Pseudomonas*	Perifollicular pustules	Hygiene; culture if refractory
Furuncle	Deep follicle abscess; *S. aureus*	Tender boil	Warm compress, incision and drainage; methicillin-resistant *S. aureus* (MRSA) cover; nasal mupirocin for recurrence
Erysipelas	Upper dermis + lymphatics; *S. pyogenes*	Sharp border	Antibiotics
Cellulitis	Deep dermis + subcutis; streptococci, *S. aureus*; mixed in diabetes	Indistinct border	Systemic antibiotics ± MRSA; **mark the border**
Fournier gangrene	Fascia; polymicrobial	Pain out of proportion, grey skin, odour, crepitus/gas	Emergency debridement → *Ch 26*

KEY POINT Fournier gangrene is a fascial disease — skin shows a fraction of the necrosis; testes are spared (own blood supply within tunica). Imaging (CT most specific for gas) must never delay surgery. Debride to bleeding tissue, second look at 24–48 h. Details → *Ch 26*.

- **Ecthyma gangrenosum**: gram-negative (*P. aeruginosa*) bacteraemia in immunocompromised; purpuric macule → haemorrhagic bulla → black eschar; biopsy shows necrotising vasculitis with "blue haze" of organisms. Immediate IV antipseudomonal antibiotics ± debridement; poor prognosis.
- **Genital bite**: *Eikenella corrodens* → rapid, very painful necrotic ulcer; amoxicillin–clavulanate 875/125 mg twice daily + irrigation, debridement, closure.

Hidradenitis suppurativa (HS, acne inversa)

- Chronic disease of apocrine-bearing skin (axilla, groin, perineum, perianal), after puberty; female, Black ancestry; familial up to 41%.
- Now **follicle-centred**: follicular plugging → rupture of keratin/bacteria into dermis → inflammation → sterile abscesses, **sinus tracts**, hypertrophic scars.
- Complications: fistulae to urethra, bladder, rectum, peritoneum; lymphoedema; amyloidosis; hypoproteinaemia; contractures; **SCC in chronic scars**.

† Hurley stage	Features	Mainstay
I	Abscess(es), no tracts or scarring	Topical clindamycin, general measures, deroofing of nodules
II	Recurrent abscesses with tracts and scarring; separated lesions	Oral tetracycline / clindamycin + rifampicin; deroofing; adalimumab if refractory
III	Diffuse interconnected tracts across whole area	Adalimumab + **wide excision** and grafting/flap

- General: smoking cessation, weight loss, friction/moisture control, metabolic and hormonal management.
- Medical: topical clindamycin (equal to oral tetracycline in randomised trial); clindamycin/minocycline + rifampicin; dapsone, zinc, acitretin, metformin; spironolactone (women), finasteride/dutasteride (men); **adalimumab 40 mg weekly**; steroids → relapse on stopping. **Avoid lithium**.
- Surgery: unroofing; **wide excision with grafting** or CO_2 laser for chronic disease. **Repeated incision and drainage is discouraged** — it leaves the tract to recur.

Red groin rash

Feature	Candidal intertrigo	Tinea cruris	Erythrasma	Inverse psoriasis
Organism	*Candida*	*Trichophyton, Microsporum, Epidermophyton*	*Corynebacterium minutissimum*	None
Scrotum	**Involved**	**Spared**	—	May be involved
Look	Macerated red; **satellite lesions**	Raised scaly advancing border, central clearing	Red-brown scaling patch	Non-scaling plaque
Test	KOH: yeast + pseudohyphae	KOH: branching hyphae	Wood's: **coral-red**	KOH negative
Treat	Topical imidazole **≥2 weeks**; fluconazole 150 mg occasionally; test for diabetes	Topical azole/allylamine; deep follicular → terbinafine 250 mg/day 1 week; no steroids	Antibacterial soap, clindamycin 1%, miconazole; oral erythromycin if severe	Low-potency steroid

> **KEY POINT** Antifungal spectrum: azoles cover yeast and dermatophyte (use when unsure); nystatin misses tinea; allylamines are weaker on *Candida*. A negative KOH from an untreated lesion → rethink the diagnosis.

Infestations

Feature	Pediculosis pubis	Scabies
Organism	*Phthirus pubis*	*Sarcoptes scabiei*; crusted (Norwegian) in immunosuppressed
Sign	Lice/nits on hair; eyelashes	Nocturnal itch; **serpiginous burrows**; finger webs, genitals
First-line	Permethrin 5% to hair-bearing areas, remove nits, **repeat at 1 week**	Permethrin 5% whole body overnight, **repeat at 1 week**, treat household
Alternatives	Oral ivermectin (refractory, eyelash); malathion 0.5%	Ivermectin 200 µg/kg ×2; lindane disfavoured (CNS toxicity)
Extra	Screen for gonorrhoea/chlamydia	Postscabies glans nodules → intralesional triamcinolone; itch persists weeks

Sexually transmitted infections (herpes, syphilis, condyloma, molluscum) → *Ch 29*.

9 · PREMALIGNANT AND MALIGNANT LESIONS

Penile intraepithelial neoplasia (PeIN) / squamous cell carcinoma in situ (SCCis)

- Full-thickness intraepidermal carcinoma; classified like invasive SCC as **HPV-related** (types 16, 8) or **non-HPV** (LS, chronic LP). Other risks: immunosuppression, radiation, arsenic, thermal injury.
- **Bowen disease**: solitary sharply demarcated red scaly plaque (shaft: thicker, verrucoid); seventh decade; indolent.
- **Erythroplasia of Queyrat**: SCCis of the glans/mucosa of the uncircumcised — velvety red plaque.
- **Bowenoid papulosis**: multiple small red-brown papules in sexually active young adults (third decade); HPV-16; discontinuous atypia; **regresses in men**; female partners need cervical screening.
- Diagnosis: biopsy **several areas** to exclude dermal invasion.

Lesion	Treatment
SCCis — accessible (scrotum)	Excision with **5-mm margin** (favoured)
SCCis — tissue preservation (glans)	Mohs micrographic surgery, laser, cryoablation; topical 5-fluorouracil (5-FU) or imiquimod 5% (higher recurrence). † Glans resurfacing for extensive/refractory glans disease
Bowenoid papulosis	Observation in reliable young man; or 5-FU 0.5%, tazarotene, imiquimod, cryo, CO_2 laser, electrocoagulation — relapse common
Verrucous carcinoma (Buschke–Lowenstein)	Local excision/Mohs; **no primary radiotherapy** (anaplastic transformation)
Pseudoepitheliomatous keratotic micaceous balanitis (PEKMB)	Biopsy (exclude verrucous/SCC); excision/ablation, 5-FU; refractory → glans resurfacing + split-skin graft

> **KEY POINT** HPV mapping: 6/11 → condyloma and verrucous carcinoma (low-grade, locally destructive, rarely metastasises); 16 → bowenoid papulosis and SCCis. Invasive penile SCC → *penile cancer chapter*.

Other genital malignancies

Tumour	Key features	Treatment
Basal cell carcinoma	Rare genitally (scrotum, vulva); pearly papule with telangiectasia	Excision/Mohs; advanced → hedgehog inhibitors (vismodegib, sonidegib)
Kaposi sarcoma	Human herpesvirus 8 (HHV-8); classic (elderly Mediterranean), AIDS (CD4 <500; genital ~3%), transplant; glans lesion can obstruct meatus	Restore immunity (antiretroviral therapy; switch calcineurin → mTOR inhibitor); solitary → excision/laser/intralesional vinblastine; regional → radiotherapy 15–30 Gy; disseminated → chemotherapy
Melanoma	<1% of penile cancers; irregular pigmented/amelanotic lesion	Wide excision / partial penectomy early; advanced → BRAF/MEK inhibitors, anti-CTLA-4, anti-PD-1
Extramammary Paget disease (EPD)	Sharp red plaque at apocrine sites; Paget cells (CEA, EMA +); underlying carcinoma **10–30%** (urethra, prostate, bladder, rectum)	**Mandatory search for internal malignancy**; excision/Mohs; imiquimod, 5-FU, radiotherapy
Cutaneous T-cell lymphoma (mycosis fungoides)	Pruritic patches (buttock, suprapubic) → plaques → tumours; Sézary if blood	Ultrapotent steroid, nitrogen mustard, phototherapy, total-skin electron beam, systemic

CEA = carcinoembryonic antigen; EMA = epithelial membrane antigen; mTOR = mammalian target of rapamycin; MEK = MAP-ERK kinase; CTLA-4 = cytotoxic T-lymphocyte antigen-4; PD-1 =

programmed death-1.

10 · BENIGN MALE GENITAL LESIONS

Lesion	Site / look	Mimic	Management
Pearly penile papules	Rows of white dome/filiform papules on corona; 14–48% young men; angiofibromas	Condyloma (no HPV, no partner risk)	Reassure; CO_2 laser/cryo if cosmetic
Ectopic sebaceous glands (Fordyce spots)	Pin-sized flesh-coloured shaft papules	Warts	Reassure
Angiokeratoma of Fordyce	1–2 mm red-purple scrotal papules; may bleed	Fabry disease (similar lesions)	Observe; laser if bleeding
Zoon balanitis	Shiny, moist, well-circumscribed red glans plaque; elderly uncircumcised; plasma cells	Erythroplasia of Queyrat, EPD	**Biopsy → circumcision (curative)**; else steroids, calcineurin inhibitors, laser
Sclerosing lymphangitis	Firm cord at coronal sulcus after vigorous sex	—	Self-limiting; ± urethritis screen
Median raphe cyst	Ventral penis near glans; no urethral communication	—	Excision under local

KEY POINT Zoon balanitis vs erythroplasia of Queyrat: both are red glans plaques in uncircumcised men — one benign and cured by circumcision, the other SCCis. Hence a persistent red glans plaque always needs biopsy.

11 · PIGMENTARY DISORDERS

- Vitiligo: Wood's lamp accentuates depigmentation; narrow-band UVB / 308-nm excimer laser for localised genital disease.
- † Autoimmune melanocyte loss; depigmented sharply bordered macules, Koebner phenomenon; genital vitiligo is not premalignant but must be distinguished from LS (vitiligo has normal skin texture, no atrophy or scarring). Topical steroid or calcineurin inhibitors ± phototherapy.

12 · MCQ ONE-LINERS

1. Size cut-off separating papule from nodule and vesicle from bulla: 0.5 cm.
2. Erosion heals without scar; ulcer (involves dermis) scars.
3. Prepuce enhances topical steroid penetration as an occlusive dressing.
4. Key defect in atopic dermatitis: filaggrin.
5. Commonest allergic contact allergen: nickel.
6. Severe allergic contact dermatitis: 3-week prednisone taper.
7. Most recurrent erythema multiforme minor is triggered by HSV.
8. SJS commonest cause: drugs; genital involvement ~60%.
9. FDE recurs at the same site, typically glans, within 24 h of re-exposure.
0. Psoriasis histology: "sandwich sign".
1. Biologic with best data for genital psoriasis: ixekizumab (IL-17).
12. PUVA contraindicated in genital psoriasis: genital SCC risk.
13. Balanitis circinata: HLA-B27 reactive arthritis; histology like psoriasis.
14. LP primary lesion: polygonal violaceous flat-topped papule.
15. LS is associated with HPV-independent penile SCC.
16. LS: clobetasol for 3 months then maintenance; recurrence 40–50%.
17. Late stage of male LS with stricture: balanitis xerotica obliterans.
18. Extensive seborrhoeic dermatitis: test for HIV.
19. Behçet: HLA-B51; genital ulcer most specific sign.
20. Behçet 2014 criteria threshold: ≥4 points.
21. PG biopsy taken from the ulcer edge; avoid debridement.
22. Fournier gangrene: testes spared; second look at 24–48 h.
23. HS: avoid repeated incision and drainage; adalimumab is effective biologic.

24. Lithium exacerbates hidradenitis suppurativa.
25. Erythrasma fluoresces coral-red under Wood's lamp.
26. Scrotal involvement favours Candida over tinea cruris.
27. Nystatin does not treat dermatophytes.
28. Genital bite organism: Eikenella corrodens; treat with amoxicillin–clavulanate.
29. Ecthyma gangrenosum: Pseudomonas bacteraemia, black eschar.
30. Permethrin is repeated at 1 week to kill newly hatched lice/mites.
31. Erythroplasia of Queyrat: SCCis of glans in uncircumcised men.
32. Bowenoid papulosis: HPV-16, young adults, may regress.
33. Buschke–Lowenstein: HPV 6/11; radiotherapy relatively contraindicated.
34. EPD: search for underlying urogenital or rectal carcinoma.
35. Zoon balanitis: plasma cell balanitis, cured by circumcision.

31 Tuberculosis and Parasitic Infections of the Genitourinary Tract

Reading time ≈ 61 min (slow pace)

1 · UROGENITAL TUBERCULOSIS — ORGANISM AND PATHOGENESIS

- **Urogenital tuberculosis (UG TB)** — preferred term over "genitourinary TB" because the urinary tract is involved far more often than the genital tract. Untreated, it causes irreversible damage (renal failure, infertility). TB is the second "great imitator" (after syphilis).
- **Organism**: *Mycobacterium tuberculosis* complex (MTBC), acid-fast bacilli. Commonest human isolate *M. tuberculosis*, then *M. bovis*; *M. africanum* in West Africa. Species are clinically identical but differ in drug susceptibility: ***M. bovis* is innately resistant to pyrazinamide** — relevant to bacille Calmette–Guérin (BCG, live attenuated *M. bovis*) infection.
- UG TB = **15–20%** of pulmonary TB in developing countries; second commonest extrapulmonary site after lymph nodes.

Infection and immunity

- Inhaled bacilli replicate in alveolar macrophages; before cellular immunity develops (up to 12 weeks) they spread via hilar nodes and blood to kidney and epididymis.
- **Granuloma**: foamy macrophages, epithelioid cells, Langhans giant cells, lymphocyte cuff. T cells secrete interleukin-2 (IL-2), tumour necrosis factor-α (TNF-α) and, most importantly, **interferon-γ (IFN-γ)**, which maintains the granuloma. If killing fails, bacilli become dormant.
- **Latent** (90–95%): calcified granuloma, not infectious; lifetime reactivation **5–15%**. **Active**: caseation → bacillary release. Reactivation triggers: old age, renal failure, diabetes, malnutrition, human immunodeficiency virus (HIV), immunosuppression.
- Isolated UG TB is generally **not infectious**, except via skin inoculation or aerosol-generating procedures on infected tissue or fluid — relevant in theatre.

Route	Frequency	Notes
Haematogenous	Principal	Seeds kidney and epididymis; rest of tract by contiguity. Latency up to decades (46 years) — so UG TB often appears without active lung disease
Ascending	Uncommon	Intravesical BCG: UG TB in **~1%** (pyelonephritis, renal abscess, ureteric obstruction, cystitis, prostatitis, epididymo-orchitis)
Contiguous	Rare	Spine → psoas → kidney; gut TB → enterorenal / enterovesical fistula

> **KEY POINT** UG TB is a blood-borne disease landing in the two best-perfused UG organs — kidney and epididymis — and spreading from there. Haematogenous seeding gives parenchymal disease; descending urinary spread gives luminal disease (distal ureteric stricture, cystitis). Healing by fibrosis turns an infection into an obstruction.

2 · CLINICAL FEATURES AND PATHOLOGY BY SITE

- Non-specific and indolent: patients are often repeatedly treated for bacterial infection or investigated for cancer. Renal TB is silent until it reaches the bladder.
- Constitutional symptoms in **< 20%** — absence does not exclude TB. **Lower urinary tract symptoms dominate** (dysuria, storage symptoms); haematuria and flank pain ~one-third; colic < 10% (sloughed papillae, clot, caseous debris).
- **Hallmark: sterile pyuria ± haematuria** (> 90% in developing countries) — should prompt TB work-up.

Site	Frequency / route	Signature lesion
Kidney	**80%**, haematogenous	Caseating cavities → autonephrectomy (up to one-third)

Site	Frequency / route	Signature lesion
Ureter	Descending	Distal stricture at vesicoureteric junction (VUJ); beaded corkscrew ureter
Bladder	Descending	Cystitis starting at ureteric orifices → contracted "thimble" bladder (< 20 mL)
Epididymis	10–55%, haematogenous; second commonest; bilateral ~one-third	Starts in **globus minor (tail)**; posterior scrotal sinus in up to half
Vas	Contiguous	Thickened, beaded
Testis	Contiguous from epididymis; isolated rare	Hard tumour-like mass; hydrocele ~5%
Prostate	22–49%	Chronic prostatitis unresponsive to antibiotics
Seminal vesicles	Via vas or ejaculatory ducts	Low-volume ejaculate, azoospermia, haemospermia — infertility may be first symptom
Urethra	1.9–4.5% (relatively resistant)	With prostatic disease; urethroscrotal fistula
Penis	Exceedingly rare	Ulcer → cavernosal nodules

Kidney

- Without imaging change: granulomatous interstitial nephritis (reversible renal failure), glomerulonephritis, amyloidosis, miliary TB.
- **Localised renal TB**: periglomerular granulomas coalesce → caseation → cavities → papillary necrosis, abscess, flank sinus. Calyceal inflammation → calcification → infundibular stenosis.
- **Autonephrectomy**: caseo-cavernous (cavities ± calcification) or fibrotic (shrunken, calcified **"putty" kidney**). End-stage renal failure ~7%.
- Persistent squamous metaplasia of the pelvis → long-term risk of squamous cell carcinoma (SCC).

Ureter and bladder

- Ureter: mural granulomas → scarring → stricture, commonest at VUJ; pan-ureteric disease gives a beaded corkscrew ureter. A scarred ureter can both obstruct and reflux. **Stricture-related obstruction is an important cause of renal failure in UG TB.**
- Bladder: starts around ureteric orifices; dome most affected, trigone spared. After ~1 year of fibrosis it contracts (< 100 mL → severe storage symptoms; thimble < 20 mL); commoner where diagnosis is late.

Genital tract

- Epididymis: globus minor (more vascular) → caseous nodules → skin adherence → **posterior scrotal sinus**. Usually extends into testis to form a tumour-like mass.
- Prostate: haematogenous → peripheral lesions, calcified hard gland; urinary route → mimics bacterial prostatitis. **Clue**: quinolones are active against MTBC, so short prostatitis courses give a partial response then relapse.
- Penis: **papulonecrotic tuberculid** is a hypersensitivity reaction — culture and polymerase chain reaction (PCR) negative; diagnosed by response to empirical treatment.

3 · DIAGNOSIS

ALGORITHM
Hierarchy: culture or MTBC DNA positive → diagnosis. Negative → caseating granulomas on tissue in right context. Negative → compatible picture + exposure + response to empirical treatment. Always look for coexisting pulmonary TB (up to one-fifth): chest radiograph in every patient.

Urine culture — gold standard

- **Early-morning (first-void) urine, 3–5 consecutive days**, cultured promptly — acidic urine inhibits mycobacteria. Culture any tissue too.

- Sensitivity 10–90% (intermittent shedding). Ziehl–Neelsen smear of urine for acid-fast bacilli (AFB): sensitivity **< 50%**.
- Media: **Löwenstein–Jensen** (solid, 4–6 weeks; cheap) plus liquid Mycobacteria Growth Indicator Tube (MGIT, ~10 days). Only culture gives susceptibility and monitors treatment.

Nucleic acid amplification tests (NAAT)

- Result in 1–2 days; detect MTBC below culture threshold. Urinary inhibitors reduce sensitivity.
- **GeneXpert MTB/RIF**: automated cartridge; detects MTBC **and rifampicin resistance** — a surrogate for multidrug-resistant TB (MDR-TB = resistance to isoniazid [INH] + rifampicin), as > 90% of rifampicin-resistant strains are INH-resistant.
- World Health Organization (WHO): part of the initial work-up of all TB. United States: adjunct; a negative NAAT does not exclude extrapulmonary TB.
- **Cannot monitor treatment**: dead bacilli still shed DNA.

Histology and endoscopy

- Biopsy when cultures negative; consistent with TB in only one-fifth to one-third. Adding NAAT to tissue improves yield. Epididymal nodule fine-needle aspiration cytology diagnostic in ~two-thirds.
- Cystoscopy: hyperaemia, ulcers (may mimic cancer), tubercles, **"golf-hole" ureteric orifice** (retracted, gaping orifice from ureteric fibrosis) → image the upper tract. Bladder biopsy yield 19–52% in culture-negative cases; biopsy if malignancy possible.

Tuberculin skin test (TST) and interferon-γ release assays (IGRA)

- Both show **exposure only** — cannot separate latent from active TB; a positive does not rule TB in, a negative does not rule it out. Use: screening for infection. TST is usually positive in UG TB.
- TST (Mantoux) read at 48–72 h; cut-offs by risk: **≥ 5 mm** contacts, immunosuppressed; **≥ 10 mm** immigrants, high-risk workers, drug injectors; **≥ 15 mm** no risk factors. False positive with BCG; false negative 10–25%.
- IGRA (QuantiFERON, T-SPOT.TB): blood test with antigens absent from BCG — **no BCG false positives**; single visit; preferred screen for those from endemic countries.

Imaging

Modality	Signs / role
Plain kidney–ureter–bladder (KUB) film	Calcification in 24–44%: punctate → globular → triangular/ring (papillary necrosis). **Lobar "putty"/"cement" kidney** — calcific rims outlining lobes, pathognomonic of end-stage renal TB. Arrowhead stone in "hiked-up" pelvis. Ureteric calcification **intraluminal** (schistosomiasis: mural)
Intravenous urography (IVU)	Former gold standard. Early: **moth-eaten calyx**; **phantom calyx** (stenosed infundibulum, calyx not opacified); cloverleaf calyces; **Kerr's kink** (hiked-up pelvis, sharp pelviureteric junction angle); **pipe-stem ureter** (rigid, straight, aperistaltic); beaded corkscrew ureter; contracted bladder. Commonest findings are **obstructive** (infundibular stenosis, hydrocalycosis, distal stricture)
Computed tomography (CT) urography	Now standard: more sensitive for calcification and wall thickening; large granuloma may mimic renal cell carcinoma; abscesses, psoas, nodes, spine. Less sensitive than IVU for earliest papillary change
Ultrasound (US)	Children, pregnancy, scrotum, transrectal. **Main use: serial monitoring of hydronephrosis during treatment**
Magnetic resonance imaging (MRI)/urography	Radiation-free; diffusion-weighted imaging separates pyonephrosis from hydronephrosis and grades fibrosis. Avoid gadolinium in renal failure

> **KEY POINT** Imaging signs are fibrosis made visible: destruction (moth-eaten calyx, cavities) heals by scarring (phantom calyx, Kerr's kink, pipe-stem ureter, golf-hole orifice, thimble bladder). Hence obstruction is the commonest finding and must be watched during treatment.

4 · MEDICAL TREATMENT

- **Why multidrug**: bacilli exist in different micro-environments and metabolic states; drugs differ in cidal action, target population, penetration and optimal pH; combinations prevent resistance.
- Baseline: liver and renal function, platelets, HIV test; hepatitis B/C and diabetes where appropriate. Directly observed therapy (DOT) ensures adherence.
- First-line drugs achieve high urinary concentrations and work in acidic urine — well suited to UG TB.

ALGORITHM

Standard 6-month short course (2HRZE / 4HR): Intensive 2 months — daily INH (H) + rifampicin (R) + pyrazinamide (Z) + ethambutol (E); stop E once fully susceptible → kills rapidly multiplying bacilli. Continuation 4 months — HR daily (or thrice weekly by DOT; twice weekly no longer recommended) → eradicates persisters. Pyridoxine throughout.

Drug	Daily adult dose	Signature toxicity	Monitor / note
INH	5 mg/kg (300 mg)	Peripheral neuropathy, hepatitis	Pyridoxine 25–50 mg
Rifampicin	10 mg/kg (600 mg)	Hepatitis, flu-like syndrome, **drug interactions**	Antiretroviral and immunosuppressant levels; rifabutin preferred with antiretrovirals (uveitis)
Pyrazinamide	Weight-banded 1000–2000 mg	Hyperuricaemia, arthralgia, hepatitis	*M. bovis* resistant
Ethambutol	Weight-banded 800–1600 mg	Optic neuritis (red–green colour loss)	Visual acuity, colour vision; **only non-hepatotoxic** first-line drug

- **Not for UG TB**: weekly rifapentine and the new 4-month rifapentine–moxifloxacin regimen (not approved for extrapulmonary TB).
- Longer courses (12–24 months, or ≥ 6 months after surgery) used by some for complicated disease (abscess, extensive scarring) because UG TB relapses more than pulmonary TB — decide with a TB specialist.
- **Monitoring**: liver enzymes (monthly if liver disease; hepatotoxicity reverses on stopping); vision on ethambutol; urine culture for response; **serial US for obstruction** — healing fibrosis can worsen strictures and bladder contraction during treatment.

Corticosteroids

- Rationale: limit immune-mediated destruction and scarring. Proven in TB meningitis (mortality) and pericarditis (constriction). In UG TB used to prevent ureteric stricture and bladder contraction on **anecdotal evidence only**; consider when a stricture worsens on treatment. Rifampicin increases steroid metabolism†, so doses need adjustment.

5 · SURGICAL TREATMENT

- ~Half of patients need surgery; more than half of operations are now **reconstructive**. Overriding goal: organ preservation.
- Indications: relieve obstruction / drain abscess; remove a non-functioning infected kidney resistant to cure; medically resistant hypertension from a non-functioning kidney; reconstruction.
- **Timing: 4–6 weeks after starting chemotherapy** — inflammation subsides, bacillary load falls, lesions stabilise (emergencies excepted).

Relief of obstruction

ALGORITHM
Uraemia or sepsis → emergency decompression → retrograde double-J stent (preferred; succeeds in < half, 41%) → fails → antegrade stent → fails → percutaneous nephrostomy (PCN; loculated systems may need more than one) → definitive repair; unsalvageable kidney → nephrectomy.

- Early stenting or PCN limits nephron loss and keeps reconstruction possible. Avoid high contrast pressure (dissemination). Removing a PCN without correcting the cause risks a tuberculous cutaneous fistula.

Nephrectomy

- Partial nephrectomy: localised polar lesion unresponsive to drugs.
- **Total nephrectomy — established**: non-functioning kidney with recalcitrant/recurrent TB despite optimal therapy; or with medically resistant hypertension (improves in ~two-thirds). **Contested**: routine removal of an asymptomatic non-functioning kidney — no supporting evidence in the modern drug era.
- Open technique: oblique extendable retroperitoneal incision (dense fibrosis); remove caseous perinephric tissue en bloc; **ligate artery and vein separately** (late arteriovenous fistula); ureter usually left; avoid pleura and peritoneum. Laparoscopy increasingly preferred in experienced hands.

Ureteric strictures

- **Lower ureteric strictures commonest**, often need open surgery; decide by length, guidewire passage, blood supply, renal function.
- Endoscopic: TB strictures are ischaemic and densely fibrotic, so results are poorer than for other strictures. Best: short, residual lumen, good function. Early double-J stenting of strictures arising on chemotherapy may stabilise them. Balloon dilatation: high failure, repeats common.
- **Image every stricture** (US or IVU) during treatment; worsening → add steroids; **no improvement after 6 weeks of chemotherapy → open surgery**.

Site / gap	Procedure	Key point
Pelviureteric junction (PUJ), short	Dismembered pyeloplasty; flap pyeloplasty if longer	Harder than congenital PUJ obstruction
PUJ unreconstructable	Ureterocalicostomy (to lower-pole calyx)	Cover lower pole with capsule or omentum to prevent stenosis
Upper/mid ureter	Excision + tension-free ureteroureterostomy; Davis intubated ureterotomy	—
Lower ureter	Excise to healthy ureter → ureteroneocystostomy	Bladder mobilisation gains 2–3 cm
Moderate gap	Psoas hitch	Up to **5 cm**; needs good capacity; avoid genitofemoral/femoral nerves
Long gap	Boari flap ± hitch	**10–15 cm**; poor in contracted bladder
Multiple / recurrent	**Ileal ureter** (isoperistaltic interposition)	When the native ureter is unusable

> **KEY POINT** Trap: in TB the bladder itself may be contracted, ruling out bladder-based bridges (psoas hitch, Boari flap) — assess capacity before choosing; ileal ureter ± augmentation then becomes the option.

Contracted bladder

- Indication: intolerable frequency, urgency, pain, haematuria, typically capacity **< 100 mL**. Operate after chemotherapy has controlled active disease.

- Half the bladder diseased → ileal augmentation; severely contracted → ileocaecal or sigmoid augmentation; **thimble bladder (< 20 mL) → orthotopic substitution**. Usual bowel rules: check renal function, detubularise to a low-pressure reservoir, teach clean intermittent self-catheterisation†, long-term follow-up. Complications: mucus, electrolyte (hyperchloraemic acidosis) disturbance, infection. Exclude an obstructed or refluxing ureter and reimplant at augmentation†.
- Urethra: bladder neck contracture → incision; stricture → endoscopic; fistula → suprapubic diversion, delayed repair.

Genital surgery

- Only after failed medical therapy (abscess, persistent sinus, mass). Testis spared → **epididymectomy** (ligate vas, start at globus minor, preserve testicular blood supply); testis involved → scrotal orchidectomy. The vas is diseased distal to the external ring, so ligation there suffices.
- Infertility: obstructive (epididymal/vasal/ejaculatory duct) azoospermia often irreversible despite cure; offer sperm retrieval with assisted reproduction†.

6 · BCG-RELATED UROGENITAL INFECTION

- Ascending UG TB in ~1% after intravesical BCG for non-muscle-invasive bladder cancer: granulomatous prostatitis, epididymo-orchitis, pyelonephritis/renal abscess, ureteric obstruction; rarely systemic BCG sepsis. Mantoux is unhelpful (BCG positive).
- Management†: local symptomatic cystitis → withhold BCG, symptomatic care, quinolone; persistent or organ infection → INH + rifampicin (+ ethambutol) for 3–6 months; **pyrazinamide is useless (*M. bovis*)**; systemic BCG sepsis → triple therapy + steroids, stop BCG permanently. → *bladder cancer chapters*

7 · RELAPSE, SPECIAL SITUATIONS

- Relapse after UG TB **6.3–22%** (vs 2–6% pulmonary), mean time ~5 years — a diseased kidney holds many hard-to-sterilise foci. Follow-up: some advise **10 years**. Culture any recurrence for susceptibility.

Situation	Change	Duration
MDR-TB (INH + R resistant)	Second-line; BPaL (bedaquiline + pretomanid + linezolid), all-oral, not yet approved for extrapulmonary TB	> 18 months; 6–9 months with BPaL
Extensively drug-resistant (XDR) TB	MDR + fluoroquinolone + bedaquiline/linezolid resistance; 5–6 drugs	≥ 2 years
Pregnancy	Treat promptly (TB riskier than drugs). HRE + pyridoxine; pyrazinamide omitted in United States (WHO includes it). Breastfeeding allowed	9 months (without Z)
HIV on effective antiretroviral therapy (ART)	Standard daily; rifabutin with protease inhibitors	6 months (9 if not on ART)
Renal transplant	Rifamycin with drug-level monitoring, or rifamycin-free	≥ 18 months if rifamycin-free

- **HIV**: active TB risk 15–20-fold; test every TB patient. UG TB is more disseminated and bilateral but with **less fibrosis, fewer strictures and contracted bladders** (fibrosis needs an intact immune response); mortality higher.
- **Transplant**: allograft TB within 6 months, presents with fever, imaging normal; **urine AFB culture is the key test**.

8 · SCHISTOSOMIASIS (BILHARZIA)

Organisms and life cycle

Species	Distribution	Target / venous home	Snail
S. haematobium	Africa, Arabian Peninsula	Vesical and pelvic venous plexus — UG tract	***Bulinus***
S. mansoni	Africa, Middle East, South America	Mesenteric veins — liver, gut	*Biomphalaria*
S. japonicum	China, South-East Asia	Mesenteric veins — liver, gut	*Oncomelania*

ALGORITHM
Fork-tailed cercariae penetrate intact skin in fresh water → schistosomulae → lungs → liver sinusoids → paired adults (female in male's gynaecophoric canal) migrate to vesical plexus (80–110 days; live 3–5 years) → eggs (~140 µm, **terminal spine**) cross bladder wall into urine → in fresh water hatch to miracidia → *Bulinus* snail → asexual multiplication → cercariae.

- **Eggs retained in tissue cause the disease**; adults evade immunity via the tegument and cannot multiply in humans.
- Risk follows freshwater contact (children 5–15 years heaviest burden); dams and irrigation increase transmission.

Pathology

- **Egg granuloma** (eosinophils, lymphocytes, plasma cells) → coalescent polypoid masses (active stage) → fibrosis and calcification → **"sandy patches"** (yellow granular lamina propria of calcified eggs) → obstruction, metaplasia, cancer.
- Calcified, non-compliant bladder and ureter raise upper-tract pressure → hydroureter precedes hydronephrosis; tubular (concentrating) function lost before glomerular.
- Obstruction overwhelmingly **distal** (intramural, juxtavesical, lower third); meatus rarely (~1%).
- Hydroureter patterns: **segmental** (fibrosis/sandy patch, little hydronephrosis); **tonic** (hypertrophied, trabeculated, impaired peristalsis — reversible once obstruction relieved); **atonic** (thin, fibrotic, aperistaltic — failed).
- Bacterial urinary tract infection (UTI) common; ***Salmonella*** shelters in the worm tegument → chronic *Salmonella* bacteriuria/bacteraemia; antibiotics alone fail — treat both infections.

Bladder cancer

- International Agency for Research on Cancer **Group 1 carcinogen**. Pathway: hyperplasia → squamous metaplasia → dysplasia → carcinoma. Mechanisms: chronic inflammation, VEGF, bacterial nitrosamines (from nitrates) alkylating DNA (p53 mutations).
- Schistosomal cancer: younger (40–50 years), **SCC 60–90%** (often well-differentiated, exophytic), adenocarcinoma 5–15%, posterior wall commonest. Mass drug administration (MDA) in Egypt shifted SCC towards urothelial carcinoma — evidence of causation. Managed surgically → *bladder cancer chapters*.

Clinical features

- **Acute** (travellers): cercarial dermatitis ("swimmer's itch"); **Katayama fever** 2–8 weeks later (fever, urticaria, eosinophilia) — rare with *S. haematobium*.
- **Chronic**: egg-laying from 8–12 weeks → **painless terminal haematuria**, dysuria, frequency (in endemic areas male haematuria seen as "puberty"); anaemia, proteinuria.
- Stages: **active** (polyps, haematuria) → **late chronic active** (schistosomal contracted bladder, capacity to ~50 mL, trigone spared) → **quiescent** (eggs fall, symptoms fade, silent obstruction progresses) → **chronic inactive** (no viable eggs; 40–60% with obstructive uropathy first present here).

- **Female genital schistosomiasis** (one-third to three-quarters of infected women): friable sandy patches of cervix and vagina — contact bleeding, dyspareunia, infertility; may increase HIV susceptibility. **Male genital**: haemospermia (eggs may appear in semen before urine), epididymitis or funiculitis mistaken for tumour, infertility (often reversible).

> **KEY POINT** Trap: a negative urine egg count does not exclude severe disease — the most damaged kidneys belong to egg-negative, often asymptomatic, inactive-stage patients.

Diagnosis

Test	Meaning	Limitation
Urine egg microscopy — **gold standard**	Active infection	Excretion peaks at noon: collect **09:00–15:00**; concentrate. Negative first 8–12 weeks and in light or inactive disease
Rectal snip (before bladder biopsy) — squash preparation	Active/past; shows egg viability	Invasive; rectal snip avoids bladder biopsy risks
Serology (FAST-ELISA + Western blot)	Exposure (> 90% sensitive; positive from 4–6 weeks)	Persists after cure — **never use to judge cure**
Circulating anodic/cathodic antigen	Active infection (live worms)	Urine CCA strip poor for *S. haematobium*
PCR	Most sensitive	Impractical in the field

- Imaging: US (wall thickening, polyps, hydronephrosis); plain film — **calcified bladder like a "fetal head in the pelvis"**, mural ureteric calcification; IVU — ureteric striations (earliest), filling defects, tonic vs atonic ureter; voiding cystourethrogram — reflux ~25%. Cystoscopy: sandy patches, polyps, ulcers, golf-hole or stenotic orifices, tumours.

Treatment

- **Praziquantel (PZQ)** — only WHO drug: **40 mg/kg once**, or 20 mg/kg twice 6–8 h apart. Cure/egg reduction 60–90%; even partial response cuts worm burden. Mild transient side effects. Pregnancy: United States Food and Drug Administration (FDA) category B; many programmes exclude, though adverse effects are rarely reported. Katayama fever: add steroids.
- PZQ is weak against schistosomulae → repeat after several weeks to catch maturing juveniles. Artemisinins act on schistosomulae (chemoprophylaxis) — avoid wide use where malaria is co-endemic.
- Active-stage polypoid ureteric obstruction resolves with PZQ alone (renal function recovers in 1–2 months; childhood polyps in 2–6 weeks). Late fibrotic stenosis (up to 80% of obstructed ureters) responds poorly → surgery.

Complication	Surgery
Distal / intramural ureteric stenosis	Reimplantation with antireflux tunnel (Politano–Leadbetter variants); balloon dilatation (restenosis common); meatoplasty for isolated meatal stenosis
Long or multifocal stenosis	Boari flap; ileal ureter (isoperistaltic); ileal conduit; long-term nephrostomy if unreconstructable
Contracted bladder	Ileocystoplasty or diversion, always with PZQ
Chronic deep ulcer	Partial cystectomy (fulguration fails)

9 · FILARIASIS

Organisms and pathology

- Lymphatic filariasis (LF): mosquito-borne (*Culex* in urban India) *Wuchereria bancrofti* (> 90%), *Brugia malayi*, *B. timori*.

- Adults live in lymphatics 5–7 years; **only *W. bancrofti* lives in male genital lymphatics** — genital filariasis = bancroftian (Brugia affects the leg below the knee). Microfilariae are **nocturnally periodic** (peak midnight).
- *Wolbachia* endosymbiont (needed for worm embryogenesis) — killed by doxycycline.
- Repeated infection over years needed → disease in long-term residents; genital disease rare before teens. Lymphatic dilatation → dying worms, Wolbachia and bacterial infection inflame → scarring, obstruction → hydrocele, elephantiasis.

Manifestations

- **Acute adenolymphangitis**: fever + **retrograde** lymphangitis (spreads away from node; bacterial lymphangitis ascends).
- **Funiculoepididymitis**: acute painful cord-like swelling, recurrent; testis spared, so sterility rare; often mistaken for tumour → unnecessary orchidectomy.
- **Hydrocele**: commonest chronic male lesion; may be huge. Suggestive: cord/epididymal nodules, endemic residence, thick fibrous tunica with cholesterol or calcium.
- **Genital elephantiasis**: late; lymph scrotum (ruptured vesicles leaking lymph, superinfection).
- **Chyluria**: rupture of dilated retroperitoneal lymphatics into the renal collecting system → milky urine, fat and protein loss, malnutrition; intermittent, may remit; earlier than elephantiasis.

Diagnosis

- Night (midnight) Giemsa blood smear for microfilariae — positive in a minority. **Circulating antigen** card/strip test (*W. bancrofti* only; none for Brugia). PCR; antibody (IgG4) shows exposure only. US "**filarial dance sign**" (live adults in scrotal lymphatics). Lymphoscintigraphy maps lymphatic damage.
- "Burned-out" disease: lymphoedema/hydrocele with negative smear and antigen — negative tests do not exclude LF.
- Chyluria work-up†: milky urine clearing with ether/chloroform, Sudan III fat stain, urinary triglyceride and chylomicrons; chylous clot; cystoscopy to lateralise milky efflux from a ureteric orifice; lymphangiography / lymphoscintigraphy or MR lymphangiography to show pyelolymphatic fistula; exclude non-filarial causes (TB, tumour, trauma).

Treatment

Drug	Action	Regimen / caution
Diethylcarbamazine (DEC)	Microfilaricidal, modest on adults	Drug of choice for active LF: 6 mg/kg/day (2 mg/kg three times daily); single-day course ≈ 12-day course. **Contraindicated if onchocerciasis or *Loa loa* possible** (blindness, encephalopathy)
Albendazole	Micro- and macrofilaricidal	400 mg twice daily 21 days; single dose in MDA
Ivermectin	Microfilaricidal	150–400 µg/kg once; caution with *Loa*
Doxycycline	Anti-*Wolbachia*; sterilises/kills adults	200 mg daily 4–8 weeks; improves lymphoedema and hydrocele; not in pregnancy or children

- Treat all with microfilaraemia, even if asymptomatic.
- MDA: annual albendazole + ivermectin (Africa, where onchocerciasis/*Loa* co-endemic) or albendazole + DEC (elsewhere, incl. India); triple therapy (IDA) more suppressive.
- Lymphoedema: hygiene, prevent bacterial infection, elevation, compression, physiotherapy.
- **Hydrocele**: aspiration recurs; large/symptomatic → **excision of the sac** (procedure of choice; eversion with partial excision alternative); ligate leaking lymphatics. Funiculoepididymitis: excise nodules preserving testis and cord. Elephantiasis: reduction scrotoplasty for selected; lymphadenectomy worsens drainage.
- **Chyluria management†**: conservative first — low-fat, high-protein diet with medium-chain triglycerides (absorbed via portal vein, bypassing lymphatics), fluids, DEC; most remit.

Persistent/nutritional loss → **renal pelvic instillation sclerotherapy** (povidone-iodine or silver nitrate via ureteric catheter) of the side identified at cystoscopy; failure → **chylolymphatic (renal hilar lymphatic) disconnection**, open or laparoscopic: stripping hilar lymphatics around vessels, ureter and pelvis; lymphovenous anastomosis alternative.

- Onchocerciasis ("hanging groin"): ivermectin then doxycycline; DEC blinds. Loiasis (haematuria ~30%): encephalopathy risk with high microfilaraemia.

10 · HYDATID DISEASE OF THE KIDNEY AND OTHER PARASITES

Echinococcosis

- *Echinococcus granulosus* (dog tapeworm); eggs ingested in pastoral areas. Kidney is the **third site** (2–3%) after liver and lung. Slow growth (1–2 cm/year) → mass, pain; **hydatiduria** (grape-skin daughter cysts in urine) and colic when it ruptures into the collecting system (communicating cyst). Function usually preserved.
- Imaging: thick-walled spherical cyst, calcified rim, daughter cysts / floating membranes ("water-lily" sign†); appearance stages the cyst. Serology adjunctive (60–90%). Avoid diagnostic aspiration (anaphylaxis, seeding)†.
- **Albendazole 400 mg twice daily for 1–6 months**; surgery for size or location. PAIR (puncture, aspiration, injection, re-aspiration) suits liver, **not kidney** (risk to collecting system).
- Surgical precautions: perioperative albendazole (± PZQ) to reduce secondary seeding; **rupture → anaphylaxis**. †Pack the field with hypertonic (20%) saline–soaked swabs; aspirate cyst, then inject scolicidal agent only if no collecting-system communication (chemical injury); remove intact (pericystectomy, cystectomy) or partial nephrectomy; nephrectomy only for a destroyed kidney; close cyst–calyx communications; anaesthetist ready for anaphylaxis.

Others

- **Enterobiasis** (pinworm): perianal pruritus; ectopic migration in women → vulval/cervical granulomas, salpingitis, tubo-ovarian abscess; rarely epididymis. Albendazole 400 mg or mebendazole 100 mg once, repeat at 2 weeks; treat household.
- **Amoebiasis** (*Entamoeba histolytica*): renal involvement, painful genital/perianal ulcers. Tinidazole or metronidazole, then luminal agent (paromomycin or iodoquinol).
- **Trichomoniasis**: sexually transmitted protozoan → *Ch 29*. Fungal urinary infection → *Ch 26*.

11 · MCQ ONE-LINERS

1. Commonest site of UG TB: kidney (80%); commonest genital site: epididymis.
2. Epididymal TB starts in the globus minor (tail).
3. Hallmark urine finding in UG TB: sterile pyuria.
4. Urine for AFB culture: early-morning samples on 3–5 consecutive days.
5. GeneXpert detects MTBC plus rifampicin resistance (MDR surrogate).
6. NAAT cannot monitor treatment — dead bacilli shed DNA.
7. IGRA has no BCG false positives; neither IGRA nor TST separates latent from active.
8. Pathognomonic end-stage renal TB on KUB: lobar calcified "putty" kidney.
9. Phantom calyx = stenosed infundibulum, non-opacified calyx.
0. TB ureteric calcification intraluminal; schistosomal mural.
1. Commonest site of TB ureteric stricture: VUJ.
2. Thimble bladder: capacity < 20 mL → orthotopic substitution.
3. Golf-hole ureteric orifice: retracted orifice from ureteric fibrosis.
4. Only non-hepatotoxic first-line anti-TB drug: ethambutol.
5. *M. bovis* (and BCG) is innately pyrazinamide-resistant.
6. Elective surgery: 4–6 weeks after starting anti-TB therapy.
7. Stricture not improving after 6 weeks of chemotherapy → open surgery.
8. TB nephrectomy: ligate renal artery and vein separately (AV fistula).
9. UG TB relapse up to 22%; follow up ~10 years.
20. HIV-positive UG TB: less fibrosis, fewer strictures, higher mortality.
21. *S. haematobium* snail host: *Bulinus*; egg has terminal spine.

22. Schistosomal bladder cancer: mainly SCC, younger patients.
23. "Fetal head in the pelvis": calcified schistosomal bladder.
24. Praziquantel 40 mg/kg single dose for *S. haematobium*.
25. *Salmonella* sheltering in the schistosome tegument causes chronic bacteriuria — treat both.
26. Genital filariasis is caused by *W. bancrofti* only.
27. Filarial dance sign: live adult worms on scrotal US.
28. DEC contraindicated with onchocerciasis or *Loa loa*.
29. Persistent filarial chyluria: pelvic instillation sclerotherapy, then hilar lymphatic disconnection.
30. Renal hydatid: albendazole ± surgery; PAIR not used for kidney.

★ Putting it together: infection decisions

Reading time ≈ 6 min (slow pace) · Integrates Ch 26–31 · quick-reference for ward, clinic and viva

A · URINARY TRACT INFECTION (UTI): MATCH THE DRUG TO THE SITE

Situation	Choice	Principle
Uncomplicated cystitis	Nitrofurantoin 5 days, fosfomycin 3 g single dose, trimethoprim–sulfamethoxazole 3 days (if local resistance <20%), pivmecillinam	Narrow, urine-confined, flora-sparing drugs
Pyelonephritis, complicated UTI, men	Tissue-penetrating agent (fluoroquinolone, cephalosporin; carbapenem for resistant organisms) guided by culture	Nitrofurantoin and fosfomycin do not reach tissue
Extended-spectrum β-lactamase (ESBL) organism	Fosfomycin or nitrofurantoin for cystitis; carbapenem for invasive infection	Plasmid co-resistance to quinolones and aminoglycosides
Asymptomatic bacteriuria	Treat only in pregnancy and before mucosa-breaching urological surgery	Otherwise treatment causes harm and resistance

B · WHEN DOES A UTI NEED MORE THAN ANTIBIOTICS?

ALGORITHM

Image (US or CT) if: no response by 72 hours, sepsis, suspected obstruction or stone, gas-forming infection, men with febrile UTI, recurrent pyelonephritis.

Drain any obstructed infected kidney (stent or nephrostomy) ; renal abscess ≤3 cm → antibiotics alone, and abscesses >3–5 cm; treat the stone later.gt;5 cm → percutaneous drainage. Treat the stone later.

Emphysematous pyelonephritis: CT class guides care. Gas confined to the collecting system or kidney → antibiotics + drainage; extension beyond the kidney with poor prognostic features → nephrectomy (→ *Ch 26*).

Fournier gangrene: resuscitation, broad-spectrum antibiotics and **urgent radical debridement**, repeated until healthy tissue; scores (LRINEC, FGSI) help, but surgery must not wait for them.

C · RECURRENT UTI IN WOMEN: PREVENTION LADDER

ALGORITHM

Confirm with culture; exclude anatomical causes in atypical cases → behavioural measures and hydration → vaginal oestrogen (postmenopausal) → non-antibiotic options (methenamine hippurate, cranberry, D-mannose, immunostimulants) → self-start or post-coital antibiotic → continuous low-dose prophylaxis.

D · PELVIC PAIN SYNDROMES COMPARED

	Chronic prostatitis / chronic pelvic pain syndrome (CP/CPPS)	Interstitial cystitis / bladder pain syndrome (IC/BPS)
Core	Pelvic pain in men >3 months without proven infection (NIH category III)	Bladder pain related to filling, with urinary symptoms, no other cause
Phenotyping	UPOINT domains; NIH-CPSI score	Hunner lesion vs non-Hunner on cystoscopy
Treatment	Multimodal, by domain: α-blocker, antibiotic only if infection, pelvic floor physiotherapy, neuromodulators, psychological support	Behavioural → oral (amitriptyline, pentosan polysulfate) → intravesical → Hunner lesion fulguration/injection → botulinum toxin, neuromodulation → diversion last

E · GENITOURINARY TUBERCULOSIS IN FOUR LINES

ALGORITHM

Suspect: sterile pyuria, recurrent "UTI" with negative cultures, contracted bladder, beaded vas or epididymal mass.

Confirm: three early-morning urine samples for acid-fast bacilli, culture and nucleic-acid amplification (GeneXpert).

Treat: 2 months isoniazid, rifampicin, pyrazinamide, ethambutol, then 4 months isoniazid + rifampicin; watch for ureteric stricture during treatment.

Operate after at least 4–6 weeks of drugs: nephrectomy for a non-functioning kidney, reimplantation for stricture, augmentation for a contracted bladder (→ *Ch 31*).